Mapping Europe's Digital Acquis

A Granular History of EU Digital Policymaking

Timo Seidl[1*] & Nir Kosti[2, 3]

[1]Munich School of Politics and Public Policy, Technical University of Munich, Munich, Germany

[2]Geschwister-Scholl-Institut für Politikwissenschaft, Ludwig Maximilian University of Munich, Munich, Germany

[3]The Hebrew University of Jerusalem, Jerusalem, Israel

September 21, 2026

Over the last decade, digital policy has moved to the center of EU policymaking. Yet we have little systematic knowledge of how much of it the EU has produced, where it is made, or what it does. We map the EU's digital acquis from 1979 to today: we decompose every regulation, directive, and decision into nearly two million recitals and (sub-)articles and classify each with a two-stage pipeline that distills a human-validated ensemble of large language models into open encoder models. The share of digital provisions in newly adopted legislation rose from under one percent in the 1980s to roughly one in ten, and most sit outside the flagship digital acts. Testing two prominent claims, we show that digital legislation has become more assertive over time, with its obligations and prohibitions falling increasingly on private actors, and that it is more rights-oriented than the rest of EU law yet still more market- than rights-oriented, with the rights it invokes being mostly civil. We release our provision-level dataset, classification models, and a Python package for building granular legislative corpora: open infrastructure for measuring not just how much legislation there is, but what it says and does.

Keywords: Digitalization, European Union, Digital Policymaking, Digital Sovereignty, Computational Text Analysis

Words: 8126 (9839 including references)

*Corresponding author: Timo Seidl, timo.seidl@hfp.tum.de. We thank participants at the 38th Annual SASE Conference in Bordeaux, the 13th Biennial Conference of the Standing Group on the European Union in Catania, and the 32nd International Conference of Europeanists in Dublin for helpful comments and suggestions. We document our use of large language models in Online Appendix Part F, following the GUIDE-LLM reporting checklist (Feuerriegel et al. 2026).

1 Introduction

Over the last decade, digital policy has matured into one of the most prominent and consequential areas of EU policymaking. The EU's data protection rules became the poster child for the 'Brussels Effect' (Bradford 2020). Digital sovereignty has become one of Brussels' favorite buzzwords (e.g., Council of the European Union 2025). And many of the EU's recent flagship policies — from the Digital Services and Digital Markets Acts to the AI Act and Data Governance Act — are quintessentially digital regulations, underscoring that the Commission has increasingly 'prioritised digital policy as a cornerstone of its strategic agenda' (Thüer et al. 2026, 1). Yet, despite its prominence, we know surprisingly little about EU digital policy as a whole. How much of it is there — and since when? Where is it made: in the flagship acts everyone can name, or scattered across the wider body of EU law? And what does it do: how does it regulate, whom does it bind, and whose rights does it invoke?

While a rapidly growing number of articles, books, and special issues have shed light on the dynamics and drivers behind the evolution of EU digital policymaking (Adler-Nissen and Eggeling 2026; Bellanova, Carrapico, and Duez 2022; Falkner et al. 2024; Farrand 2026; Newman 2020), this literature suffers from two shortcomings. The first is *coverage*: most studies offer rich accounts of particular acts, legislative processes, or policy subfields. Individually, however, such studies (often of recent cases) cannot reveal the broader patterns and trends of EU digital policymaking, or the different, sometimes contradictory, developments that characterize it. The second is *resolution*: digital policy does not come neatly packaged in legislation labeled 'digital'. Its 'transversal' character has made it play out across a 'myriad of parallel policy discussions and governance arenas' (Bonnamy and Perarnaud 2024, 15): digital provisions surface in acts that are not primarily digital — in a battery regulation that prescribes interoperable digital product passports, or in electricity-market rules that govern the algorithm setting wholesale power prices across Europe. The legal act, in short, is not the right level of analysis. To find digital policy, one has to look for it where it lives: inside the acts, at the level of individual provisions.

In this paper, we provide such a granular mapping of the EU's *digital acquis* from 1979 — when the Commission first identified the 'challenge of new information technologies' (European Commission 1979) — to today. By granular, we mean that our unit of analysis is the individual provision — a recital or a (sub-)article — rather than the legal act (Section 3). By digital policy, we mean deliberate public interventions that shape the course and character of digitalization by influencing the development, deployment, governance, and use of digital technologies (Section 2). We further characterize provisions by their regulatory tone — whether they oblige, prohibit, or permit — and the actor they regulate, and by their rights profile: how far they are oriented toward rights or toward markets, and which rights — civil, political, or social — they invoke (Section 4). Such a mapping is, at bottom, a measurement exercise: creating valid measures from these concepts demands data organized in the right way, and instruments capable of applying them at scale (Adcock and Collier 2001; Birkenmaier, Lechner, and Wagner 2024; Laurer et al. 2025).

We therefore take individual provisions, not whole acts, as our unit of analysis (see also Brandsma et al. 2025; Frantz and Siddiki 2022). With `eurlex-builder`, an open-source Python package we built for this purpose, we retrieve every EU regulation, directive, and decision from the official EUR-Lex APIs, parse their different document formats, and decompose each act into its recitals and (sub-)articles — alongside act-level metadata and the citation ties between acts. The result is a corpus of 146,234 acts and 1,956,898 provisions. Measurements at this level aggregate freely upward: to acts, to policy fields, to the corpus as a whole. We develop a two-step classification pipeline. First, an ensemble of five large language models (Benoit et al. 2025; Halterman and Keith 2026) classifies a large sample of provisions, validated against human coding; we then distill these labels into fine-tuned open encoder models that classify the full corpus cheaply and reproducibly, with no proprietary API required.

Our findings show that the share of digital provisions in newly adopted legislation rose from under half a percent in the 1980s (0.3%) to roughly one in ten today (10.5%). Crucially, while the well-known acts are an important part of the digital acquis, they only tell part of the story. For example, an inventory of EU digital laws curated by the Centre for European Policy Studies (CEPS) (Zenner, Marcus, and Sekut 2025) captures just 17.1% of all digital provisions. Digital provisions also sit in legislation whose primary purpose is not digital — from vehicle type-approval rules granting regulators access to software and algorithms to agricultural rules requiring the use of satellite data to monitor farmland. In fact, significant parts of EU digital policy are made outside conventional digital policy arenas.

Our measures also allow us to move from description to adjudication, putting two claims from the literature on digital policymaking to a systematic test. The first is that EU digital policy has adopted an increasingly 'assertive regulatory tone' (Farrand 2026, 70). Classifying the deontic force of every article, we find that it has: digital legislation has become markedly more assertive over time, and its obligations and prohibitions increasingly fall on private actors. The second claim is that the EU has built a rights-driven digital empire (Bradford 2023). Scoring how far each provision is oriented toward rights and toward markets, and classifying the rights that provisions invoke, we find that digital law is more rights-oriented than the rest of EU law but still more market- than rights-oriented, and that among the provisions that invoke a right, civil rights dominate social and political ones (Marshall 1992). Together, these analyses show how digital policy has evolved, where it can be found, what and whom it regulates, and whose rights it invokes.

We make three contributions. Empirically, we provide the most comprehensive mapping to date of the decades-long evolution of EU digital policy. Methodologically, we contribute to the recent turn from measuring the volume of legislation and regulation to measuring its substance (Section 3): moving beyond act-level topic classification, we measure legislative content at the level of individual provisions. The pipeline that applies these measures across two million provisions — distilling a validated ensemble of large language models into open encoder models — is efficient, high-performing, and transferable to any body of law. Infrastructurally, we release the provision-level dataset, the fine-tuned classifiers, and a

Python package (`eurlex-builder`) for building comparable legislative corpora — open resources that let other researchers extend, re-score, and repurpose the map we draw here.

Our paper proceeds as follows. We first define digital policy and trace its historical evolution (Section 2), then make the case for measuring it at the level of provisions (Section 3) and introduce the two debates in the literature we want to put to a systematic test (Section 4). We then describe our corpus and classification pipeline, and how we validated it (Section 5). The findings follow in two steps (Section 6): we first map the digital acquis — its growth, its dispersion, and its place within EU law — and then test two prominent claims about its nature: that EU digital policymaking has become more assertive, and that it is defined by rights. We conclude (Section 7) by discussing the implications and limitations of our approach.

2 EU Digital Policymaking: Definition and Evolution

Before recounting the history of EU digital policy, we need to define what counts as a digital policy — and what does not (Birkenmaier, Lechner, and Wagner 2024, 266). We define digital policies as deliberate public interventions that shape the course and character of digitalization by influencing the development, deployment, governance, and use of digital technologies.[1] Digital policies can involve various instruments, from rules and decisions, to programs and strategies, to investments and procurement; these, in turn, can affect all layers of the digital stack, from the physical (infrastructures, production), to the code (designs, standards), to the information layer (content, data) (Haggart, Scholte, and Tusikov 2021; Hiltunen 2024; Sheikh 2022).

This definition of digital policy is deliberately broad, but not so broad as to be meaningless or synonymous with technology policy. What distinguishes digital policy from technology policy more broadly is its substantive engagement with digital technologies — computation, data, networks, platforms, software — and with the activities these reconfigure. Battery regulation, for example, is technology policy but not digital policy, even though batteries are integral to many digital devices. Semiconductor regulation, by contrast, is both, since semiconductors constitute the physical layer of the digital stack. This means that the EU has been making digital policy *avant la lettre* — long before the term came into common use — under labels such as 'information society services', 'telematics', and 'electronic communications'. We treat these earlier framings as digital policy too, since the underlying interventions engage digital technologies, even if the vocabulary has shifted.

In the EU, as elsewhere, digital policymaking gradually developed from a series of isolated acts into a coherent policy field over the course of several decades (Küsters and Sottilotta 2026; Nebbiai 2026; Newman 2020). After some early — and unsuccessful — initiatives in the area of 'computer policy' (Van

[1] We define digitalization as a socio-technical process whereby economic, political, and social activities are fundamentally reconfigured by digital technologies, transforming, among many things, how people consume and distribute information, find jobs and do their work, connect with friends and talk to strangers, buy and sell things, or express themselves politically or artistically.

Laer 2010), Europe only began to seriously discuss the 'challenge of new information technologies' in the late 1970s (European Commission 1979). In the ensuing decades, initiatives remained fragmented, with the most important developments being the experimentation with industrial policies in the 'IT' or 'telematics' sector (Cram 1994; Sandholtz 1992), competition policies and liberalization in the telecommunications sector (Clifton, Díaz-Fuentes, and Revuelta 2010), and extensive debates on Europe's approach to the 'information society' (Shahin and Finger 2009) — debates that culminated in the 1995 Data Protection Directive (Newman 2008) as well as the 2000 E-Commerce Directive (Stengg 2024).

After 2000, the EU's digital strategy became embedded in the Lisbon Agenda's pursuit of Europe as the world's 'most competitive and dynamic knowledge-based economy', and at around 2010 the 'digital single market' frame became dominant (European Commission 2010; Mărcuț 2017, 48). In the years after and against the background of the rise of Big Tech and growing concerns around platform power, surveillance, and lack of competitiveness, digital policy began to play an increasingly prominent role in the EU's overall policy agenda. The General Data Protection Regulation (GDPR) is undoubtedly the most important initiative of this era. It almost single-handedly created the EU's reputation for being a global rule-setter for the digital age (Bradford 2020) and is the 'crown jewel and a foundational step' (Nebbiai 2026, 708) in Europe's 'digital constitutionalism' (De Gregorio 2022). However, despite serving as both an inspiration and template for most subsequent digital policies (Papakonstantinou and De Hert 2024), the GDPR was still framed as advancing the dual objectives of creating a digital single market and protecting fundamental rights in the digital age (Newman 2020; König 2022; Streinz 2021). Digital policy was not yet framed in terms of sovereignty and security, autonomy and assertiveness. This began to change in the 2020s — when digital sovereignty became the rhetorical center of gravity and 'guiding principle' (Falkner et al. 2024, 2100) of the Commission's increasingly ambitious digital agenda (Thüer et al. 2026).

Against the background of the Covid-19 pandemic, Big Tech dominance, and growing geopolitical tensions, digital policymaking became increasingly shaped by what Küsters and Sottilotta (2026) call the 'digital trilemma' between competitiveness and innovation, security, and the protection of fundamental rights — giving digital policy broader significance beyond single market regulation and fundamental rights protection. Under the banner of digital sovereignty and across a range of policy areas, Europe began 'reasserting control over the digital' (Falkner et al. 2024, 2107) and connecting digital policy more explicitly with (economic) security considerations (Bellanova, Carrapico, and Duez 2022; Farrand 2026). At that time, the Commission also began to 'act-ify' (Papakonstantinou and De Hert 2024) digital policies, departing from standardized naming conventions and using more easily identifiable names like Digital Markets Act, Digital Services Act, Data Governance Act, Data Act, or Artificial Intelligence Act. In short, in the 2020s, digital policy moved to the center of Europe's broader policy agenda while moving in a 'more decisive, assertive and strategic' direction (Stengg 2024, 279).

3 The Case for Granularity in Studying EU Digital Policymaking

While the existing literature has painted a rich picture of the nature and evolution of EU digital policymaking, it has largely done so through qualitative case studies of single policies or policy areas. Such studies offer depth and nuance, but they run into two problems. The first is *coverage*: case studies tend to concentrate on a small number of flagship acts, leaving the longer tail of less prominent policies unexamined. Moreover, because individual digital policies are situated in broader relational webs and historical trends (Koniaris, Anagnostopoulos, and Vassiliou 2018; Heidebrecht 2024; Hurka, Haag, and Kaplaner 2022), studying them in isolation risks missing the broader forest for some particularly prominent — and mostly recent — trees. The second is *resolution*: because most case studies operate at the level of whole acts, we typically learn about provisions in digital legislation rather than about digital provisions in legislation (see also Brandsma et al. 2025). Yet the legal making of the digital economy happens throughout the acquis, not only in legislation labeled digital (see also Hiltunen 2024). The two problems are independent: even projects that assemble comprehensive inventories of EU digital laws (Zenner, Marcus, and Sekut 2025) still miss the digital provisions in legislation that is not primarily digital. Resolution does not follow from coverage.

Scholars also disagree about the nature of EU digital policymaking: some highlight the rights-related aspects of the EU's regulatory approach (Bradford 2023; Roberts et al. 2023), others stress its market-creating ones (König 2022; Krarup and Horst 2023) — objectives that can, of course, co-exist (Newman 2020; Streinz 2021). Which aspects are present and which are dominant — over time, across sub-policy areas, and even within individual acts — is precisely the kind of question that requires broad coverage at granular resolution.

To remedy these problems, we develop a granular approach to the study of EU digital policymaking that starts at the level of (sub-)articles and recitals.[2] Every act is a mixture of provisions that often reflect different goals and priorities. By disaggregating acts into provisions, we can analyze acts in a less coarse-grained way without losing the ability to re-aggregate from individual provisions to acts and even the topology of the digital acquis as a whole. This *aggregational flexibility* — the ability to move between the micro level of provisions, the meso level of acts, and the macro level of policy fields — is the central advantage of a granular approach (for a similar, sentence-level approach, see Brandsma et al. 2025).

Beyond the level at which we measure legislative text, there is the question of which of its properties we measure. The dominant approach quantifies regulation at the level of the act, along structural dimensions: the number of acts and amendments, their length in words or pages (Kosti and Levi-Faur 2019). Such measures have brought many insights but are silent on content — on what legal acts do: whether they impose a duty or grant a discretion, whom they target, and what substantive goods or rights they extend. A recent turn in the study of legislation and regulation has begun to open up this content (Brandsma et al. 2025; Damonte and Bazzan 2024; Hinterleitner, Knill, and Steinebach 2024;

[2] In principle, this approach generalizes to other levels of granularity, such as paragraphs or sentences. This is particularly relevant for documents not organized into recitals and articles.

Kosti 2024b; Radaelli and Rangoni 2026; Shapiro 2024; Siddiki and Frantz 2024). Our approach builds on and extends this turn. Where existing work typically extracts one content type from one unit — for example, regulatory obligations from the sentences of EU law via rule-based patterns (Brandsma et al. 2025) — we classify several dimensions of a provision's policy content: what it is about, its regulatory tone, the actor it regulates, and the rights it invokes. The granular level at which we operate is what makes this possible: deontic force and rights content live at the level of the individual provision, not the act.

4 What Digital Policy Does

Beyond mapping where digital policy is made and how it has grown, we want to know what it does. We therefore engage two prominent debates about the nature of EU digital policymaking: whether it has become increasingly assertive, and whether it amounts to a rights-driven digital empire (Bradford 2023). Each debate tells us what to measure — the first, the deontic force of provisions and the actors they bind; the second, their orientation toward rights or markets and the rights they invoke.

4.1 Regulatory Tone

Regulatory tone captures how assertive a provision is: whether it obliges an actor to do something, prohibits them from doing it, or permits it — put simply, whether a provision prescribes an action or leaves discretion. An obligation imposes a positive duty: the actor is required to act, and it is inaction that constitutes non-compliance. A prohibition imposes a negative duty: the actor is required to refrain from acting, and it is action that constitutes non-compliance. A permission, by contrast, imposes no duty at all; it confers discretion, a right, or a power, leaving the actor free to act or abstain without sanction. This distinction maps onto the three basic categories of 'deontic logic' (Von Wright 1951; Crawford and Ostrom 1995; Frantz and Siddiki 2022; Siddiki and Frantz 2024) and reflects the layered character of the law, which can either confer rights, privileges, or powers on an actor, or impose duties upon them (Coode 1845); it also underpins a broad literature on legislative content, including work on delegation that contrasts 'must' as mandatory with 'may' as permissive (e.g., Kosti 2024a; Vannoni, Ash, and Morelli 2021). Regulatory tone thus captures the variegated effects of legal provisions — rather than treating regulation as a uniform constraint — and it applies to any body of law: every operative provision has a tone, digital or not.

In the digital domain, EU policymaking is widely said to have taken on a more 'assertive regulatory tone' (Farrand 2026, 70) — 'moving from voluntary codes to binding regulations, replacing light-touch requirements regarding standards with prescriptive requirements' (Farrand 2026, 121; see also Falkner et al. 2024; Stengg 2024). Yet this claim has not been systematically tested: has EU digital policy really become more assertive — and if so, on whom does the assertiveness fall? Measuring the respective shares of obligations, prohibitions, and permissions across digital and non-digital articles lets us answer

the first part of this question. Answering the second requires knowing not only what a provision demands, but from whom.

The same obligation carries a very different politics when it binds a member-state administration than when it binds a platform company or an EU agency. We therefore also code the *regulated actor* of every article: whether its primary addressee is an EU institution, a member state, a private actor, or an individual. Crossing tone with actor turns an aggregate trend into a distributional statement: if the assertiveness thesis holds, we should see a rising share of obligations and prohibitions falling in particular on private companies, while member states may get both new obligations and new permissions to regulate.

4.2 Rights Profile

The *rights profile* of a provision, in turn, captures two things. First, its orientation: how strongly the provision is oriented toward protecting or extending rights, and how strongly toward making, correcting, or steering markets. We rate the two separately, so that provisions engaging both, or neither, remain distinguishable; the difference between the two ratings places each provision on a continuum from market-oriented to rights-oriented. Rating them separately does not presuppose that markets and rights trade off against each other: a provision can be strongly oriented toward both, and how often that happens is an empirical question. Second, the type of rights invoked, drawing on T.H. Marshall's (1992) famous typology of citizenship rights: civil rights (liberty, free expression, due process), political rights (participation in political power as voter or officeholder), and social rights (welfare, inclusion, and the means to participate fully in society). Like regulatory tone, the rights profile is a general property of legislation: any provision can be rated for its market and rights orientation, which is what allows us to compare digital with non-digital law, and any rights-invoking provision can be classified by the kind of right it extends.

In the digital domain, the rights profile speaks directly to the EU's reputation. The EU is often said to differentiate itself from the American market-led and the Chinese state-led models through its focus on 'promoting ethical outcomes through protecting fundamental rights' (Roberts et al. 2023, 79): what its many digital rules 'have in common is their focus on enhancing rights—be it the fundamental rights of internet users, the democratic rights of digital citizens, [or] the social rights of platform workers' (Bradford 2023, 108–9). Yet how well this reputation captures EU digital policy is disputed. On one reading, EU digital policy retains a common market-building thrust, even in its data and consumer protection rules (König 2022); similarly, rights and ethical aims are often pursued through market integration rather than against it, so that the two are entangled rather than opposed (Krarup and Horst 2023). Different types of rights, moreover, are often lumped together in such accounts.

The rights profile addresses both gaps. It allows us to empirically test how rights-oriented the EU's digital acquis actually is and how its position on the market-rights continuum has evolved over time. It also lets us characterize the kind of digital citizenship the EU implicitly constructs: a civil-rights-dominant

model focused on individual autonomy and procedural protections (informational self-determination, due-process rights against algorithmic decisions); a political-rights-tilted model emphasizing democratic participation (electoral integrity, political-ad transparency); or a social-rights-inflected model that secures connectivity and inclusion alongside individual liberty (affordable connectivity, accessibility, platform-worker protections).

5 From Concepts to Measurements

Creating valid measures from the concepts developed in Section 2, Section 4.1, and Section 4.2 demands two things (Adcock and Collier 2001; Birkenmaier, Lechner, and Wagner 2024; Laurer et al. 2025): data organized to match those concepts, and instruments capable of applying them at scale. This section provides both. We first introduce our corpus and the coding scheme that translates concepts into provision-level categories. We then describe the two-stage classification pipeline that applies the scheme to the corpus at scale and how we validated our approach.

5.1 Building a Granular Corpus

Constructing a provision-level corpus requires us to reliably extract recitals and (sub-)articles from acts. To do this at scale and across the EU's full legislative history, we developed `eurlex-builder`, an open-source Python package we release alongside this paper. To our knowledge, this is the first open-source tool to extract EU legislative text at configurable granularity across EUR-Lex's heterogeneous document formats — be it recitals, whole articles, numbered paragraphs (Art. 5(1)), lettered points (Art. 5(1)(a)) or, in the case of communications, paragraphs. Existing tools such as the `eurlex` R package (Ovádek 2021) handle document retrieval but not structured extraction or translation while datasets such as EUPLEX (Hurka, Haag, and Kaplaner 2022) or EUPROPS (Golub 2024) provide valuable additional indicators or curation but do not allow researchers to assemble bespoke corpora. By contrast, `eurlex-builder` is a configurable Python pipeline built on the official EUR-Lex SPARQL and REST APIs that allows users to build their own datasets, either based on an explicit CELEX list (fixed mode) or a descriptive search by date range, document type, and optional EuroVoc keyword filter, together with the desired sub-article granularity, language fallback, and translation settings.[3]

Our own corpus comprises every directive, regulation, and decision adopted by the EU between 1979 and April 2026. We focus on these three binding instruments, rather than, for example, communications, although `eurlex-builder` allows for paragraph-level extraction of communications as well. We decompose these acts into recitals — which explain the context, objectives, and reasoning behind a

[3] `eurlex-builder` automatically detects and parses the different HTML structures EUR-Lex has used over the decades, falling back to PDF extraction via Docling for documents without machine-readable HTML. Non-English documents are translated using Helsinki-NLP Opus-MT models. The pipeline is fully resumable via checkpointed DuckDB state and supports parallel fetching. The main output is four Parquet tables: works (act-level metadata), text_units (recitals, articles at the chosen granularity, and annexes), relations (inter-document ties), and eurovoc (subject descriptors). The package is available on PyPI and at github.com/tseidl/eurlex-builder; releases are archived on Zenodo (DOI: 10.5281/zenodo.21496963).

piece of legislation — and articles — its operative provisions, which establish the legally binding rules. The articles themselves we further decompose into numbered sub-paragraphs, since these often carry distinct meaning. For example, only some numbered paragraphs of a given article may address digital policy at all; and different paragraphs of the same article often carry different deontic content — establishing a procedural mechanism in paragraph 1, imposing a substantive obligation in paragraph 2, and granting a discretionary power in paragraph 3. The resulting corpus contains 1,956,898 provision-level units across 146,234 acts: 1,065,116 recitals and 891,782 (sub-)article units. In addition, the corpus contains act-level cross-references — citations, amendments, legal-basis references, and repeals. We release all data, the eurlex-builder package, and the fine-tuned encoder models described below as open infrastructure for future research. The extraction pipeline, the canonical corpus filters, and the treatment of non-English documents are documented in Online Appendix A1–A3; the one substantive exclusion — individual enforcement decisions such as merger clearances and state-aid rulings — is examined in E8, which shows that no finding depends on it.

5.2 Operationalization

The granular nature of our dataset allows us to reconstruct the policy content of individual provisions. First, we translate our definition of digital policy from Section 2 into a binary classification rule. A coding instruction — which doubles as the prompt for the LLMs — details when a provision should be classified as digital policy, including definitions, inclusion and exclusion rules, and examples. The same logic — translating a substantive concept into a coding rule with definitions, boundary conditions, and examples — applies to the other classification tasks we use, although we only apply them to subsets of our corpus: regulatory tone, and by extension regulated actor, only to articles, since recitals lack a clear deontic character; rights typology only to provisions already classified as digital policy. The complete coding instructions, which operationalize the conceptualization developed in Section 2, Section 4.1, and Section 4.2, can be found in Appendix B2.

Table 1 Concepts and their measurement.

Concept	Values	Applied to	Produced by
Digital policy	digital, not digital	all provisions	encoder models
Regulatory tone	obligation, prohibition, permission, none	articles	encoder models
Regulated actor	EU institutions, member states, private actors, individuals, other	articles	encoder models
Rights type	civil, political, social, none	digital provisions	ensemble of LLMs
Market and rights orientation	two ratings from 0 to 9; position = rights – market	digital provisions; non-digital sample	ensemble of LLMs

Note: The encoder models are distilled from the labels of a five-model ensemble, which classifies rights type directly; the orientation ratings come from a three-model ensemble (Figure 1).

We classified our corpus in a two-step procedure, described in Figure 1. Broadly following the framework proposed by Halterman and Keith (2026), we begin by translating our conceptualizations into a structured codebook written for both human coders and an ensemble of five LLMs (Benoit et al. 2025): four large open-weight models (Qwen3-235B-A22B-Instruct, Llama-3.3-70B-Instruct, gpt-oss-120b, and DeepSeek-V4 Pro) and one proprietary model (GPT-5.2) (Yang et al. 2025). The full instructions are reproduced in Online Appendix B2. We then refine these coding instructions iteratively, validating along three axes: how closely the LLMs agree with human coders, how closely different LLMs agree with one another, and how stable a model's classifications are across small variations in the prompt (Barrie, Palaiologou, and Törnberg 2024) (Online Appendix C4). Once the instructions stabilized, we froze them and had both the human coders and the LLM ensemble code a fresh sample of provisions, re-running the same three checks. As Table 2 reports, the LLMs agree strongly with the human coders on every task: for digital policy the ensemble's consensus reaches Krippendorff's $\alpha = 0.90$, essentially the inter-coder ceiling of 0.91, and agreement remains high for tone, regulated actor, and rights type. Two external checks corroborate the pipeline: the classifier recovers every act in an expert-curated inventory of EU digital laws, and an act's digital share predicts the EU's own independent subject indexing with an AUC of 0.95 (Online Appendix D1 and D3).

Table 2 Validation of the classification pipeline.

	Agreement (proportion / α)			
Task	Inter-coder	Ensemble	Encoder	Encoder F1
Digital policy	.97/.91	.96/.90	.96/.90	.95
Regulatory tone	.94/.90	.93/.88	.92/.86	.94
Regulated actor	.92/.89	.88/.83	.86/.80	.85
Rights type	.94/.88	.93/.84	—	—

Note: Cells show the proportion of agreement / Krippendorff's α: between the two human coders, between the ensemble's consensus and the reference coder, and between the encoder and the reference coder. No encoder was trained for rights type, which the ensemble classifies directly (Table 1). Encoder F1 is measured against ensemble labels on a held-out test split.

We then use the validated ensemble to label a sizeable, stratified subset of the corpus and distill these consensus labels into a smaller encoder model, which classifies the full corpus and produces the measurements we analyze below. Crucially, the encoder models faithfully replicate the consensus codings of their teacher models, reaching held-out F1 scores from 0.85 (regulated actor) to as high as 0.95 (digital policy). The exception is rights type: because the task applies only to the roughly 85,000 digital provisions, we apply the LLM ensemble directly (the dashed path in Figure 1) — at that scale, distillation would add a training and validation step without a real gain, especially given how few provisions carry a coded right. We release the encoders, together with the provision-level dataset, on Hugging Face (huggingface.co/collections/tseidl/eu-digital-acquis) so that other researchers can apply

our classifications to similar corpora. Our approach thus combines a human-validated use of generative LLMs to create high-quality training data with a highly performant, computationally efficient encoder fine-tuned on those labels.

The market and rights ratings follow the same direct path, with one further difference. Our classification tasks assign each provision one label from a small set of classes (Table 1). The market and rights ratings, by contrast, are continuous, and we measure them accordingly: three open models with accessible token probabilities (Qwen3-235B-A22B-Instruct, Gemma 4 31B, GLM 5.2) score every digital provision twice on a 0–9 scale — once for its market orientation, once for its rights orientation. Rather than taking the digit a model outputs, we compute the expected value over its token probabilities at the score position — a technique that outperforms both direct numeric scoring and pairwise-comparison scaling (Licht et al. 2025). Averaging across the three models yields the two orientation scores; their difference places each provision on the market-rights continuum. We validate the ordering these ratings produce against human pairwise comparisons (Online Appendix C7).

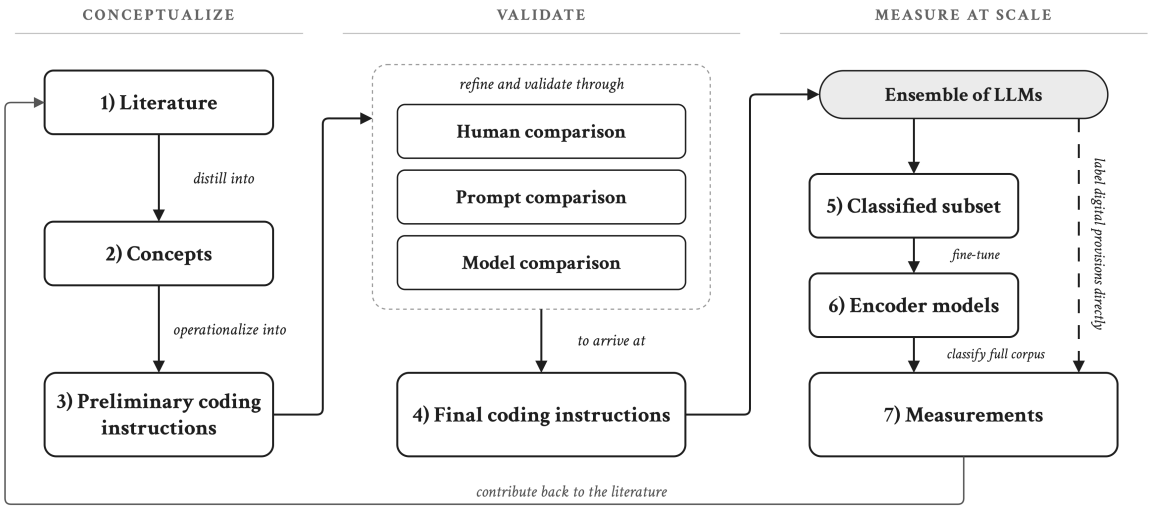


Figure 1 From concepts to measurements: the classification pipeline.

6 Findings

6.1 Mapping the digital acquis

How much digital policy has the EU produced, and since when? The trend line in Figure 2 traces the share of provisions classified as digital policy across every regulation, directive, and decision the EU adopted between 1979 and April 2026. Throughout the 1980s, fewer than one in two hundred provisions were related to digital policy (0.3%); by the 2020s, it was roughly one in ten (10.5%), with single months reaching 26%.[4] This growth has two sources: a growing number of acts that are both largely

[4] The classifier recognizes the early decades' vocabulary, including 'electronic data interchange systems' and the 'telematics network' (Online Appendix E7), and agreement with human coding remains high across periods (C6). Classifications

and substantially digital, exemplified by the landmark laws of the last decade: GDPR, DMA, DSA, the AI Act; but also, and crucially, a dense thicket of digital provisions that has grown below them, inside legislation whose primary purpose is not digital. A dedicated body of digital legislation has formed in the EU, with the share of newly adopted acts that are at least half digital rising nearly tenfold, from 0.5% in the 1990s to 4.9% in the 2020s.

But this dedicated body of legislation is only one part of the EU's digital acquis. In fact, the ten flagship acts labeled in Figure 2 together account for just 3.0% of all digital provisions, and even the 163 matched acts from the CEPS' curated inventory of EU digital laws (Zenner, Marcus, and Sekut 2025) capture only 17.1% (see Online Appendix D1). Most digital provisions are found in neither.[5] Nearly half of all digital provisions (47.2%) sit in acts that are less than half digital, partly as substantive rules in their own right (e.g., darknet-market monitoring under the EU Drugs Agency regulation), partly as references to core digital policies (e.g., data protection requirements in rail passenger rights). They are scattered across the full breadth of EU law — visible only to a net cast as wide as the acquis and as fine-grained as its provisions. This is where our granular approach earns its keep.

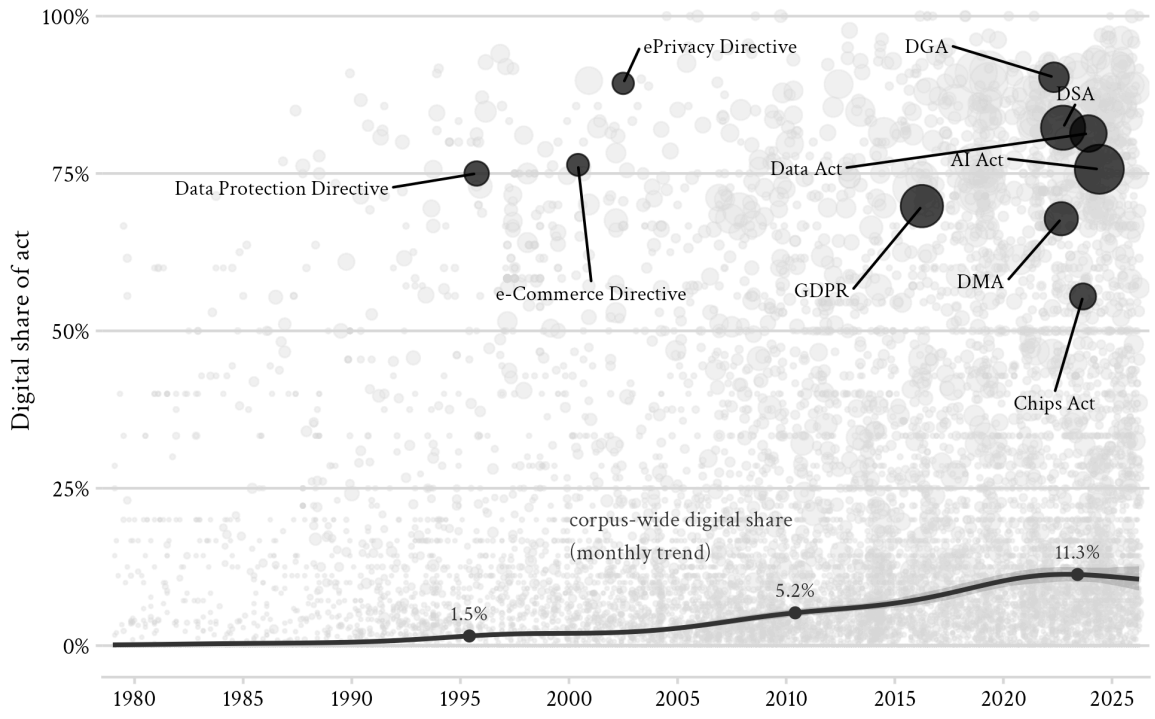


Figure 2 The rise of digital policy in EU legislation, 1979–2026. Points are individual acts (with at least five classified provisions) by the share of their provisions classified as digital; labeled points are selected landmark digital acts, sized by their number of digital provisions. The line is the corpus-wide digital share of newly adopted provisions: a binomial GAM trend fit to monthly counts, with a 95% act-level cluster-bootstrap band (Online Appendix E1).

are stable across prompt variations (C4), and the trend persists with alternative encoder configurations (E5) and with enforcement decisions included (E8).

[5]Heuristic filters and an LLM-based audit put formulaic references at between five and ten percent of dispersed digital provisions; removing all of them lowers the share of digital provisions outside the expert list by only about one percentage point (Online Appendix E6). Including or excluding one-off enforcement decisions likewise does not change the findings (Online Appendix E8).

Digital policy is not only growing more frequent; it is also referenced ever more frequently by the rest of the acquis. Citations between acts are a standard measure of how law is embedded in the wider legal order (Fjelstul 2019; Hurka, Haag, and Kaplaner 2022; Koniaris, Anagnostopoulos, and Vassiliou 2018), and because every act in our corpus carries a digital share, we can estimate how many of them point to digital content (Online Appendix E2). By 2025, an estimated one in seven references in newly adopted law does — about three times digital policy's share of the cumulative stock of citable provisions, and disproportionate even when benchmarked against the vintages and instrument types actually cited (Figure 3). Digital law, in other words, has become more central to the wider acquis. Data protection accounts for much of this connection: among references from essentially non-digital acts to acts that are at least half digital, 57.8% point to data-protection legislation. Other prominent connections concern the shared digital systems through which policies are administered (Online Appendix E2).

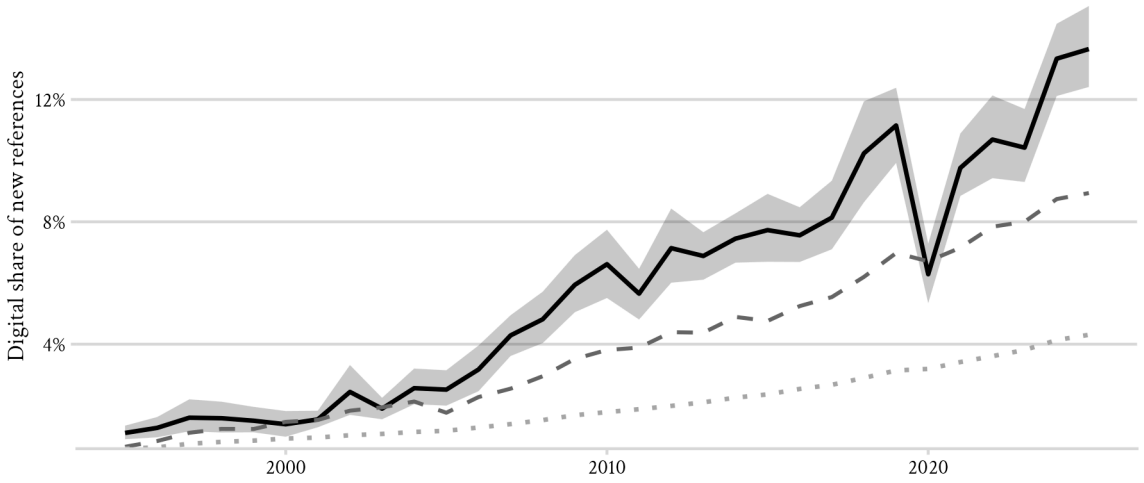


Figure 3 Does new EU law cite digital policy disproportionately? The line shows the share of each year's new references that concern digital policy, 1995–2025 (solid line; the band is a 95% interval from resampling citing acts). It is compared against two benchmarks: what citing blindly from the accumulated stock of provisions would produce (dotted line), and what citing blindly among the provisions of the vintages and instrument types actually cited would produce (dashed line). Citations are recorded between acts, so we read a cited act's digital share as the probability that a reference to it concerns digital policy (e.g., citing the GDPR counts as 0.7 of a digital reference).

6.2 What digital policy does

So far, we have provided a granular description of EU digital policy. This has allowed us to trace patterns across EU law that are difficult to see when looking only at whole acts or selected cases. It is an approach, we believe, from which the study of EU policymaking, digital and otherwise, has much to gain. But our approach does not just allow for better description; it also opens the door to asking old questions in new ways. We illustrate this potential by taking up two of them, at the center of current debate about what digital policy *does*: how assertively it regulates and whom it binds, and whose rights it invokes.

Beginning with the first question: the share of obligations and prohibitions — as opposed to permissions — is higher for digital than for non-digital provisions (Figure 4, Panel A). This corroborates the observation in the literature that the EU's digital policy has taken on a more 'assertive regulatory

tone' (Farrand 2026, 70), with more provisions whose deontic force establishes obligations or prohibitions rather than permissions. Moreover, the assertive share of digital articles has been growing over time, rising to over 90 percent of tone-bearing articles (and over 80 percent of all articles) in the 2020s. The wider acquis moved differently: among non-digital articles, the assertive share of tone-bearing articles drifted slightly downward, and the assertive share of all articles grew far more slowly than in digital law. By the 2020s, digital articles were roughly seven percentage points more assertive than non-digital ones among tone-bearing articles, and 24 points more assertive across all articles (on the choice of denominator, see Online Appendix E3).

Panels B and C show whom this growing assertiveness targets, focusing on private actors and member states (most remaining obligations bind the EU's own institutions; Online Appendix E3). The share of articles imposing obligations or prohibitions on private actors has grown markedly, and much faster in digital law, where it approaches forty percent of all tone-classified articles by the mid-2020s. The share binding member states, by contrast, has stagnated in the wider acquis and declined in digital law since the mid-2000s. The EU's growing digital assertiveness, in other words, is increasingly aimed at the private actors that build and run the digital economy rather than at member states. Thus, if the EU has been 'reasserting control over the digital' (Falkner et al. 2024, 2107), it is first and foremost control over private companies active in digital markets.

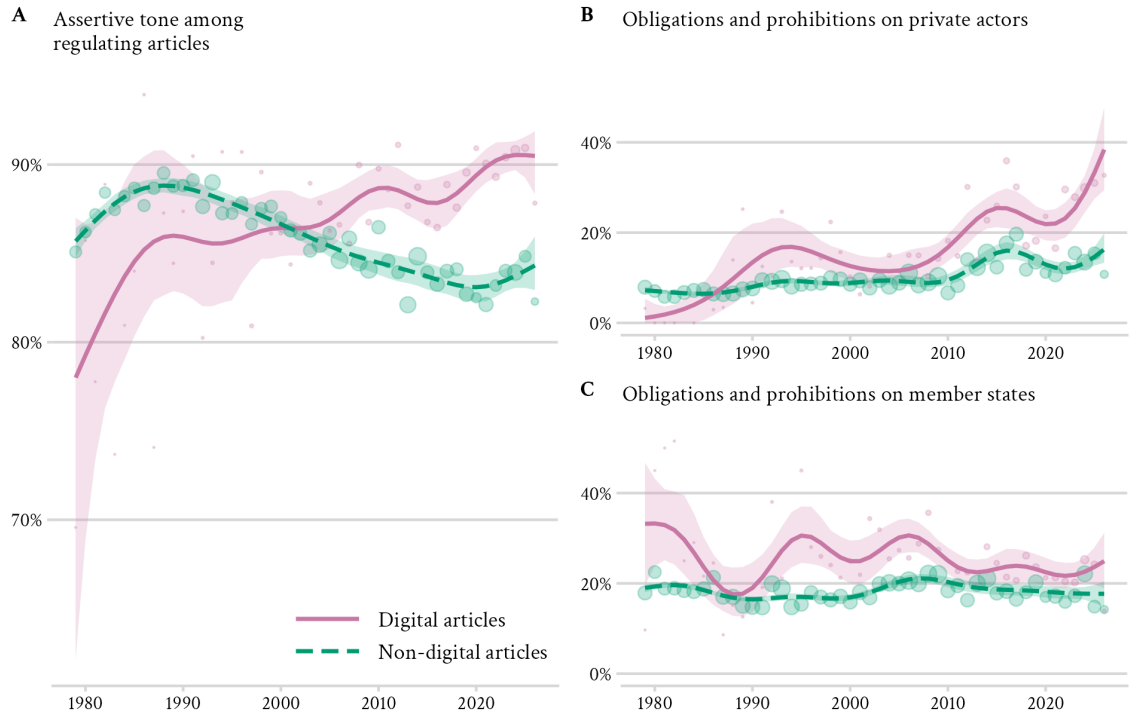


Figure 4 Regulatory tone in digital and non-digital articles, 1979–2026. Panel A: among articles with deontic content (obligation, prohibition, or permission; articles classified as none excluded), the share classified as obligation or prohibition. Panels B and C: articles imposing obligations or prohibitions on private actors (B) and on member states (C), as a share of all tone-classified articles — here, articles without deontic content remain in the denominator, since the question is how much of the law binds each actor; B and C share the same scale. Lines: binomial GAM per group fit to yearly counts; bands: 95% act-level cluster bootstrap; point size: number of articles.

The second question concerns the EU's reputation as a rights-driven digital empire (Bradford 2023), and the objection that market-building remains the common thrust of EU digital policy, with rights pursued through market integration rather than against it (Section 4.2). Our findings speak to both. Market orientation is the more pronounced of the two: digital provisions receive an average market rating of 3.9 against a rights rating of 2.6, and only 29.9% score higher on rights than on markets (25.4% and 21.5% if the difference must exceed half a point or a full point; Figure 5, Panel A). Yet rights orientation is stronger than in the rest of EU law. Non-digital provisions[6] receive similar market ratings (3.6) but much lower rights ratings (0.7), and only 8.9% lean toward rights. EU digital law is thus more rights-oriented than the rest of EU law, and still more market- than rights-oriented.

The balance between rights and market orientation differs across landmark acts (Figure 5, Panel B). Data protection law leans toward rights: the 1995 Directive, the GDPR, the ePrivacy Directive. But even these laws have a market side (König 2022): the GDPR, the most rights-oriented of the flagship acts, has a mean market rating of 3.0, close to the average for non-digital law (3.6). At the other end, the DMA and the Chips Act, policies that correct and steer markets, are the most market-oriented. The AI Act and the DSA are the most balanced, with rights- and market-oriented provisions side by side (see also Krarup and Horst 2023). Over time, the two orientations have moved closer together (Figure 5, Panel A). In 1995–1999, digital provisions were still far more market- than rights-oriented, with a market rating of 5.2 against a rights rating of 2.1. Since 2020 the two stand at 3.8 and 2.8, and the share of provisions leaning toward rights has grown from 13.2% to 31.3%, with a high point of 38.1% in 2015–2019 that does not depend on the GDPR alone (rights minus market: −0.70, and −0.79 without it).

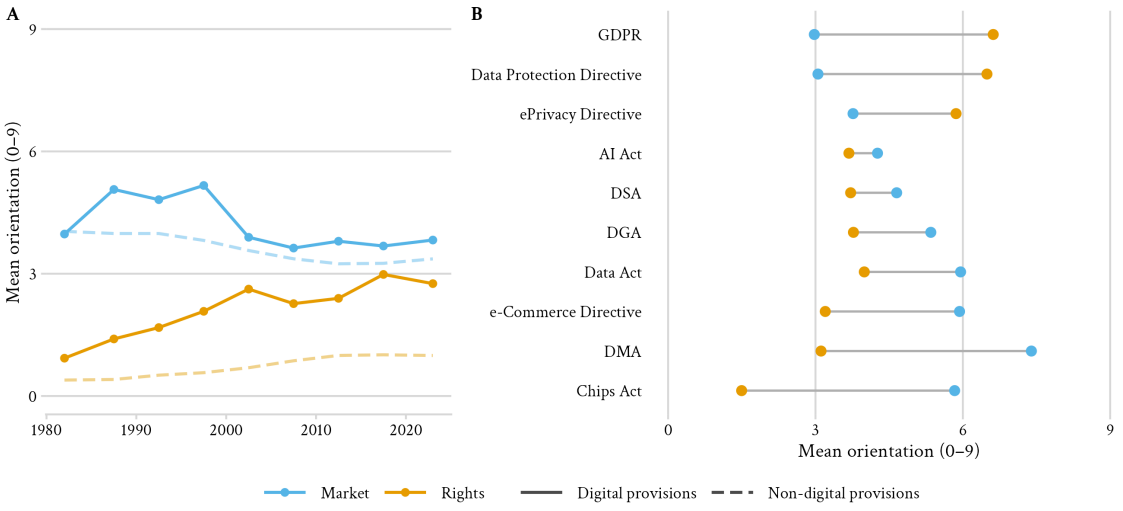


Figure 5 The market–rights balance of the digital acquis. Panel A: mean market and rights orientation (0–9) by five-year period of adoption; solid lines show digital provisions, faint dashed lines a random sample of non-digital provisions rated in the same way, whose rights rating stays below 1.1 in every period; 1979–1984 is the first period and 2020–2026 the last. Panel B: landmark digital acts; each segment runs from the mean market rating (blue) to the mean rights rating (orange) of the act's digital provisions. Ratings are expected values over token probabilities, averaged over three models (Online Appendix C7).

[6]The comparison sample comprises 100,000 provisions drawn at random, in equal numbers per decade, from the provisions our classifier labels as not digital, and rated with the same instructions and models (Online Appendix C7). Non-digital law contains a different mix of administrative, procedural, and substantive provisions, so the comparison is a broad benchmark rather than a like-for-like contrast.

What kinds of rights are at stake? Here we turn from how provisions are oriented to the rights they explicitly invoke. Among the 13.9% of digital provisions that invoke a right, civil rights dominate: 85.3% concern civil rights, against 12.5% social and 2.2% political rights. In digital law, civil rights are mainly those of data subjects, social rights concern universal service and accessibility, and political rights concern political advertising and media pluralism (Online Appendix E4). The EU's political digital rights largely amount to protecting electoral processes from data-driven targeting, rather than conferring participatory rights to co-shape digital economies and societies. The digital citizenship the EU constructs is thus first and foremost a civil one (Marshall 1992), centered on individual autonomy and procedural protection, with social rights playing a supporting role and political rights a marginal one.

These findings speak to both sides of the debate over what EU digital policy is *for*. The rights-driven reading (Roberts et al. 2023; Bradford 2023) is right about a real and durable layer of the acquis, one that sets digital law apart from the rest of EU law. But the balance between rights and markets varies widely across the digital acquis, and the rights invoked are mostly civil. The market-centered reading is right that market orientation remains the more pronounced of the two, including in acts often invoked as paragons of a rights-driven model (König 2022; Krarup and Horst 2023). Yet this market-governing acquis has grown more rights-oriented since the late 1990s — and its growing assertiveness, as we saw, falls increasingly on the firms that run the digital economy.

7 Conclusion

How much digital policy has the EU produced — where is it made — and what does it do? We have pursued these questions across every regulation, directive, and decision the EU adopted between 1979 and April 2026, decomposing them into 1,956,898 provisions and classifying each with a comprehensively validated pipeline of large language models and encoders. Digital policy, we find, has grown from a negligible share of newly adopted provisions in the 1980s (0.3%) to roughly one in ten today (10.5%). It is made not only in the flagship acts: most digital provisions lie outside the expert inventory of digital laws, and nearly half sit in legislation that is less than half digital. New EU law also increasingly cites it. We also find that digital policy has become more assertive: obligations and prohibitions dominate its articles and fall, increasingly, on private firms rather than member states. On rights, we find that EU digital law is more rights-oriented than the rest of EU law but still more market- than rights-oriented, and that the rights it invokes are mostly civil rather than political or social.

Our approach has several limitations which future research should address. First, we implicitly treat every provision as if it had the same legal weight. However, a single article granting a right may matter more than dozens of other provisions. This is connected to a second limitation: we do not know how provisions relate to each other. Our citation analysis still relies on act-level links, although every reference sits in a specific provision and many name the article they refer to. Resolving them to the level of provisions — which provision cites which, and where in the corpus both sit — would allow us

to understand not just the role of certain acts in the overall acquis, but of specific provisions: which are most central, and which forgotten even as their acts remain prominent? Finally, we describe patterns but do not explain them. Why does digital law disperse across the acquis the way it does? Why does its growing assertiveness fall on firms rather than member states? And why are the rights it invokes mostly civil, with political rights almost absent? Answering these questions requires the kind of process-tracing and case knowledge that only qualitative scholarship provides — the very scholarship whose claims motivated our measures in the first place.

Our approach can also be used to answer further questions about digital policy itself. For example, the market orientation we measure can be decomposed into distinct modes of market governance — creating markets, correcting them, steering them. Paragraph-level extraction of Commission communications makes it possible to trace whether what digital strategies announce ever arrives in binding provisions, or how it changes on the way. Or one could ask whether the rights invoked in recitals are actually matched by operative articles. Yet nothing in our approach is specific to digital policy. The dimensions we measure — what a provision is about, how assertively it regulates, whom it binds, whose rights it invokes — characterize any body of law, and the corpus, classifiers, and tools we release are built to be reused. Turned on other fields, such as environmental policy, the same approach can show how the Union builds regulatory capacity across domains; turned on other jurisdictions, such as the United Kingdom, it can support comparison across time, fields, and legal systems.

Funding Statement

This work was supported by the European Union's Horizon Europe research and innovation programme under the Marie Skłodowska-Curie grant agreement No 101148743 (EUREG), and by the LMU Postdoc Support Fund. The funders had no role beyond financial support.

Disclosure of interest

The authors report there are no competing interests to declare.

Data availability statement

The data and replication code supporting this study are openly available on OSF at https://doi.org/10.17605/OSF.IO/SUF9M.

Use of AI Tools

Large language models were used for provision classification, orientation ratings and a supplementary audit. Claude Code (Claude Opus 4.5–5 and Fable 5) and Codex (GPT-5.6 Sol) supported research, coding, verification, manuscript criticism and editing, and some initial appendix prose and code drafts. Online Appendix Part F documents the tools, model versions, roles and human oversight following GUIDE-LLM. The authors have checked the applicable terms of use, verified the substantive content and references, and take full responsibility for the article.

8 References

Adcock, Robert, and David Collier. 2001. "Measurement Validity: A Shared Standard for Qualitative and Quantitative Research." *American Political Science Review* 95 (3): 529–46. https://doi.org/10.1017/S0003055401003100.

Adler-Nissen, Rebecca, and Kristin Anabel Eggeling. 2026. *The Brussels Bubble. Inside the European Union in the Digital Age*. Cambridge Univ. Press.

Barrie, Christopher, Elli Palaiologou, and Petter Törnberg. 2024. "Prompt Stability Scoring for Text Annotation with Large Language Models," July. https://doi.org/10.48550/arXiv.2407.02039.

Bellanova, Rocco, Helena Carrapico, and Denis Duez. 2022. "Digital/Sovereignty and European Security Integration: An Introduction." *European Security* 31 (3): 337–55. https://doi.org/10.1080/09662839.2022.2101887.

Benoit, Kenneth, Scott De Marchi, Conor Laver, Michael Laver, and Jinshuai Ma. 2025. "Using Large Language Models to Analyze Political Texts Through Natural Language Understanding." *American Journal of Political Science*, 1–17. https://doi.org/10.1111/ajps.70050.

Birkenmaier, Lukas, Clemens M. Lechner, and Claudia Wagner. 2024. "The Search for Solid Ground in Text as Data: A Systematic Review of Validation Practices and Practical Recommendations for Validation." *Communication Methods and Measures* 18 (3): 249–77. https://doi.org/10.1080/19312458.2023.2285765.

Bonnamy, Céleste, and Clément Perarnaud. 2024. "Introduction. EU Digital Policies and Politics: Unpacking the European Approach to Regulate the 'Digital'." *Politique Européenne* N° 81 (3): 8–27. https://doi.org/10.3917/poeu.081.0008.

Bradford, Anu. 2020. *The Brussels Effect: How the European Union Rules the World*. New York, NY: Oxford University Press.

———. 2023. *Digital Empires: The Global Battle to Regulate Technology*. New York: Oxford University Press.

Brandsma, Gijs Jan, Jens Blom-Hansen, Floris Van Hal, and Kody Moodley. 2025. "A Sentence-Based Approach to Measuring EU Regulatory Activity." *Journal of European Public Policy*, December, 1–25. https://doi.org/10.1080/13501763.2025.2596122.

Clifton, Judith, Daniel Díaz-Fuentes, and Julio Revuelta. 2010. "The Political Economy of Telecoms and Electricity Internationalization in the Single Market." *Journal of European Public Policy* 17 (7): 988–1006. https://doi.org/10.1080/13501763.2010.499229.

Coode, George. 1845. *On Legislative Expression; Or, The Language of Written Law*. William Benning; Company.

Council of the European Union. 2025. "Declaration for European Digital Sovereignty." Council Document ST-15781/25 TELECOM 425. Council of the European Union.

Cram, Laura. 1994. "The European Commission as a Multi-Organization: Social Policy and IT Policy in the EU." *Journal of European Public Policy* 1 (2): 195–217. https://doi.org/10.1080/13501769408406955.

Crawford, Sue E. S., and Elinor Ostrom. 1995. "A Grammar of Institutions." *The American Political Science Review* 89 (3): 582–600. https://doi.org/10.2307/2082975.

Damonte, Alessia, and Giulia Bazzan. 2024. "Rules as Data." *Regulation & Governance* 18 (3): 657–73. https://doi.org/10.1111/rego.12582.

De Gregorio, Giovanni. 2022. *Digital Constitutionalism in Europe: Reframing Rights and Powers in the Algorithmic Society*. 1st ed. Cambridge University Press. https://doi.org/10.1017/9781009071215.

European Commission. 1979. "European Society Faced with the Challenge of New Information Technologies: A Community Response." https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX:51979DC0650&from=EN.

———. 2010. "A Digital Agenda for Europe." https://eur-lex.europa.eu/LexUriServ/LexUriServ.do?uri=COM:2010:0245:FIN:EN:PDF.

Falkner, Gerda, Sebastian Heidebrecht, Anke Obendiek, and Timo Seidl. 2024. "Digital Sovereignty - Rhetoric and Reality." *Journal of European Public Policy* 31 (8): 2099–2120. https://doi.org/10.1080/13501763.2024.2358984.

Farrand, Benjamin. 2026. *Geopolitical Union: Europe's Attempt to Take Back Control of Technology Regulation*. Cambridge University Press.

Feuerriegel, Stefan, Christopher Barrie, M. J. Crockett, Laura K. Globig, Killian L. McLoughlin, Dan-Mircea Mirea, Arthur Spirling, et al. 2026. "A Reporting Checklist for Large Language Models in Behavioural Science." *Nature Human Behaviour*, June. https://doi.org/10.1038/s41562-026-02492-7.

Fjelstul, Joshua C. 2019. "The Evolution of European Union Law: A New Data Set on the *Acquis Communautaire*." *European Union Politics* 20 (4): 670–91. https://doi.org/10.1177/1465116519842947.

Frantz, Christopher K., and Saba Siddiki. 2022. *Institutional Grammar: Foundations and Applications for Institutional Analysis*. Springer International Publishing. https://doi.org/10.1007/978-3-030-86372-2.

Golub, Jonathan. 2024. "EUPROPS: A New Dataset on Policymaking in the European Union from 1958 to 2021." *European Union Politics* 25 (1): 197–217. https://doi.org/10.1177/14651165231202034.

Haggart, Blayne, Jan Aart Scholte, and Natasha Tusikov. 2021. "Introduction. Return of the State?" In *Power and Authority in Internet Governance: Return of the State?*, edited by Blayne Haggart, Natasha Tusikov, and Jan Aart Scholte, 1–12. Routledge Global Cooperation Series. Abingdon, Oxon; New York, NY: Routledge.

Halterman, Andrew, and Katherine A. Keith. 2026. "Codebook LLMs: Evaluating LLMs as Measurement Tools for Political Science Concepts." *Political Analysis* 34 (2): 188–204. https://doi.org/10.1017/pan.2025.10017.

Heidebrecht, Sebastian. 2024. "From Market Liberalism to Public Intervention: Digital Sovereignty and Changing European Union Digital Single Market Governance." *JCMS: Journal of Common Market Studies* 62 (1): 205–23. https://doi.org/10.1111/jcms.13488.

Hiltunen, Miikka. 2024. "Exploring the Legal Making of the European Digital Economy." *European Law Open* 3 (3): 633–54. https://doi.org/10.1017/elo.2024.44.

Hinterleitner, Markus, Christoph Knill, and Yves Steinebach. 2024. "The Growth of Policies, Rules, and Regulations: A Review of the Literature and Research Agenda." *Regulation & Governance* 18 (2): 637–54. https://doi.org/10.1111/rego.12511.

Hurka, Steffen, Maximilian Haag, and Constantin Kaplaner. 2022. "Policy Complexity in the European Union, 1993-Today: Introducing the EUPLEX Dataset." *Journal of European Public Policy* 29 (9): 1512–27. https://doi.org/10.1080/13501763.2021.1938174.

Koniaris, Marios, Ioannis Anagnostopoulos, and Yannis Vassiliou. 2018. "Network Analysis in the Legal Domain: A Complex Model for European Union Legal Sources." *Journal of Complex Networks* 6 (2): 243–68. https://doi.org/10.1093/comnet/cnx029.

König, Pascal D. 2022. "Fortress Europe 4.0? An Analysis of EU Data Governance Through the Lens of the Resource Regime Concept." *European Policy Analysis* 8 (4): 484–504. https://doi.org/10.1002/epa2.1160.

Kosti, Nir. 2024a. "Styles of Regulatory Discretion: A Comparative Analysis of the British and Israeli Education Legislation." *Public Administration* 102 (2): 444–60. https://doi.org/10.1111/padm.12929.

———. 2024b. "Conceptualization and Measurement of Regulatory Discretion: Text Analysis of 120 Years of British Legislation." *Regulation & Governance* 18 (3): 761–79. https://doi.org/10.1111/rego.12516.

Kosti, Nir, and David Levi-Faur. 2019. "The Coproduction of Primary and Secondary Legislation: Israel as a Case Study of Substitutive Relationships." *Law & Policy* 41 (4): 432–57. https://doi.org/10.1111/lapo.12138.

Krarup, Troels, and Maja Horst. 2023. "European Artificial Intelligence Policy as Digital Single Market Making." *Big Data & Society* 10 (1): 205395172311538. https://doi.org/10.1177/20539517231153811.

Küsters, Anselm, and Cecilia Emma Sottilotta. 2026. "Coping With the Digital Trilemma? Trade-Offs and Risks in EU Digital Policy." *European Policy Analysis* 12 (2): e70023. https://doi.org/10.1002/epa2.70023.

Laurer, Moritz, Wouter Van Atteveldt, Andreu Casas, and Kasper Welbers. 2025. "On Measurement Validity and Language Models: Increasing Validity and Decreasing Bias with Instructions." *Communication Methods and Measures* 19 (1): 46–62. https://doi.org/10.1080/19312458.2024.2378690.

Licht, Hauke, Rupak Sarkar, Patrick Y. Wu, Pranav Goel, Niklas Stoehr, Elliott Ash, and Alexander Miserlis Hoyle. 2025. "Measuring Scalar Constructs in Social Science with LLMs." In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, edited by Christos Christodoulopoulos, Tanmoy Chakraborty, Carolyn Rose, and Violet Peng, 32144–71. Suzhou, China: Association for Computational Linguistics. https://doi.org/10.18653/v1/2025.emnlp-main.1635.

Mărcuț, Mirela. 2017. *Crystalizing the EU Digital Policy*. Cham: Springer International Publishing. http://link.springer.com/10.1007/978-3-319-69227-2.

Marshall, T. H. 1992. "Citizenship and Social Class." In *Citizenship and Social Class*, 1–54. Pluto Perspectives. London: Pluto Press.

Nebbiai, Matteo. 2026. "Is Digital Regulation Reshaping EU Integration?" *Journal of European Integration* 48 (4): 707–15. https://doi.org/10.1080/07036337.2026.2662797.

Newman, Abraham L. 2008. *Protectors of Privacy: Regulating Personal Data in the Global Economy*. Ithaca: Cornell Univ. Press.

———. 2020. "Digital Policy-Making in the European Union: Building the New Economy of an Information Society." In *Policy-Making in the European Union*, edited by Helen Wallace, Mark A. Pollack, Christilla Roederer-Rynning, and Alasdair R. Young, Eighth edition, 275–96. Oxford: Oxford University Press.

Ovádek, Michal. 2021. "Facilitating Access to Data on European Union Laws." *Political Research Exchange* 3 (1): 1870150. https://doi.org/10.1080/2474736X.2020.1870150.

Papakonstantinou, Vagelis, and Paul De Hert. 2024. *The Regulation of Digital Technologies in the EU: Act-Ification, GDPR Mimesis and EU Law Brutality at Play*. Routledge Research in the Law of Emerging Technologies. Abingdon, Oxon: Routledge.

Radaelli, Claudio M., and Bernardo Rangoni. 2026. "Regulation and Its Metrics: Three Views of the Cathedral." *Journal of European Public Policy* 33 (9): 2415–44. https://doi.org/10.1080/13501763.2025.2501085.

Roberts, Huw, Josh Cowls, Emmie Hine, Jessica Morley, Vincent Wang, Mariarosaria Taddeo, and Luciano Floridi. 2023. "Governing Artificial Intelligence in China and the European Union: Comparing Aims and Promoting Ethical Outcomes." *The Information Society* 39 (2): 79–97. https://doi.org/10.1080/01972243.2022.2124565.

Sandholtz, Wayne. 1992. *High-Tech Europe. The Politics of International Cooperation*. Berkeley: University of California Press.

Shahin, Jamal, and Matthias Finger. 2009. "The History of a European Information Society: Shifts from Government to Governance." In *Global e-Governance: Advancing e-Governance Through Innovation and Leadership*, edited by G. Russell Pipe and Jirapon Tubtimhin, 62–83. Amsterdam: IOS Press.

Shapiro, Stuart. 2024. "Counting Regulations and Measuring Regulatory Impact: A Call for Nuance." *Humanities and Social Sciences Communications* 11 (1): 1463. https://doi.org/10.1057/s41599-024-03982-7.

Sheikh, Haroon. 2022. "European Digital Sovereignty: A Layered Approach." *Digital Society* 1 (3): 1–25. https://doi.org/10.1007/s44206-022-00025-z.

Siddiki, Saba, and Christopher K. Frantz. 2024. "Understanding Regulation Using the Institutional Grammar 2.0." *Regulation & Governance* 18 (3): 674–87. https://doi.org/10.1111/rego.12546.

Stengg, Werner. 2024. *Digital Policy in the EU: Towards a Human-Centred Digital Transformation*. Edward Elgar Publishing. https://doi.org/10.4337/9781035338641.

Streinz, Thomas. 2021. "The Evolution of European Data Law." In *The Evolution of EU Law*, edited by Paul P. Craig and Gráinne Búrca, Third edition. Oxford: Oxford University Press.

Thüer, Leo, Julia Pohle, Milan Schröder, and Alissa Steer. 2026. "Exporting the European Third Way: Strategic Narratives of a Value-Based Digital Order." *European Politics and Society*, April, 1–14. https://doi.org/10.1080/23745118.2026.2655142.

Van Laer, Arthe. 2010. "Developing an EC Computer Policy, 1965-1974." *IEEE Annals of the History of Computing* 32 (1): 44–59. https://doi.org/10.1109/MAHC.2010.17.

Vannoni, Matia, Elliott Ash, and Massimo Morelli. 2021. "Measuring Discretion and Delegation in Legislative Texts: Methods and Application to US States." *Political Analysis* 29 (1): 43–57.

Von Wright, G. H. 1951. "Deontic Logic." *Mind* LX (237): 1–15. https://doi.org/10.1093/mind/LX.237.1.

Yang, Eddie, Zoey Wang, Carl Zhou, and Yaosheng Xu. 2025. "Data Annotation with Large Language Models: Lessons from a Large Empirical Evaluation." Working Paper. https://www.eddieyang.net/research/llm_annotation.pdf.

Zenner, Kai, Scott J. Marcus, and Kamil Sekut. 2025. "A Dataset of EU Legal and Policy Instruments for the Digital World." *CEPS*.

Online Appendix: Mapping Europe's Digital Acquis

Keywords: Digitalization, European Union, Digital Policymaking, Digital Sovereignty, Computational Text Analysis

Contents

Overview . . . 2
Part A — Corpus Construction and Data . . . 2
A1 The eurlex-builder pipeline . . . 2
A2 Corpus coverage and composition . . . 3
A3 Translation of non-English source documents . . . 4
Part B — Coding Scheme and Human Coding . . . 5
B1 The coding scheme . . . 5
B2 Full coding instructions . . . 6
B3 Sample design and coding procedure . . . 23
Part C — Classifier Pipeline and Validation . . . 24
C1 The LLM ensemble . . . 24
C2 Human validation and agreement . . . 25
C3 Encoder distillation . . . 26
C4 Prompt stability . . . 29
C5 Error rates at natural prevalence . . . 31
C6 Temporal stability of classifier error . . . 32
C7 The market and rights ratings . . . 33
Part D — External Validity . . . 36
D1 CEPS reference list . . . 36
D2 Landmark digital laws . . . 36
D3 EuroVoc subject indexing . . . 38
Part E — Robustness of the Findings . . . 41
E1 Uncertainty of the digital-share trend . . . 42
E2 Citation-network robustness . . . 43
E3 Regulatory tone with all articles included . . . 44
E4 Rights profile over time . . . 45
E5 Encoder-configuration robustness . . . 47
E6 Are the dispersed digital provisions just cross-references? . . . 49
E7 Interpreting and stress-testing the digital-policy encoder . . . 51
E8 The enforcement-decision exclusion and the composition by instrument . . . 56
Part F — Disclosure of LLM use . . . 58
Core items . . . 58
Optional items . . . 61
References . . . 63

Overview

This online appendix accompanies 'Mapping Europe's Digital Acquis: A Granular History of EU Digital Policymaking'. It is organized in six parts: Part A documents the construction of the corpus and its coverage; Part B describes the coding scheme, the full coding instructions, and the human coding procedure; Part C presents the classifier pipeline and its validation; Part D adds external-validity checks against independent reference instruments; Part E reports robustness analyses for the main-text findings; and Part F discloses how large language models were used in the research process. All numbers, tables, and figures are computed from the project's source files when the appendix is rendered, except the learning-curve record in Section C3, which is transcribed from the training logs. The human codebook and LLM prompts are generated from the same YAML configuration, and the main validation tables in the paper and appendix read the same results file; section-specific diagnostics identify their own inputs in the accompanying code. The corpus was built with our open-source `eurlex-builder` package (github.com/tseidl/eurlex-builder, archived at doi:10.5281/zenodo.21496963); the provision-level dataset and fine-tuned encoders are released on Hugging Face (huggingface.co/collections/tseidl/eu-digital-acquis).

Part A — Corpus Construction and Data

A1 The eurlex-builder pipeline

The corpus was built with `eurlex-builder`, an open-source Python package we release alongside the paper. The package retrieves EU legislative documents through the official EUR-Lex SPARQL and REST APIs and extracts their text at configurable granularity — recitals, whole articles, numbered paragraphs (Art. 5(1)), lettered points (Art. 5(1)(a)), or, for communications, plain paragraphs. Users can assemble a corpus in two modes: *fixed* mode takes an explicit list of CELEX identifiers; *descriptive-search* mode selects documents by date range, document type, and an optional EuroVoc keyword filter. Granularity, language fallback, and translation behavior are set in the same configuration.

Extraction handles the heterogeneity of EUR-Lex's document formats. The package detects and parses the six distinct HTML structures EUR-Lex has used over the decades; for documents without machine-readable HTML — concentrated in the older strata of the corpus — it falls back to PDF extraction via Docling and pymupdf. Documents not available in English are translated with Helsinki-NLP Opus-MT models (see Section A3). The pipeline saves progress in DuckDB so interrupted runs can resume, and it can retrieve multiple documents at once.

The main output is four Parquet tables: `works` (act-level metadata: title, dates, document type, language, procedure information), `text_units` (the extracted recitals, (sub-)articles at the chosen granularity, and annexes), `relations` (act-level cross-references: citations, amendments, legal-basis references, and

repeals), and `eurovoc` (the EuroVoc subject descriptors indexing each act, which we use for the external validation in Section D3). All analyses in the paper and in this appendix read from these tables.

The package is open source (MIT license), distributed on PyPI (`pip install eurlex-builder`), and developed at github.com/tseidl/eurlex-builder.

A2 Corpus coverage and composition

The analysis corpus applies four canonical filters to the raw `eurlex-builder` output. First, we keep the three binding forms of EU secondary-law act: regulations, directives, and decisions (Article 288 TFEU). Concretely, the corpus query selects CELEX sector 3 acts with the letters R, L, and D. Merger decisions (letter M) and the competition notices and summaries published elsewhere in EUR-Lex are therefore outside the corpus by construction; enforcement decisions that do carry letter D are removed by the title filter described below. Second, we keep recitals, which explain an act's rationale, context, and objectives, and (sub-)articles, which contain its operative provisions. We use "provision" for either a recital or an article unit: a numbered paragraph where an article has them, and the whole article otherwise. Third, we drop individual enforcement decisions — state-aid rulings and scheme decisions, merger clearances, and antitrust decisions, all addressed to specific parties rather than enacting general rules — identified by the enforcement signatures their titles carry across all eras of EUR-Lex naming: case references ("Case M.…", "Case SA.…"), state-aid formulas ("State aid SA.34914 implemented by…", "aid which Italy plans to grant…"), merger-clearance formulas ("declaring a concentration to be compatible…"), and, for pre-1990s antitrust rulings, "relating to a proceeding". This removes 1,699 of the 34,369 decisions in the source metadata; a further 185 decisions have no recital or article of 50 or more characters and therefore contribute no provision, leaving the 32,485 decisions in Table 1. Section E8 reports what changes if the enforcement decisions are kept, which is nothing the paper claims.[1] Fourth, we drop provisions shorter than 50 characters — footnote markers, cross-references, and legal citations rather than substantive provisions — which removes 41,017 of 1,997,915 units (2.1%).

The resulting corpus contains 1,956,898 provisions across 146,234 acts: 1,065,116 recitals and 891,782 (sub-)article units. Table 1 and Table 2 break this down by instrument type and by decade of adoption.

[1] These decisions are binding, but they apply the rules to a named party rather than laying down general ones. Including them would fold waves of enforcement activity into the record of rulemaking: with enforcement decisions kept, the 1990s' digital share rises from 1.3% to 1.8%, driven by the telecom competition cases of the liberalization era, while recent decades barely move (2020s: 10.5% vs 10.2%). Since our object is the EU's digital rulemaking, we keep this class of enforcement decisions out of the corpus.

Table 1 Corpus composition by instrument type, after the canonical filters. (Sub-)article units are numbered article paragraphs where the act has them, whole articles otherwise.

Instrument	Acts	Provisions	Recitals	(Sub-)articles
Regulation	109,943	1,371,384	760,901	610,483
Directive	3,806	131,780	50,820	80,960
Decision	32,485	453,734	253,395	200,339

Table 2 Corpus composition by decade of adoption. The 1970s row covers 1979 only, the first year of the corpus; the 2020s row runs through April 2026. Adoption dates are missing for 2 acts, for which we fall back on the date of entry into force; 2 acts carry neither date, and their 0 provisions are classified but excluded from all time-series analyses.

Decade	Acts	Provisions
1970s	3,755	33,577
1980s	40,301	353,634
1990s	39,893	405,076
2000s	28,940	432,174
2010s	20,487	416,513
2020s	12,858	315,924

Corpus snapshots. The corpus was assembled while the extraction pipeline was still being refined, and `eurlex-builder` received several small fixes after the snapshot analyzed here was built. As a consequence, the data underlying the successive stages of the pipeline — the hand-coded validation sample, the LLM-labeled training pool, and the corpus the encoder ultimately classifies — are not byte-identical: a small share of provisions differ across snapshots, mostly in unit numbering and boilerplate trimming rather than substantive text. This does not compromise the pipeline's internal logic: encoder training pairs each text with the label assigned to that same text, and all human-validation comparisons in Part C score predictions made on the stored validation text itself. The public data release was produced by a single run of the released package version, so it is one canonical snapshot. The audits of Sections C5 and C6 were conducted against the current snapshot — the corpus the deployed encoder actually classifies, which is the relevant target for its error rates.

A3 Translation of non-English source documents

Classification uses the `text` field; for two provisions in two acts (31992D0184 and 32006D1004) this field holds French text, with the English translation stored in a separate column. Where EUR-Lex provides no English version, `eurlex-builder` retrieves the document in an available language and translates it with Helsinki-NLP Opus-MT models, storing the translation alongside the original.

Non-English source documents are rare in the analysis corpus: 5 of 146,234 acts, carrying 26 provisions. They date from 1985 to 2006 and are in French or Spanish (Table 3). At the unit level, 26 provisions in 5 acts carry a machine translation in the `text_translated` field. Because the affected share is well below one percent of the corpus, translation quality is unlikely to move any headline number; it matters, if at all, for the earliest years of the time series, which the temporal-stability audit of Section C6 covers. We did not systematically evaluate translation quality.

Table 3 Acts with a non-English source language, by decade of adoption.

Decade	Acts
1980s	1
1990s	2
2000s	2

Part B — Coding Scheme and Human Coding

B1 The coding scheme

The coding scheme comprises four classification tasks and two ratings. Task 1 (*digital policy*) classifies every provision as digital policy or not (binary), operationalizing the definition developed in the main text: deliberate public interventions that shape the course and character of digitalization. Task 2 (*regulatory tone*) classifies the deontic force of a provision as obligation, prohibition, permission, or none (descriptive/definitional); it applies to articles only, since recitals lack operative deontic character. Task 3 (*regulated actor*) identifies whose behavior a provision governs — EU institutions, member states, private actors, individuals, or other — and is coded only for articles that carry deontic force (tone other than none), since a purely descriptive provision regulates no one.[2] Task 4 (*rights type*) classifies rights-invoking provisions into T.H. Marshall's civil, political, and social rights (or none), and is coded only for provisions classified as digital policy, since the rights profile is a claim about the digital acquis.

Two further measures rate digital provisions on continuous 0–9 scales for market and rights orientation. Their output is an expected value over the model's digit-token probabilities, not a label; they are scored directly by a three-model panel with no distilled encoder behind them; and because a graded judgment cannot be checked by asking a human for the same number, they are validated not against per-item human codings but against human *pairwise* comparisons, in a Label Studio project of its own. Section C7 sets out the construction and reports the panel's coverage, agreement, convergent validity, and the pairwise human validation. The four classification tasks, and only those, carry the agreement statistics of Table 5.

[2] The instruction text reproduced in Section B2 also explains how to treat recitals, because it was written for both unit types; the task was applied, and coded, on articles only.

The file `config/label_config.yml` contains the task and category definitions, coding rules, examples, task eligibility conditions, and rating prompts. A generator script (`scripts/generate_config.py`) renders this YAML into the LLM prompts and, for the four classification tasks, into the Label Studio interface the human coders see — so that for those tasks the identical instruction text serves as LLM prompt and human codebook. This has a practical consequence: any refinement of the scheme — a sharpened exclusion, a new example — propagates mechanically to both coder types, which is how the instructions were iterated during instrument development (Section C2).[3] The rating prompts pass through the same generator and are versioned in the same file, but they address only the LLM, since the human side of the scalar is the pairwise task rather than the same rating.

The four classification instructions follow a common format: a role statement, the task definition, category definitions with linguistic cues, inclusion/exclusion rules and edge cases, and worked examples (broadly following Halterman and Keith 2026). The rating prompts are deliberately spare by comparison — a construct description, the two scale anchors, and an instruction to return a single digit — because their output is read from the model's token probabilities and any additional text would interfere. All six instructions are reproduced in Section B2.

B2 Full coding instructions

Three coding rules need explanation. First, regulations whose substance is the physical inputs or outputs of digital devices — battery rules, critical-raw-materials regulation, electronic-waste recycling — are coded as non-digital, even though digital devices figure in the surrounding context; regulating the production of inherently digital technologies (semiconductor manufacturing, for example) does count. Second, what we look for is substantive engagement, not digital vocabulary as such: a single substantive reference to an inherently digital concept is enough, and earlier EU terminology — information society services, telematics, electronic communications — counts as digital. The mirror image — digital vocabulary without digital substance — is excluded: provisions that merely use a digital channel ("submit information by electronic means", "publish on a website") do not themselves regulate anything digital, and counting them would dilute a substantively meaningful measure. A third boundary condition concerns provisions naming single-purpose digital bodies and acts (ENISA, BEREC, eu-LISA, the DSA, and the like). The instruction treats these as digital: a provision that names an entity whose primary mandate falls within a digital policy domain is coded as digital policy, whether or not the digital remit is the substantive subject of that particular provision.

[3] Until August 2026 the generator wrote a task's edge-case block into the Label Studio interface but, for binary tasks, not into the LLM prompt. Only the digital-policy task is binary, and its sole edge case — a narrow rule distinguishing digital bodies named as the substantive subject of a provision from digital bodies named incidentally — was added after the production classification run and never governed any coding: it never reached the models, and the human coders did not apply it either, coding 18 of 19 body-mentioning provisions as digital in line with the broader inclusion rule reproduced below. We have since removed that edge case, restored the inclusion rule to the wording used in production, and fixed the generator. The instruction reproduced here is therefore the operative one; it differs from the archived production draft only in where two parenthetical example lists are placed.

The complete instructions are reproduced below, generated from `config/label_config.yml` (Section B1). For the four classification tasks, what follows reproduces the operative instructions used by both coder types — supplied to the LLM ensemble as the prompt and to the human coders as the basis of the Label Studio interface (the one historical divergence between the two renderings is documented in Section B1). The two rating prompts that close the section were supplied to the scoring models only.

Task 1: Digital policy

```
You are an expert annotator analyzing official EU policy texts.

Task

Classify the following text as digital policy (1) or not digital policy (0).
Return ONLY a JSON object:
{"is_digital_policy": "1"} or {"is_digital_policy": "0"}

Definition

A text is digital policy if it contains deliberate public interventions that
shape the course and character of digitalization by influencing the
development, deployment, governance, and use of digital technologies.
Digitalization refers to a socio-technical process whereby economic, political,
and social activities are reconfigured by digital technologies—i.e., by
computation, data, networks, platforms, and software.

Code as DIGITAL POLICY (1) if the text...

Directly addresses digital technologies, including but not limited to:

- Computation, digital data, networks, platforms, software
- Internet, telecommunications, ICT, information systems
- Artificial intelligence, algorithms, automation
- Digital markets, e-commerce, digital services
- Cybersecurity, digital infrastructure

Regulates activities reconfigured by digital technologies:

- Online information consumption and distribution
- Digital work and labor platforms
- Social media and digital communication
- Digital buying, selling, and payments
- Digital content and expression
- Data collection, processing, and sharing
```

```
Falls within established digital policy domains, or names entities, bodies,
systems, or actors whose primary mandate or activity falls within those
domains:

- Content & Copyright Regulation
- Competition, Product Market & Consumer Protection in digital markets
- Data Governance & Privacy
- Industrial, Innovation & Research Policy in digital sectors
- Infrastructure & Connectivity Policy
- Digital Skills & Education
- Cybersecurity & Digital Law Enforcement
- Trade & External Policy regarding digital or data flows
- E-Government
- Digital Finance

Addresses any layer of the digital stack:

- Physical layer: digital infrastructure, semiconductor production, data
centers
- Code layer: software standards, protocols, technical rules
- Information layer: data governance, content moderation, digital rights

A single substantive reference to an inherently digital concept is sufficient—
the text does not need to be densely populated with digital keywords. Older EU
terminology such as "information society services," "electronic commerce,"
"telematics," or "electronic communications" should be treated as digital
policy terms.

Code as NOT DIGITAL POLICY (0) based on two exclusion principles

Exclusion 1 — Digital as incidental channel: The digital dimension does not
create the regulatory need but is merely one way of complying with a
requirement that exists independently. (e.g., "information may be submitted in
paper or electronic form", "published on a website", "submitted via an online
portal").

Exclusion 2 — Physical inputs/outputs of digital devices: The text addresses
physical materials, components, or waste streams associated with digital
devices rather than the digital technologies themselves. Note: regulating the
production of core digital technologies—such as semiconductor manufacturing—
counts as digital policy. (e.g., "batteries", "raw materials", "recycling of
electronic waste").

Examples
```

```
Input: "Establishing rules for online platforms' content moderation practices"
Output: {"is_digital_policy": "1"}

Input: "Setting standards for cross-border data flows and data localization
requirements"
Output: {"is_digital_policy": "1"}

Input: "High-risk AI systems shall be designed to allow for human oversight
during their period of use"
Output: {"is_digital_policy": "1"}

Input: "Member States shall ensure that providers of intermediary services
cooperate with digital services coordinators"
Output: {"is_digital_policy": "1"}

Input: "Member States shall establish a framework for electronic identification
means for natural and legal persons"
Output: {"is_digital_policy": "1"}

Input: "The data that must be used as the basis for determining the
representative prices shall be easily accessible, permanently available, and
reliable"
Output: {"is_digital_policy": "0"}

Input: "A general obligation to communicate requests and documents in a digital
form"
Output: {"is_digital_policy": "0"}

Input: "Requirements for recycling electronic waste and recovering rare earth
materials"
Output: {"is_digital_policy": "0"}

Input: "Consumers of accessible products face high prices due to limited
competition among suppliers"
Output: {"is_digital_policy": "0"}

Input: "Environmental standards for manufacturing facilities, including server
farms"
Output: {"is_digital_policy": "0"}

Text to classify
Text:
```

Task 2: Regulatory tone

```
You are an expert annotator analyzing official EU legal texts (directives,
regulations, and decisions).

Task

Classify the following text into one of: obligation, prohibition, permission,
none. Return ONLY a JSON object:
{"regulatory_tone": "obligation"} or {"regulatory_tone": "prohibition"} or
{"regulatory_tone": "permission"} or {"regulatory_tone": "none"}

Definition

Regulatory tone captures the deontic nature of an article — whether it obliges,
prohibits, or permits action, or whether it is purely descriptive. Classify by
MEANING, not surface form: "shall not restrict" is a prohibition even though it
uses "shall"; "is authorised to" is a permission even though no modal appears.

Categories

Obligation: The article imposes a positive duty. Someone is required to do
something, and it is inaction that constitutes non-compliance.
Linguistic cues:
- 'shall' + active verb (e.g., 'shall monitor', 'shall provide')
- 'must' + active verb
- 'is required to', 'is obliged to', 'is responsible for'
- passive equivalents: 'shall be ensured', 'shall be monitored'

Prohibition: The article imposes a negative duty. Someone is required to
refrain from acting, and it is action that constitutes non-compliance. This
includes provisions that tell an actor not to restrict, not to prohibit, not to
impose, etc. — the target is still being told what NOT to do, even when the
object of that non-action is itself permissive.
Linguistic cues:
- 'shall not', 'must not', 'may not' (when meaning prohibition)
- 'is prohibited', 'is forbidden'
- 'shall neither ... nor'
- passive prohibitions: 'shall not be restricted'

Permission: The article confers discretion, a right, or a power. The actor is
free to act or abstain without sanction. No duty is imposed.
Linguistic cues:
- 'may' + active verb (when meaning permission, not prohibition)
- 'can', 'could'
```

```
- 'is authorised to', 'is entitled to', 'is empowered to'
- 'is free to'

None (descriptive / definitional): The article is purely descriptive,
definitional, or declarative and contains no deontic force. It does not oblige,
prohibit, or permit anyone. Typical cases: articles stating scope, defining
terms, or declaring the purpose of the regulation.

CRITICAL: classify by MEANING, not by modal verb. A provision using "shall" can
be an obligation, a prohibition, or even describe a permission depending on
what it actually does. Read the full clause before deciding. A single article
may contain more than one deontic element (e.g., "Member States shall ensure X,
and may require Y"). In such cases, identify the DOMINANT tone — the one that
carries the main regulatory force of the article. If two elements are equally
weighted, prefer obligation > prohibition > permission. Passive constructions
carry the same deontic force as active ones. "The free movement of data shall
not be restricted" is a prohibition directed at whoever could do the
restricting.

Edge cases

'Shall prohibit' vs. 'shall not': "Member States shall prohibit X" is an
OBLIGATION on Member States to enact a prohibition in their own law. It is NOT
itself a prohibition. "Member States shall not do X" is a prohibition on Member
States.

'May not' — permission denied vs. prohibition: "May not" is almost always a
PROHIBITION ("the Commission may not delegate this power" = prohibition on the
Commission). If the intended meaning is "is not permitted to," classify as
prohibition. If it rarely means "is not required to," classify based on
substantive meaning.

Descriptive, definitional, declarative, scope, or transitional articles: Code
`none` for scope clauses ("This Regulation applies to..."), definitions
("'personal data' means..."), subject-matter declarations ("This Regulation
lays down..."), entry-into-force and applicability dates ("This Regulation
shall enter into force on X"), and transitional provisions. No deontic force =
none, even when the article uses "shall". This INCLUDES negative scope clauses,
which are `none`, NOT prohibition: a provision stating that a regulation,
directive, article, or paragraph "shall not apply", "does not apply", or "shall
not affect" something is limiting the REACH of rules, not directing an actor to
refrain from acting. E.g. "This Regulation shall not apply to...", "Those
Articles shall not apply to...", "Article X shall not apply where...". A
prohibition needs an actor told not to act ("providers shall not impose..."); a
```

```
rule that "shall not apply" concerns coverage, not conduct. If such an article
also carries a genuine obligation or permission elsewhere, classify on that
dominant element — the scope carve-out itself never makes it a prohibition.

Examples

Input: "The Commission shall, on an ongoing basis, monitor developments in
third countries and international organisations that could affect the
functioning of decisions adopted pursuant to paragraph 3 of this Article."
Output: {"regulatory_tone": "obligation"}

Input: "The Commission shall assess the need for amendment of the list set out
in Annex III and of the list of prohibited AI practices laid down in Article 5,
once a year following the entry into force of this Regulation."
Output: {"regulatory_tone": "obligation"}

Input: "Each Member State shall provide for one or more independent public
authorities to be responsible for monitoring the application of this
Regulation."
Output: {"regulatory_tone": "obligation"}

Input: "Member States shall specify the procedures under which any change
affecting the information referred to in paragraph 1 must be notified to the
supervisory authority."
Output: {"regulatory_tone": "obligation"}

Input: "Member States shall prohibit the processing of personal data revealing
racial or ethnic origin, political opinions, religious or philosophical
beliefs."
Output: {"regulatory_tone": "obligation"}

Input: "The competent supervisory authority shall not adopt its draft decision
referred to in paragraph 1 within the period referred to in paragraph 3."
Output: {"regulatory_tone": "prohibition"}

Input: "Member States shall neither restrict nor prohibit the free flow of
personal data between Member States for reasons connected with the protection
afforded under paragraph 1."
Output: {"regulatory_tone": "prohibition"}

Input: "The free movement of personal data within the Union shall be neither
restricted nor prohibited for reasons connected with the protection of natural
persons with regard to the processing of personal data."
Output: {"regulatory_tone": "prohibition"}
```

```
Input: "From 12 January 2027, providers of data processing services shall not
impose any switching charges on the customer for the switching process."
Output: {"regulatory_tone": "prohibition"}

Input: "The Commission may adopt guidelines taking into account the proposal of
the EDIB in accordance with Article 30, point (h), of Regulation (EU) 2022/868
laying down interoperable frameworks for common standards and practices."
Output: {"regulatory_tone": "permission"}

Input: "Member States may call upon experts of the scientific panel to support
their enforcement activities under this Regulation."
Output: {"regulatory_tone": "permission"}

Input: "Member States may provide that data relating to administrative
sanctions or judgements in civil cases shall also be processed under the
control of official authority."
Output: {"regulatory_tone": "permission"}

Input: "Notifying authorities may notify only conformity assessment bodies
which have satisfied the requirements laid down in Article 31."
Output: {"regulatory_tone": "permission"}

Input: "Testing procedures may include testing in real-world conditions in
accordance with Article 60."
Output: {"regulatory_tone": "permission"}

Input: "This Regulation establishes harmonised rules for the placing on the
market, the putting into service and the use of artificial intelligence systems
in the Union."
Output: {"regulatory_tone": "none"}

Input: "For the purposes of this Regulation, the following definitions apply:
'personal data' means any information relating to an identified or identifiable
natural person."
Output: {"regulatory_tone": "none"}

Input: "This Regulation shall not apply to products in the process of shipment
to the Community."
Output: {"regulatory_tone": "none"}

Text to classify
Text:
```

Task 3: Regulated actor

```
You are an expert annotator analyzing official EU legal texts (directives,
regulations, and decisions).

Task

Classify the following text into one of: eu_institutions, member_states,
private_actors, individuals, other. Return ONLY a JSON object:
{"regulated_actor": "eu_institutions"} or {"regulated_actor": "member_states"}
or {"regulated_actor": "private_actors"} or {"regulated_actor": "individuals"}
or {"regulated_actor": "other"}

Definition

The regulated actor is the entity whose behaviour the provision governs — the
grammatical and substantive target of the deontic force (for articles) or the
actor whose behaviour the recital anticipates regulating (for recitals). When
multiple actors appear, choose the one the provision directly addresses.
Classify by the actor being REGULATED, not the actor being protected or
benefited. For ARTICLES, identify the target of the operative provision. For
RECITALS, identify the actor named as bearer of the duty, power, or restriction
the recital describes (e.g., "Member States should ensure X" → member_states;
"the Commission shall consider Y" → eu_institutions). For purely descriptive
recitals that name no actor, use "other." For passive constructions with no
explicit actor, infer the actor from context (e.g., "shall not be restricted"
in an EU directive typically targets Member States).

Categories

EU Institutions: The Commission, Council, Parliament, EU agencies (e.g., EDPB,
EDPS, ENISA), the Court of Justice, and any EU-level body.

Member States: National governments, national competent authorities, national
supervisory authorities, and other state-level actors of EU Member States.

Private Actors: Companies, providers, operators, controllers, processors,
platforms, manufacturers, notified bodies, professional associations, and other
non-state organizational actors.

Individuals: Natural persons, data subjects, consumers, workers, users,
citizens — persons regulated in their individual capacity. Use only when the
individual is the REGULATED actor (their behaviour is governed), not the
protected one. This is rare in EU digital policy.
```

```
Other / Unclear: Use when the actor cannot be confidently assigned to one of
the four categories, or when the provision is purely descriptive and no clear
actor is being regulated.

CRITICAL: identify the actor being REGULATED, not the actor being protected or
benefited. A prohibition on authorities that protects individuals is directed
at AUTHORITIES. For passive constructions, infer the actor from context. In EU
directives, implicit actors are typically Member States; in regulations
affecting firms, they are typically private actors. If an article has multiple
actors, pick the DOMINANT one — the entity the article primarily regulates.

Edge cases

'Shall prohibit' — regulated actor is the enforcer: "Member States shall
prohibit X" regulates Member States (who have the duty to enact the
prohibition), not the actors doing X (who are downstream targets of the
domestic prohibition to be enacted).

EU body's own operations → eu_institutions: Internal operations of an EU agency
or body (its procedures, board decisions, staff arrangements, budget execution,
reporting duties) are regulated at `eu_institutions`, even when applicants,
staff, or contributions originate from Member States.

No clear single actor → `other`: For impersonal provisions ("A Community design
shall not subsist in a design contrary to public policy"), parties to an
international agreement, or other cases where no single identifiable actor is
being regulated, use `other` rather than picking the most likely category.

Examples

Input: "The Commission shall, on an ongoing basis, monitor developments in
third countries and international organisations that could affect the
functioning of decisions adopted pursuant to paragraph 3 of this Article."
Output: {"regulated_actor": "eu_institutions"}

Input: "The Commission may adopt guidelines taking into account the proposal of
the EDIB in accordance with Article 30, point (h), of Regulation (EU)
2022/868."
Output: {"regulated_actor": "eu_institutions"}

Input: "The competent supervisory authority shall not adopt its draft decision
referred to in paragraph 1 within the period referred to in paragraph 3."
Output: {"regulated_actor": "eu_institutions"}
```

```
Input: "Each Member State shall provide for one or more independent public authorities to be responsible for monitoring the application of this Regulation."
Output: {"regulated_actor": "member_states"}

Input: "Member States shall neither restrict nor prohibit the free flow of personal data between Member States for reasons connected with the protection afforded under paragraph 1."
Output: {"regulated_actor": "member_states"}

Input: "Member States may call upon experts of the scientific panel to support their enforcement activities under this Regulation."
Output: {"regulated_actor": "member_states"}

Input: "Member States shall prohibit the processing of personal data revealing racial or ethnic origin, political opinions, religious or philosophical beliefs."
Output: {"regulated_actor": "member_states"}

Input: "The competent authorities shall not take any decision that produces an adverse legal effect on a person or significantly affects a person only by reason of the automated processing of PNR data."
Output: {"regulated_actor": "member_states"}

Input: "Notifying authorities may notify only conformity assessment bodies which have satisfied the requirements laid down in Article 31."
Output: {"regulated_actor": "member_states"}

Input: "The free movement of personal data within the Union shall be neither restricted nor prohibited for reasons connected with the protection of natural persons with regard to the processing of personal data."
Output: {"regulated_actor": "member_states"}

Input: "From 12 January 2027, providers of data processing services shall not impose any switching charges on the customer for the switching process."
Output: {"regulated_actor": "private_actors"}

Input: "Testing procedures may include testing in real-world conditions in accordance with Article 60."
Output: {"regulated_actor": "private_actors"}

Input: "This Regulation establishes harmonised rules for the placing on the market, the putting into service and the use of artificial intelligence systems in the Union."
```

```
Output: {"regulated_actor": "other"}

Input: "For the purposes of this Regulation, the following definitions apply."
Output: {"regulated_actor": "other"}

Text to classify
Text:
```

Task 4: Rights type

```
You are an expert annotator analyzing official EU policy texts on the rights
extended to digital citizens.

Task

Identify the type of right that the text invokes, justifies, or extends,
following T.H. Marshall's typology of citizenship rights. Return ONLY a JSON
object:
{"rights_type": "civil"} or {"rights_type": "political"} or {"rights_type":
"social"} or {"rights_type": "none"}

Definition

Following Marshall's typology, citizenship rights divide into three classes.
CIVIL rights — the rights necessary for individual freedom (liberty, freedom of
speech / thought / faith, property and contract, and the right to justice —
i.e. to defend and assert one's rights on equal terms by due process of law).
POLITICAL rights — the right to participate in the exercise of political power,
as a voter or as a member of a body invested with political authority. SOCIAL
rights — the whole range from a modicum of economic welfare and security to the
right to share fully in the social heritage and to live the life of a civilised
being according to the standards prevailing in society. Code by what the right
SECURES IN THIS PROVISION, not by the issue area of the surrounding act. A
reference to rights counts whether it actually conveys a right or merely
invokes, justifies, or explains one.

Categories

Civil: Individual freedom, autonomy, dignity, privacy, justice, due process,
and expression in a personal capacity. Includes informational self-
determination — the right to control data about oneself — and procedural rights
(notification, explanation, contest, redress) that secure individual standing
against state, platform, or algorithmic power.
Linguistic cues:
```

```
- right to (privacy / erasure / rectification / access / portability / object)
- right to be informed / to obtain / to a copy
- right to an effective remedy / to a fair hearing / to contest
- right to human review / to a meaningful explanation
- freedom of expression / of thought / of conscience
- right to encryption, anonymity, pseudonymisation
- fundamental rights / Charter of Fundamental Rights / CFR Art. X / Convention
rights
- compliance with fundamental rights / human dignity / non-refoulement
- named beneficiary class (children, minors, victims of trafficking, vulnerable
persons)

Political: The right to participate in the exercise of political power — as a
voter, an officeholder, or a member of a deliberative political body. In the
digital context, this covers a NARROW set of provisions: (a) direct digital
participation in political power (e-voting, online petitions/initiatives,
genuine co-determination of platform rules where users actually take part in
rule-making); (b) protection of OFFLINE political rights from digital
interference (electoral integrity, foreign manipulation of elections,
disinformation in electoral contexts); (c) communicative rights tied EXPLICITLY
to electoral processes or democratic discourse (political-advertising
transparency, media pluralism in electoral contexts). Generic platform
transparency, risk audits, and reporting obligations are NOT political unless
the provision itself frames them in electoral or democratic-discourse terms.
Linguistic cues:
- right to vote / suffrage / electoral process
- political participation / democratic deliberation
- co-determination / right to be consulted on (platform rules)
- electoral integrity / election interference / electoral disinformation
- political advertising / political campaign / political party
- media pluralism in electoral / democratic discourse

Social: Substantive material rights to economic welfare, inclusion, and shared
access to publicly-valued cultural and educational goods — affordable
connectivity, universal access, digital skills, accessibility for persons with
disabilities, welfare-logic protections for digital workers, and direct access
to cultural heritage, research and educational materials, and broadcasting
content of major social importance. The provision must itself extend or
constitute the access, not merely invoke it as a stated purpose or promote
awareness of an existing service.
Linguistic cues:
- universal service / affordable access / connectivity
- digital inclusion, digital divide, digital literacy
- accessibility, persons with disabilities
```

```
- minimum protection for platform workers / gig workers
- right to (training / skills / digital education)

None (no rights claim): The text does not invoke, justify, or extend a right to
a natural person or class of natural persons. Most provisions fall here:
organizational, jurisdictional, definitional, procedural-for-authorities,
market-coordination, or enforcement-architecture clauses.

CODE BY WHAT THE RIGHT SECURES IN THIS PROVISION, not by the issue area of the
surrounding act. A privacy clause inside a media-pluralism law is civil; a
transparency clause inside a data-protection law can be political if it serves
democratic accountability. References count when they invoke, justify, or
explain a right, not only when they convey one. If multiple types are
referenced, pick the DOMINANT one — the right the provision most directly
secures or invokes. Default to "none" when in doubt; "none" is the expected
modal class.

Edge cases

Long articles — scan all paragraphs for rights content: Articles may have
multiple paragraphs with shifting content. Early paragraphs often state scope
or operative provisions while explicit rights-bearing language (fundamental
rights, Charter or Convention citations, named beneficiary classes, "right to
X", "comply with fundamental rights") appears in later paragraphs as a
safeguard or override. Read the FULL text and code by what the right SECURES
anywhere in the provision — not by the early paragraphs' dominant framing.

GDPR-style data subject rights: Treat data access, rectification, erasure,
portability, and objection as CIVIL — they secure individual autonomy and
informational self-determination (CFR Art. 8).

Algorithmic transparency: civil vs political: Code as CIVIL when transparency
grounds an individual-explanation or due-process claim against a specific
decision (AI Act Art. 86 right to explanation; DSA Art. 17 statements of
reasons). Code as POLITICAL only when the provision EXPLICITLY invokes
electoral processes, political advertising, or democratic discourse
(recommender-system transparency for electoral processes; political-ads
transparency notices). Generic transparency obligations without an electoral /
democratic framing are regulatory architecture — see "Generic transparency vs
democratic transparency" below.

Generic transparency vs democratic transparency: Generic platform transparency,
mandatory risk audits, systemic-risk reporting, and reporting obligations
confer no participatory right and are typically regulatory architecture — code
```

```
NONE. Code POLITICAL only when the provision explicitly frames the
transparency, audit, or report in terms of electoral integrity, political
advertising, or democratic discourse. Test: would the same mechanism applied to
dish-soap advertising still feel political? If no, it's not political.

Freedom of expression online: Default to CIVIL (personal liberty). Code as
POLITICAL only when the provision explicitly invokes public discourse,
elections, media pluralism, or democratic participation.

Procedural rights inside social legislation: A right to an effective remedy or
to be heard remains CIVIL even when the substantive matter is welfare-flavoured
— code by what the right secures, not the topic.

Universal service / connectivity / digital skills: SOCIAL — these extend the
means to participate fully in digital society.

Cultural heritage, research, and broadcasting access: SOCIAL — provisions that
directly extend access to cultural heritage, research / educational materials,
or broadcasting content of major social importance (copyright exceptions for
research / teaching / heritage; mass digitisation by libraries, museums,
archives; wide public access to events of major societal importance). Code
`none` when access appears only as a justifying purpose or promotion of
awareness.

Group / collective rights: Collective redress, consumer associations, worker
councils and similar collective arrangements: code as the right's dominant
flavour (welfare-logic → social; democratic participation → political).

Examples

Input: "The data subject shall have the right to obtain from the controller
without undue delay the rectification of inaccurate personal data concerning
him or her."
Output: {"rights_type": "civil"}

Input: "The provider of the high-risk AI system shall ensure that affected
persons have the right to obtain from the deployer clear and meaningful
explanations of the role of the AI system in the decision-making procedure and
of the main elements of the decision taken."
Output: {"rights_type": "civil"}

Input: "Recipients of the service shall have the right to lodge a complaint
against providers of intermediary services alleging an infringement of this
Regulation."
```

```
Output: {"rights_type": "civil"}

Input: "Providers of very large online platforms shall identify, analyse and
assess systemic risks stemming from the functioning and use of their services,
including any actual or foreseeable negative effects on civic discourse and
electoral processes."
Output: {"rights_type": "political"}

Input: "Member States shall ensure that very large online platforms publish, on
a freely accessible part of their online interface, information on the
functioning of their recommender systems used in connection with electoral
processes."
Output: {"rights_type": "political"}

Input: "Each political advertisement shall be accompanied by a transparency
notice clearly identifying the sponsor, the source of funding, and the
election, referendum or legislative process to which it relates."
Output: {"rights_type": "political"}

Input: "The Union shall promote citizens' participation in democratic life,
including through digital means, and shall safeguard media pluralism in the
digital environment."
Output: {"rights_type": "political"}

Input: "Member States shall ensure that all consumers in their territory have
access to an adequate broadband internet access service at an affordable
price."
Output: {"rights_type": "social"}

Input: "Manufacturers shall ensure that ICT products and services are designed
and produced so that they are accessible to persons with disabilities."
Output: {"rights_type": "social"}

Input: "Platform workers shall have the right to a minimum level of social
protection comparable to that afforded to employees in standard employment
relationships."
Output: {"rights_type": "social"}

Input: "This Regulation shall apply from 12 January 2027."
Output: {"rights_type": "none"}

Input: "The Commission shall adopt implementing acts setting out the technical
specifications referred to in Article 6."
Output: {"rights_type": "none"}
```

```
Input: "For the purposes of this Regulation, "online platform" means a hosting
service that, at the request of a recipient of the service, stores and
disseminates information to the public."
Output: {"rights_type": "none"}

Text to classify
Text:
```

Market-orientation rating

```
You are an expert annotator analyzing official EU policy texts.

Task

Rate how strongly the text is oriented toward MARKETS: whether it builds,
opens, expands, or maintains markets — removing barriers to services and data
flows, harmonising national rules into a larger single market, establishing
property rights, or setting the rules of open competition and a level playing
field; whether it corrects specific market failures — market power, negative
externalities, information asymmetries, undersupplied public goods; or whether
it steers markets toward chosen sectors, technologies, or capabilities through
industrial policy, procurement, or investment.

Scale

0 = no market content at all
9 = the text is entirely about making, correcting, or steering markets

Respond with ONLY a single digit from 0 to 9.
```

Rights-orientation rating

```
You are an expert annotator analyzing official EU policy texts.

Task

Rate how strongly the text is oriented toward RIGHTS: whether it protects,
extends, or operationalizes rights of natural persons — fundamental rights,
data protection and privacy, procedural rights and redress, protections framed
as individual entitlements, accessibility and inclusion, worker protections, or
democratic and participatory rights.

Scale
```

```
0 = no rights content at all
9 = the text is entirely about protecting or extending rights

Respond with ONLY a single digit from 0 to 9.
```

B3 Sample design and coding procedure

Human coding draws on a single shared sample frame (`processed-data/coded/sample.csv`): a queue of provisions from which both coders work. The frame was grown in three rounds: round v1 (400 items, 200 recitals and 200 articles) served instrument development — the iterative refinement of the coding instructions described in Section C2; round v2 (500 items) is the frozen-instruction validation sample at the core of the agreement statistics in this appendix and in Table 2 of the main text; round v3 (119 items, articles only) extends the article pool for the two article-only tasks under the same frozen instructions, bringing the number of tone-eligible coded articles to 300. Both authors coded rounds v1 and v2; round v3 was coded by one author only. Inter-coder agreement therefore rests on v2, and v3 adds items for comparing the models with the reference coder.[4]

Sampling is stratified on membership in the CEPS inventory of EU digital laws (Zenner, Marcus, and Sekut 2025): by default, half of each draw comes from provisions of CEPS-listed acts and half from the remainder of the corpus, ensuring enough positive cases for the digital-policy task at a positive rate far above the ~4% corpus base rate (a deliberate enrichment whose consequences for error-rate estimation are discussed in Section C5). In round v2, this yields 250 of 500 items from CEPS-listed acts; round v1 was additionally balanced between recitals and articles, and round v3 restricted the pool to articles before stratifying.

Both coders (the authors) coded independently in Label Studio, one project per coder, with no visibility of each other's annotations. The interface — generated from the same YAML as the LLM prompts — enforces the task gating: the article-only tasks are flagged per item type, and the rights-type panel is shown only when the item is coded as digital policy. Label Studio is the source of truth for all coding values; a management script (`scripts/manage_coding.py`) initializes the per-coder projects, pushes newly sampled items to every coder, and syncs annotations back to per-coder mirror files (merged into `processed-data/coded/human_codings.csv`), defensively clearing values for items where a task does not apply. All agreement statistics are computed from these mirrors.

[4] The shared sample frame was drawn while the pipeline was still evolving: early draws used a 30-character minimum length (the analysis corpus later settled on 50) and predate the exclusion of enforcement decisions (Section A2), and a subsequent corpus rebuild replaced a handful of source texts. As a result, 40 of the 619 coded items lie outside the final analysis corpus — most of them provisions of enforcement decisions. They remain in the agreement statistics, which assess the coding instrument rather than the corpus.

Part C — Classifier Pipeline and Validation

Following Birkenmaier, Lechner, and Wagner (2024), we assess agreement with human coders, sensitivity to prompts and model configuration, and consistency with independent reference sources.

C1 The LLM ensemble

We classify each text unit (recital or (sub-)article) with a five-model LLM ensemble, prompted with the coding instructions of Section B2. The lineup (Table 4) pairs four open-weight models with one proprietary frontier model, so that most of the pipeline runs on open models, with one closed model as an anchor. Both features follow current guidance: large-scale evaluations of LLM annotation find reliability rising with model size and recommend a floor well below the smallest model used here, and reviews of validation practice advise against measures obtainable only through closed APIs (Yang et al. 2025; Birkenmaier, Lechner, and Wagner 2024).

Table 4 The production LLM ensemble. gpt-oss-120b is a reasoning model and is run with `reasoning_effort = "low"`. The three OpenRouter-served models are pinned to the numerical precision listed: routing is constrained to providers serving those weights at that quantization, so the served model is fixed rather than left to the router's choice. For the two models reached through their developers' own APIs, serving precision is not disclosed. All calls are made through the `ellmer` R package with structured (JSON-schema-constrained) output, using the providers' batch APIs where available and synchronous calls otherwise.

Model	Developer	Weights	Access	Precision
Qwen3-235B-A22B-Instruct	Alibaba	open	Together AI → OpenRouter	FP8
Llama-3.3-70B-Instruct	Meta	open	Together AI → OpenRouter	FP8
gpt-oss-120b	OpenAI	open	Together AI → OpenRouter	MXFP4 (native)
GPT-5.2	OpenAI	proprietary	OpenAI API	not disclosed
DeepSeek-V4 Pro	DeepSeek	open	DeepSeek API	not disclosed

We assign the label receiving the most votes. If labels tie, we use GPT-5.2's choice among the tied labels. For each item, we also retain the fraction of models choosing each category and use these fractions as training targets for the encoders (Section C3). The ensemble was validated against human coders on the frozen-instruction sample (rounds v2 and v3; Section C2) and then used to label stratified corpus samples as training data for the encoders; texts shorter than 50 characters are excluded throughout (Section A2).

C2 Human validation and agreement

We validate the two classification stages that produce the paper's categorical measures — the LLM ensemble's consensus labels and the encoders distilled from them — against human coders on the frozen-instruction sample: round v2 for all four tasks, extended by round v3 for the two article-only tasks, which brings the coded article pool to 300 (Section B3). Table 5 is the appendix-detail counterpart of Table 2 in the main text and reads from the same single source (`processed-data/validation/validation_results.csv`, produced by `scripts/validate.R`): for each task it reports the eligible scope, the number of paired codings, raw agreement, Krippendorff's α, and — for the encoders — the held-out distillation-test F1.

Table 5 Validation of the classification pipeline, per task and comparison. Inter-coder compares the two human coders on their full overlap; ensemble vs. human compares the five-model consensus (production prompt) with the dominant coder for the task; encoder vs. human compares the distilled encoder with the same coder, id-disjoint. F1 is the encoder's held-out distillation-test F1 (positive-class for the binary task, weighted for the categorical tasks). Rights type has no encoder row: it applies only to the roughly 85,000 digital provisions, which the validated ensemble classifies directly at little cost, so no encoder was distilled (Section C3).

Task	Comparison	Eligible scope	n	Agreement	α	F1
Digital policy	Inter-coder	all provisions	500	0.966	0.910	—
	Ensemble vs. human		500	0.962	0.899	—
	Encoder vs. human		500	0.962	0.902	0.954
Regulatory tone	Inter-coder	articles	181	0.945	0.903	—
	Ensemble vs. human		300	0.930	0.883	—
	Encoder vs. human		292	0.918	0.862	0.936
Regulated actor	Inter-coder	tone-bearing articles	119	0.924	0.889	—
	Ensemble vs. human		211	0.877	0.826	—
	Encoder vs. human		211	0.858	0.798	0.849
Rights type	Inter-coder	digital provisions	117	0.940	0.878	—
	Ensemble vs. human		123	0.927	0.844	—

Sample sizes vary because tasks have different eligibility rules and coders did not code every eligible item. Human–human comparisons use items coded by both authors; model–human comparisons use the available codings from the reference coder for each task. The encoder rows are id-disjoint: any validation item that entered the encoder's training pool (via the ensemble-labeled training consensus) is excluded from encoder-vs-human scoring, so the encoder is only scored on items it was not trained

on; the `source` column of the results file records how many items this removes.[5] Encoder predictions come from a fresh forward pass on the stored validation texts (`scripts/predict_validation.py`), not from training-time logs.

C3 Encoder distillation

To ensure reproducibility independent of proprietary APIs, we distill the LLM ensemble's consensus into an open-source transformer. We fine-tune ModernBERT-large (≈400M parameters, 8,192-token context) on a sample of corpus items labeled by the five-model ensemble. Training proceeds as a learning curve: we classify 30K-item chunks at a time, retrain the encoder after each chunk, and stop when adding more training data no longer clearly improves the encoder's agreement with the human codings. Two design choices — where to stop adding data and, for the actor encoder, the class ratio — were therefore judged against the human-coded validation set. The check that this set played no part in is the natural-prevalence audit of Sections C5 and C6, drawn after all choices were fixed.

To address class imbalance (a 3.0% positive rate in the training-time corpus snapshot), we downsample the negative class to a 4:1 ratio within each pool, splitting 80/10/10 into train / validation / test. Input texts are tokenized to a maximum of 1,024 tokens, which covered 99.86% of that snapshot without truncation.[6]

Table 6 Fine-tuning hyperparameters of the production encoder. Learning rate and weight decay were selected by a grid search reported in Table 8. Training was performed on Apple M4 hardware (128 GB unified memory) using the MPS backend.

Parameter	Value
Epochs	5 (best epoch by validation F1 retained)
Effective batch size	32 (micro-batch 4, gradient accumulation 8)
Learning rate	2×10^{-5} (linear decay)
Weight decay	0.1
Maximum sequence length	1,024 tokens

We trialed both label-supervision strategies. Hard labels use the ensemble's majority-tiebreak consensus as a discrete target; soft labels use the per-class vote fraction as a continuous KL-divergence target.

[5] Excluding by id is not quite the same as excluding by text. EU legislation repeats standard formulas — "This Regulation shall enter into force on 1 April 1990", "This Decision is addressed to the Kingdom of the Netherlands" — so a validation item can have a word-for-word twin in the training pool under a different id, and the encoder has then effectively seen its text. Dropping every validation item whose exact text occurs anywhere in the training pool removes 39 digital-policy items, 20 tone items, and 4 actor items, and moves α from 0.902 to 0.899, 0.862 to 0.855, and 0.798 to 0.794. The effect is small because these items are formulaic and easy; for the digital-policy task, none of them was coded as digital.

[6] Of the 2,115,236 items in the May 2026 training-time corpus snapshot — distinct from the current analysis corpus described in Section A2 — approximately 3,000 (0.14%) exceed 1,024 tokens, 1,100 exceed 2,048 tokens, and 420 exceed 4,096 tokens; the median item length is 57 tokens. Distribution estimated from a 50,000-item random sample tokenized with the ModernBERT-large tokenizer. ModernBERT's RoPE positional embeddings remain valid up to 8,192 tokens, so the encoder still produces predictions on truncated items via the first 1,024 tokens of input.

Hard targets collapse a unanimous 5–0 vote and a contested 3–2 vote to the same label, whereas soft targets retain the vote fraction and train the encoder to reproduce it. This follows the central intuition of knowledge distillation: a teacher's full predictive distribution carries information about how it generalizes that its winning class discards, and higher-entropy targets can provide a richer, lower-variance training signal (Hinton, Vinyals, and Dean 2015). Relatedly, training on full distributions of human judgments rather than one-hot labels has improved generalization under distribution shift and adversarial robustness (Peterson et al. 2019). Our targets are coarser than either — five voters admit only six distinct vote fractions — and we do not assume that ensemble disagreement represents valid uncertainty by definition; whether retaining it helps is an empirical question, which the hard-versus-soft runs below answer. We also probed the calibration trade-off of the class-balance ratio with a 10:1 negative:positive run at the 88K pool.

Table 7 Learning-curve trajectory for the fine-tuned ModernBERT-large classifier. Each row is a separate training run on a pool grown by 30K corpus chunks (88K reports the two-seed average). Human α is Krippendorff's α against the n=500 human-coded validation set; Test F1 is computed on the 10% held-out split of the (balanced) training pool. The selected production configuration is bolded.

Pool	Labels	Neg:pos	Train n	Test F1	Human α	Time (min)
30K	hard	4:1	3596	0.941	0.876	31
30K	soft	4:1	3596	0.939	0.881	31
60K	hard	4:1	7060	0.935	0.872	52
60K	soft	4:1	7060	0.916	0.860	57
88K	hard	4:1	10416	0.914	0.893	87
88K	soft	4:1	10416	0.919	0.906	92
88K	hard	10:1	22915	0.905	0.867	164
88K	soft	10:1	22915	0.909	0.900	160
120K	hard	4:1	13856	0.940	0.902	131
120K	**soft**	4:1	13856	0.940	0.916	130

Two findings emerge from this trajectory. First, soft-label distillation consistently outperforms hard-label training once the training pool exceeds ~2,500 positives (88K pool onward). Second, increasing the negative-class ratio from 4:1 to 10:1 is associated with lower α on the positive-enriched validation set, consistent with a larger imbalance biasing the model toward the negative class. We therefore retain the 4:1 balance and adopt soft-label distillation on the 120K consensus pool as the production data configuration.

Holding data quantity and label supervision fixed at these values, we tune learning rate and weight decay on a 3×2 grid. Each configuration is trained for five epochs (the trainer retains the best epoch by validation F1); selection is by F1 on the held-out 10% test split of the balanced training pool. The human-coded validation set was not used to choose among the configurations in this sweep.

Table 8 Hyperparameter sweep results. Each row is a separate fine-tuning run on the 120K consensus pool with soft-label distillation and 4:1 class balancing. Test F1 is computed on the held-out 10% test split (1,732 items); human α is Krippendorff's α against the n=500 human-coded validation set, which was not used to choose among the configurations in this sweep. The production configuration (run 3) is bolded.

Run	LR	WD	Test F1	Test acc.	Human α	Human agr.	Time (min)
3	**2e-05**	**0.10**	**0.942**	**0.978**	**0.917**	**0.968**	**129**
5	5e-05	0.10	0.941	0.977	0.901	0.962	120
2	2e-05	0.01	0.937	0.975	0.923	0.970	121
1	1e-05	0.10	0.936	0.975	0.895	0.960	—
0	1e-05	0.01	0.933	0.974	0.885	0.956	—
4	5e-05	0.01	0.923	0.970	0.885	0.956	118

The learning rate of 2×10^{-5} gives the highest test F1 and human α at both weight-decay values. Weight decay of 0.1 gives higher test F1 at all three learning rates, but does not consistently give higher human α.

The selected configuration (LR = 2×10^{-5}, WD = 0.1) achieved F1 = 0.942 on the distillation test set and Krippendorff's α = 0.917 against the human-coded validation sample at model-selection time. The deployed production encoder — retrained under this configuration after the validation pipeline was hardened — reaches a held-out test F1 of 0.954 from its saved metadata and, in the fresh id-disjoint validation pass, α = 0.902 against the human benchmark (n = 500), close to the inter-coder agreement on the same task (inter-coder α = 0.910). The model is released on Hugging Face.

The tone and actor encoders

The regulatory-tone and regulated-actor encoders follow the same recipe — ModernBERT-large, soft-label distillation of the five-model consensus, 1,024-token inputs, selection by held-out test F1 with human α reserved as the external check — with two task-specific adjustments. The tone encoder uses the digital-policy encoder's configuration (learning rate 2e-05, weight decay 0.1, negative-class ratio 4:1). Test F1 varied little across the combinations in Table 9, so we retained that configuration. It reaches test F1 = 0.936 against the ensemble consensus and, id-disjoint, α = 0.862 against the human benchmark (n = 292).

The actor encoder required the opposite class-balance correction to the one reported above for the binary task: because its bottleneck class (`individuals`) is rare (~1% of tone-bearing articles), a *low* negative ratio caused rare-class over-prediction on the natural human-coded set, and raising the ratio to 12:1 improved agreement with the human codings at every step. It reaches test F1 = 0.849 and, id-disjoint, α = 0.798 (n = 211). The preferred class ratio therefore differs between tasks.

Table 9 Hyperparameter sweep for the regulatory-tone encoder (learning rate × weight decay, five epochs each, soft labels, 4:1 balancing). Test F1 is against the ensemble consensus on the held-out split; human α against the human-coded validation set is reported as an external check, not used for selection. The landscape is flat, so the production configuration of the digital-policy encoder was retained.

Run	LR	WD	Test F1	Human α	Time (min)
0	1e-05	0.01	0.939	0.841	181
2	2e-05	0.01	0.938	0.856	167
1	1e-05	0.10	0.938	0.857	801
5	5e-05	0.10	0.938	0.862	173
3	2e-05	0.10	0.936	0.867	162
4	5e-05	0.01	0.935	0.862	171

We apply the ensemble directly to digital provisions for the rights-type task because the subset is small enough to classify without training a separate encoder. We therefore save the ensemble consensus labels directly as the task's predictions (`processed-data/predictions/rights_type_llm_consensus.parquet`). The ensemble's agreement with the human benchmark on this task is reported in Table 5 there is no encoder row to report.

C4 Prompt stability

We test whether classifications change when we repeat a prompt or reword its definition without changing its meaning (Barrie, Palaiologou, and Törnberg 2024). Stability is necessary but not sufficient for validity, since a model can be reliably wrong; instability, however, would mean that our classifications depend on phrasing, so we check for it before relying on them.

We test the digital-policy task, which determines which provisions enter the subsequent digital-policy analyses. Five near-equivalent reformulations of the definition sentence join the production wording (Table 10). Everything else in the instruction is held fixed — the inclusion and exclusion rules, the ten worked examples — isolating sensitivity to the formulation of the construct sentence while holding the rest of the instrument constant. Each of the six instructions classifies the frozen 500-item validation sample three times on all five production models, 45,000 classifications in all.[7]

[7]The four open models run synchronously; GPT-5.2 runs through OpenAI's batch API. Runs are at the production temperature of zero rather than at the sampling temperatures Barrie, Palaiologou, and Törnberg (2024) use, because the object here is the pipeline as deployed; the intra-prompt margin therefore measures serving-side nondeterminism, which is the only run-to-run variation our production settings admit.

Table 10 The six statements of the digital-policy construct. The first is the production wording; the other five are near-equivalent reformulations. Only this sentence varies between the instructions; all rules, edge cases, and examples are held fixed.

Variant	Definition sentence
production	A text is digital policy if it contains deliberate public interventions that shape the course and character of digitalization by influencing the development, deployment, governance, and use of digital technologies.
constitutes	A text counts as digital policy if it constitutes a deliberate public intervention that shapes the course and character of digitalization by influencing the development, deployment, governance, or use of digital technologies.
expresses	A text is digital policy when it expresses a deliberate public intervention intended to shape digitalization through the development, deployment, governance, or use of digital technologies.
policy action	Classify a text as digital policy when its substantive policy action deliberately shapes digitalization by affecting how digital technologies are developed, deployed, governed, or used.
public measure	Digital policy comprises texts that set out deliberate public measures shaping digitalization, including the development, deployment, governance, and use of digital technologies.
substantive rule	A text qualifies as digital policy if its substantive rule deliberately influences digitalization by shaping the development, deployment, governance, or use of digital technologies.

Agreement is high both within and between prompts (Table 11). *Within* prompts — identical instruction, three runs — α never falls below 0.946 for any model–instruction pair. *Between* prompts — the majority label under each instruction, compared across the six formulations — α runs from 0.946 to 0.982. In item terms: swapping the production definition for a reformulation moves the majority classification of between 1 and 10 of the 500 items, depending on the model and the formulation (median 6). Across these near-equivalent formulations, classifications are therefore only weakly sensitive to how the definition is phrased.

Table 11 Prompt stability for the digital-policy task. Within-prompt α is the range over the six instructions of the agreement between three repeated runs of the same instruction; between-prompt α compares the majority label under each of the six instructions. Both are Krippendorff's α on the frozen 500-item validation sample.

Model	Within-prompt α (range)	Between-prompt α
Qwen3-235B	0.979–0.990	0.973
Llama-3.3-70B	0.974–0.993	0.973
gpt-oss-120b	0.946–0.964	0.946
GPT-5.2	0.964–0.982	0.975
DeepSeek-V4 Pro	0.992–1.000	0.982

Two limits are worth stating. The exercise covers the digital-policy gate only; the tone, actor, and rights instructions have not been perturbed in this way. And it varies the definition sentence, not the worked examples, which are the other component of the instruction a classification could depend on.

C5 Error rates at natural prevalence

Our initial validation sample deliberately included extra digital provisions so that we could assess how well the classifier recognizes them (Section B3). In the full corpus, however, most provisions are non-digital. Even if the classifier mistakenly labels only a small fraction of these as digital, those errors could noticeably inflate the estimated digital share. We therefore conducted a second audit, drawing provisions without deliberately increasing the digital share.

Design. We drew 500 provisions from the paper's analysis universe (Section A2), restricted to provisions carrying an encoder prediction and an adoption date. The sample included 100 randomly chosen provisions from each of five decade groups. We sampled the same number of provisions from each decade so that we could compare classifier performance over time (Section C6). To calculate error rates for the corpus as a whole, we give each decade weight according to how many provisions it contains. For example, a decade containing twice as many provisions receives twice the weight. One of the authors coded every item blind, under the production instructions of Section B2, in a Label Studio project reduced to the digital-policy task. The sample was drawn shortly before the enforcement-decision filter was redesigned (Section E8), and 29 of its items sit in acts the final filter excludes; we report on the 471 items inside the final universe (results over all 500 are similar). Finally, a leakage check: 99 of the audited texts also occur in the ensemble-labeled pool the encoder was trained on. None of the disagreements falls among them, and agreement restricted to the 372 held-out texts is 98.1% — the audit shows no sign of being flattered by training-set leakage.

Results. Coder and encoder agree on 98.5% of items (Krippendorff's $\alpha = 0.86$). The coder marks 29 provisions digital (6.2%), the encoder 24 (5.1%), overlapping on 23: precision 0.96, recall 0.79, F1 0.87 for the digital class. The failure mode this section was designed to catch — false positives swamping a small positive class at 20-to-1 odds — does not materialize: 1 of the 442 provisions the coder marks

non-digital is classified digital (specificity 0.998). Reweighting the decade strata to their corpus shares gives sensitivity 0.81 and specificity 0.997, and with them a Rogan–Gladen misclassification-corrected digital share of 5.1%, against the observed 4.3%: the encoder is, if anything, mildly conservative — its errors run toward missing digital provisions, not inventing them — so misclassification, if anything, understates rather than inflates the paper's headline share.

C6 Temporal stability of classifier error

The trend claims in the main text require classification error to be stable over time: a classifier that under-detects digital policy in older vocabulary ("telematics", "information society services") — or over-detects it in machine-translated early texts (Section A3) — would tilt the trend mechanically. The coding instructions address old terminology explicitly, and Section E1 notes that stable error transforms the series affinely (level and amplitude, not turning points); whether the error *is* stable is an empirical question, which the decade stratification of the Section C5 audit was designed to answer.

Table 12 splits that audit by decade. Agreement is 1.000 / 0.990 / 0.979 / 0.988 / 0.968 from the oldest stratum to the newest: high throughout, with no deterioration toward the past — the hardest stratum is the most recent one, where digital-adjacent negatives concentrate. Two features matter most for the trend. First, across the 196 pre-2000 items the encoder produces zero false positives, corroborating at natural prevalence what the enriched-sample split below shows: because observed shares at sub-1% prevalence are dominated by the false-positive rate, this speaks against the most damaging form of drift, an early rise produced by the classifier over-flagging old law. Second, the coder's and the encoder's decade profiles of the digital share track each other closely across the full half-century (Table 12, middle columns), which is consistent with stable error at the resolution this sample affords.

Table 12 Decade-stratified audit at natural prevalence (Section C5 sample, in-universe items): hand codings vs. the deployed encoder.

Decade	Items	Digital (coder)	Digital (encoder)	False positives	Agreement
1979–1989	97	0	0	0	1.000
1990–1999	99	3	2	0	0.990
2000–2009	96	5	3	0	0.979
2010–2019	85	10	11	1	0.988
2020–2026	94	11	8	0	0.968

Corroborating evidence from the enriched validation sample. The frozen validation sample, though digital-enriched, speaks to the same question. Splitting its 500 human-coded provisions by

the adoption era of their acts, encoder–human agreement is 1.000 / 0.976 / 0.956 / 0.936 from the oldest era to the newest — and among the 92 pre-2000 provisions the human coder classed as non-digital, the encoder produces not a single false positive. Since observed digital shares at low prevalence are dominated by the false-positive rate, this too speaks against the early rise being an artifact of the classifier over-flagging old law. Two caveats keep this preliminary. The sample is digital-enriched by design (Section B3), so the mild decline of era-specific agreement toward the present partly reflects composition — recent strata contain far more hard, digital-adjacent negatives. Neither sample lets us reliably assess how often the classifier misses early digital provisions: the enriched validation sample contains none identified by a human before 2000, and the second audit contains only 3. Missing early digital provisions would make the early digital share look smaller and the subsequent rise steeper. Assessing this would require a sample deliberately including more early digital provisions, for example from known early digital acts.

C7 The market and rights ratings

The main text reports a graded measure of how far a provision leans toward market-making or rights-protection. This measure is calculated differently from the four classification tasks.

Two separate ratings. We ask how strongly each provision is oriented toward markets (0–9) and toward rights (0–9). A single market-versus-rights scale places a provision that does both — such as a data-portability rule creating a market and an individual entitlement — in the same middle position as a provision that does neither. The market rating and the rights rating distinguish these cases. Their difference, `rights − market`, gives the position. We obtain the two ratings in separate calls, so neither depends on the other rating. In the main text a provision leans toward rights when its position is above zero (29.9% of digital provisions); requiring a margin of half a point or a full point gives 25.4% and 21.5%.

Scoring from token probabilities. A model can output "3" both when it is uncertain between 3 and 4 and when it confidently assigns 3. We use the *expected value* across the probabilities assigned to digits at the score position, the last generated token. We take the top-15 log-probabilities and renormalize them over the digits 0–9 (Licht et al. 2025). This gives a continuous rating using the estimator that Licht et al. find outperforms both direct scoring and pairwise aggregation.

Three models. We use three open-weight models (Table 13). Whether an API returns token probabilities depends on the provider serving the model. We therefore tested each model–provider pair and fixed the provider for the production run. We excluded gpt-oss-120b because it assigned 97–99% of its probability to one digit, adding nothing beyond the output digit.

Table 13 The three models used for scoring. Providers were fixed for the production run. Each model–provider pair was tested with live calls because log-probability support depends on the provider serving the model.

Model	Provider (pinned)	Mean score
Qwen3-235B	Parasail / Alibaba / AtlasCloud (FP8)	2.97
Gemma-4-31B-it	CoreWeave	3.03
GLM-5.2	StreamLake	3.72

Coverage and agreement. All 84,819 digital provisions have market and rights ratings from all three models. Agreement between the three models is substantial, though their ratings differ: pairwise correlations range from 0.84 to 0.86 for the market rating and from 0.88 to 0.90 for the rights rating. We average across the three models to obtain a market rating and a rights rating for each provision.

Consistency with the rights typology. The ratings and the rights typology of Task 4 were produced independently. Provisions classified as invoking a right should have higher rights ratings, and they do, by a wide margin (Table 14). Civil-rights provisions have positions 5.6 points higher than provisions classified as invoking no right. The four categories follow the expected order: civil above political and social, and all three above none. Both measures use the same texts and the same family of models. Their agreement therefore shows consistency; accuracy against human judgment is assessed through the pairwise validation below.

Table 14 Mean market and rights ratings by the independently coded rights typology (Task 4). Position is the difference rights − market; positive values lean toward rights.

Rights type	Provisions	Market	Rights	Position
Civil	10,046	3.07	6.71	+3.64
Political	263	3.22	5.05	+1.83
Social	1,472	4.39	5.38	+0.98
None	73,038	3.96	1.99	−1.97

Comparison sample of non-digital provisions. To put the digital ratings in context, the main text compares them with the rest of EU law. We drew 100,000 provisions at random from those the encoder labels as not digital, 20,000 from each of five periods (1979–1989, the 1990s, the 2000s, the 2010s, and 2020–2026). Sampling equal numbers per period keeps the early decades, which contain far fewer provisions, from being swamped by recent ones. The sample was rated with the same two instructions, the same three models, and the same scoring procedure as the digital provisions. Every sampled provision has both ratings from all three models. One detail differs from the digital run. The sample was rated in three batches, and after the first 20,000 provisions we allowed two further FP8 providers for Qwen3-235B (GMI Cloud and StreamLake), because rate limits at the original three

slowed the run; model, precision, and instructions did not change. Because the sample is drawn in equal numbers per period, its overall averages give each period the same weight, whereas the digital averages cover all digital provisions and therefore lean toward recent years. The period-by-period comparison in the main text's orientation figure is not affected by this difference: digital provisions have the higher rights rating in every period.

Human validation: pairwise comparison. A graded judgment cannot be validated in the same way as a categorical one: asking a human to assign a number on a 0–9 scale measures how they interpret the scale as much as the orientation being measured. We therefore validate the ordering. A coder sees two provisions side by side and chooses which leans further toward rights relative to markets, with "too close to call" always available. We drew 300 pairs from the scored corpus in four bands defined by the absolute difference in position, from small differences (0.25–1 scale points) to large ones (above 4.5), with 75 pairs per band. Text lengths were limited so both provisions fitted on one screen. Each act contributed one provision, and the coder saw no scores or identifiers. Including both small and large differences tests how well the measure distinguishes similar provisions; large differences alone would only show that clearly different provisions are ordered correctly. One of the authors coded all pairs in a dedicated Label Studio project.[8]

Table 15 reports the results. The coder recorded a tie on 64 pairs (21.3%), mostly where the positions were close. On pairs without a tie, the coder agreed with the ordering in 79.7% of cases overall. Agreement increased with the size of the gap, as expected: near chance (53.8%) in the narrowest band, where positions are nearly identical; around 79% in the middle bands; and 98.6% for differences above 4.5 points. For the 95 pairs with positions of opposite signs — one leaning toward rights, the other toward markets — the coder agreed with the ordering in 95.6% of pairs without a tie (5 ties). Two limits follow. Pairs closer than one scale point were not reliably distinguished, so small differences in position should not be interpreted. And comparing two provisions validates their ordering, not where the midpoint of the scale lies: a provision counts as rights-leaning when its rights rating exceeds its market rating, and no coder judged that boundary directly. The rights-leaning shares in the main text are therefore model-based descriptive quantities, which is why they are reported together with the stricter cut-offs.

[8] One pair in the widest band was corrected after a post-coding review found a reversed selection; the pair queue and judgments are preserved in `processed-data/coded/`.

Table 15 Pairwise human validation of the ordering by position. Bands group pairs by their absolute difference in position; concordance is the share of pairs without a tie where the coder chooses the provision with the position further toward rights.

Contrast band	Pairs	Ties	Concordance
0.25–1.0	75	23	0.538
1.0–2.5	75	23	0.788
2.5–4.5	75	16	0.797
4.5+	75	2	0.986

Part D — External Validity

D1 CEPS reference list

As an external check, we compare our classifications against an independently compiled inventory of EU digital laws maintained by the Centre for European Policy Studies (CEPS) (Zenner, Marcus, and Sekut 2025). The inventory lists 192 entries, of which 165 carry CELEX identifiers and 163 match adopted acts in our canonical corpus. The two identified entries that do not match are not outside the corpus window: the Quantum Act entry (CELEX 52025DC0363) is a Commission communication, an excluded instrument type, while the Competitiveness Fund entry (CELEX 52025PC0571) is a Commission proposal rather than an adopted act. All 163 matched acts (100%) have at least one provision classified as digital by our classifier — the recall against the expert list is complete. Nor is the signal marginal: the median CEPS act has 35% of its provisions classified as digital (mean 42%), with the lower shares belonging to instruments whose digital provisions sit inside broader regulatory packages.

Table 16 Digital-policy content in the matched CEPS inventory and the full canonical act universe.

Act set	Acts	Median digital share	Mean digital share	Majority digital
All canonical acts	146,234	0.0%	1.5%	1.1%
Matched CEPS acts	163	34.9%	41.8%	42.3%

D2 Landmark digital laws

Table 17 shows the share of provisions classified as digital in 10 landmark EU digital laws. All are classified as majority digital — from 55% (Chips Act) to 90% (DGA) — with the non-digital provisions consisting largely of procedural and institutional boilerplate (enforcement powers, delegated acts, comitology, entry-into-force clauses) that is not specific to digital policy.

Table 17 Share of provisions classified as digital policy in landmark EU digital legislation. Provision counts reflect the sub-article corpus (recitals and numbered article paragraphs).

Act	Year	Provisions	% Digital
DGA	2022	206	90.3%
ePrivacy Directive	2002	103	89.3%
DSA	2022	519	82.3%
Data Act	2023	354	81.4%
e-Commerce Directive	2000	127	76.4%
AI Act	2024	699	75.7%
Data Protection Directive	1995	164	75.0%
GDPR	2016	560	69.8%
DMA	2022	348	67.8%
Chips Act	2023	256	55.5%

The lowest shares belong to the Chips Act and the DMA: the former because much of its content is industrial-policy machinery (funding instruments and the governance of the Chips Joint Undertaking) rather than the regulation of digital technology itself; the latter reflecting its extensive enforcement architecture (investigation procedures, fines, limitation periods, judicial review) that mirrors generic EU competition law rather than digital-specific provisions.

Table 18 Four reproducibly sampled provisions from the landmark acts: two classified as digital policy and two as not digital policy. Excerpts longer than 450 characters are cut.

Class	Act	Type	Excerpt
Not digital	AI Act	Article	The Commission is empowered to adopt delegated acts in accordance with Article 97 in order to amend Annex V by updating the content of the EU declaration of conformity set out in that Annex, in order to introduce elements that become necessary in light of technical progress.
Not digital	Data Act	Article	When a Member State considers that a common specification does not entirely satisfy the essential requirements laid down in paragraph 1, it shall inform the Commission thereof by submitting a detailed explanation. The Commission shall assess that detailed explanation and may, if appropriate, amend the implementing act establishing the common specification in question.
Digital	DMA	Article	The undertaking providing core platform services may present, with its notification, sufficiently substantiated arguments to demonstrate that, exceptionally, although it meets all the thresholds in paragraph 2, due to the circumstances in which the relevant core platform service operates, it does not satisfy the requirements listed in paragraph 1. Where the Commission considers that the arguments submitted pursuant to the first subparagraph by…
Digital	AI Act	Article	The AI Office may invite all providers of general-purpose AI models to adhere to the codes of practice. For providers of general-purpose AI models not presenting systemic risks this adherence may be limited to the obligations provided for in Article 53, unless they declare explicitly their interest to join the full code.

D3 EuroVoc subject indexing

As a further external check, we validate the classifier against **EuroVoc**, the European Union's official multilingual subject thesaurus. Acts in the EU's document repository (Cellar) are indexed by EuroVoc descriptors that summarise their subject matter. Because EuroVoc indexing is produced independently of our pipeline — it was never used to train, prompt, or tune the classifier — it constitutes a genuine external reference, complementing the curated CEPS inventory of Section D1: where CEPS is an expert-curated list of some two hundred digital laws, EuroVoc covers the whole corpus but tags at the level of the act and by topical keyword. We curated 34 EuroVoc descriptors covering digital-policy topics and compare the classifier with these tags in both directions.

For each act we compute its digital share: the fraction of its provisions (recitals and articles) that the encoder classifies as digital policy. An act "carries a digital descriptor" if at least one of its EuroVoc tags is in the curated set. Of the 146,234 acts with classifier predictions, 99,776 (68.2%) also carry EuroVoc indexing; the remainder are concentrated before 2000 (see the coverage caveat below).

Forward direction: do classifier-digital acts carry EuroVoc digital tags?

Among indexed acts (unindexed acts are missing data, not negatives), the classifier's continuous digital share predicts the presence of a curated digital descriptor with an area under the receiver-operating-characteristic curve (AUC) of 0.951. The statistic has a direct interpretation as a ranking: draw one EuroVoc-tagged act and one untagged act at random, and 95% of the time the classifier's digital share ranks the tagged act higher, counting ties as half (50% would be a coin flip). No single descriptor drives the result: a jackknife over the curated set — dropping each of the 34 descriptors in turn and recomputing — leaves the AUC between 0.947 and 0.957. Binning makes the same alignment visible (Table 19): acts the classifier rates as non-digital almost never carry a digital tag (0.1%), while acts it rates as at least half digital carry one most of the time (58.4%).

Table 19 Forward validation, indexed acts only (acts without EuroVoc indexing are treated as missing, not as untagged). Acts are binned by the classifier's digital share; each row reports the share of acts in that bin carrying at least one curated digital EuroVoc descriptor.

Classifier digital share	Acts (n)	% with digital EuroVoc tag
0% (no digital provisions)	90,143	0.1
Under 50% (partly digital)	8,069	8.1
50% or more (majority digital)	1,564	58.4

How digital are acts with digital EuroVoc tags?

We invert the question and ask how the classifier rates the 1,698 acts that EuroVoc tags as digital *and* that fall within our prediction set (Table 20). The classifier rates these acts as substantially digital: the median digital share is 50%, 92.5% have at least one provision flagged digital, 80.6% reach a 20% share, and 53.8% reach a 50% share. Only 7.5% receive a zero share — which we examine next.

Table 20 Reverse validation. Classifier digital-share statistics for the acts that EuroVoc tags as digital and that fall within the prediction set.

Statistic	Value
n acts	1,698
Median digital share	50%
≥ 1 digital provision	92.5%
Share ≥ 20%	80.6%
Share ≥ 50%	53.8%
Exactly 0%	7.5%

Where the two instruments disagree

The 7.5% of EuroVoc digital-tag carriers that the classifier rates at a zero digital share mostly reflect EuroVoc's broader document-indexing purpose rather than classifier error: acts where a digital descriptor records a topical association rather than substantive digital-policy content. A manual reading also finds a small number of plausible classifier misses among them — an e-commerce import instrument, and acts whose digital content sits in annexes that our corpus excludes — so the zero-share cell should not be read as pure over-tagging. These acts are not an artefact of old or poorly-digitised texts — their median year is around 2016. Two patterns dominate. First, Common Foreign and Security Policy (CFSP) sanctions instruments tagged "telecommunications" or "data protection," where the digital reference is incidental: restrictions on telecom-equipment exports to a sanctioned regime, or personal-data processing in the administration of an asset freeze. Second, customs "classification of certain goods in the Combined Nomenclature" regulations, where telecommunications equipment appears as a traded good in the tariff nomenclature rather than as an object of regulation. Thinner additional patterns include VAT directives tagged "electronic commerce" and financial regulations tagged "electronic money."

The classifier recovers the major telecommunications instruments. The major instruments of EU electronic-communications regulation are all correctly rated as predominantly or substantially digital (Table 21). The European Electronic Communications Code, the ePrivacy Directive, the 2022 Roaming Regulation, and the Open Internet (net-neutrality) Regulation all exceed a 69% digital share; the 2002 Framework Directive and the BEREC Regulation sit at around half and one-third, respectively, their non-digital remainder consisting of institutional and comitology provisions. The line the classifier draws is not between policy fields but between provisions: it separates telecommunications *as the object a provision regulates* from telecommunications *as an item a provision happens to mention* — a line in a tariff nomenclature, one entry in a list of controlled goods. The distinction is not that sanctions or trade instruments are disqualified as digital policy. Where a measure is aimed at digital activity, our rules count it and the classifier follows: the CFSP regime on cyber-attacks (Council Decision (CFSP) 2019/797) is rated 47.6% digital, against 2.6% for the general Russia sanctions regulation (Regulation (EU) No 833/2014), where digital goods appear among many controlled items without the provision

turning on their digital character. Act-level subject indexing, which tags whole acts, cannot draw that line at all.

Table 21 Telecommunications policy is caught. Classifier digital share for landmark EU electronic-communications instruments — all EuroVoc-tagged digital and all rated substantially digital, in contrast to the over-tagged sanctions and customs acts discussed in the text.

Act	CELEX	% Digital
European Electronic Communications Code	32018L1972	73
ePrivacy Directive	32002L0058	89
Roaming Regulation (2022)	32022R0612	87
Open Internet Regulation (net neutrality)	32015R2120	85
Framework Directive on e-communications (2002)	32002L0021	53
BEREC Regulation	32018R1971	35

The two instruments thus converge in both directions. Where they diverge, the divergence largely reflects their different purposes: EuroVoc is a controlled document-indexing thesaurus built for retrieval — it tags whole acts by subject association — while our classifier measures digital-policy content at the level of individual provisions. EuroVoc tags cannot replace provision-level classification: they include related subjects, are sparse before 2000, and do not locate digital content within an act.

Coverage caveat. EuroVoc indexing in Cellar is sparse before 2000 — a genuine indexing gap concentrated in routine 1980s agricultural regulations — so the validation set is effectively post-2000; coverage is essentially complete thereafter.

Part E — Robustness of the Findings

This part collects robustness analyses for the main-text findings.

E1 Uncertainty of the digital-share trend

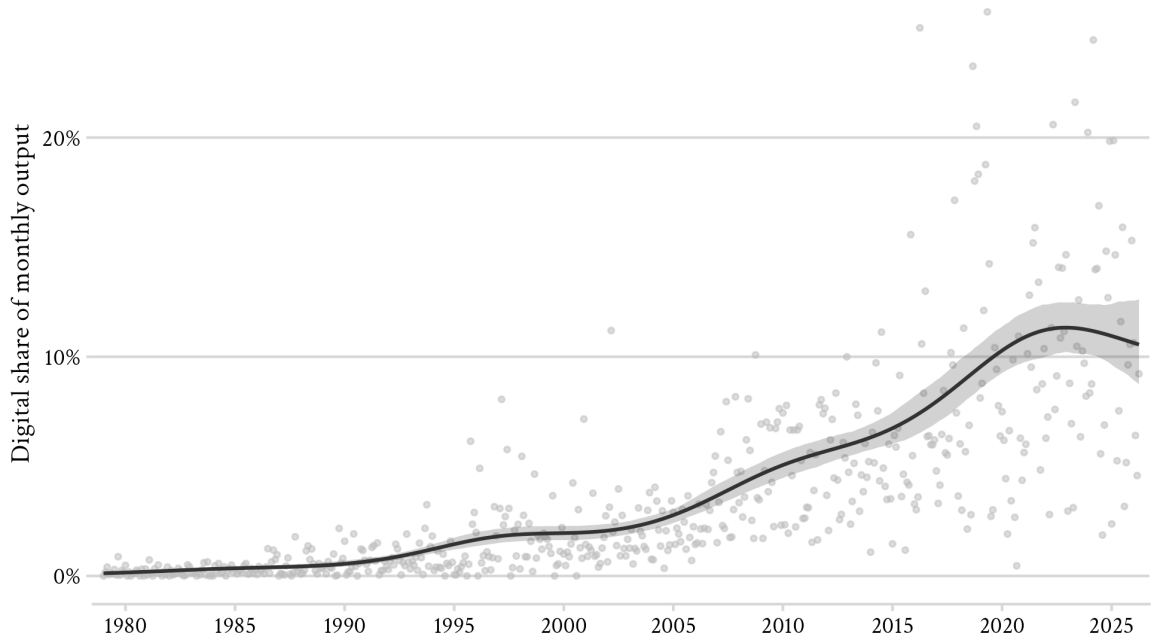


Figure 1 **The digital-share trend with a 95% act-level cluster-bootstrap band.** The corpus-wide digital-share trend of Figure 2 in the main text, shown here with its underlying monthly observations and the pointwise 95% interval. The band is computed by resampling acts (not provisions) with replacement, re-aggregating to monthly counts, and refitting the GAM trend on each replicate.

The trend line in Figure 2 of the main text is reported without its underlying monthly observations; Figure 1 adds them and makes explicit where uncertainty enters a census. Our corpus is a census of adopted legislation, not a sample: the share of provisions adopted in a given month that are digital is a population quantity, and intervals around the monthly points would have no sampling interpretation. Uncertainty enters only with the trend line, which makes a claim of a different kind — about the underlying intensity of digital lawmaking, of which the actually adopted acts are one realization. The band quantifies this superpopulation uncertainty (Abadie et al. 2020): how much the fitted trend would vary had the legislative process produced a somewhat different draw of acts.

Provisions arrive in groups within legal acts: the median month draws 31% of its digital provisions from a single act (95th percentile: 72%)[9], so the effective number of independent observations is closer to the number of acts than the number of provisions. We therefore resample whole acts: we fit a binomial GAM to monthly counts (basis dimension k = 10) and take pointwise 2.5th and 97.5th percentiles across 500 bootstrap replicates that draw acts with replacement.

The band is read accordingly. The band widens when digital provisions are concentrated in a few large acts and near the end of the series. Features that exceed its width, such as the step around 2018, are unlikely to be artifacts of which acts happened to be adopted when; the precise level of the post-2020 plateau, by contrast, remains genuinely uncertain. Classifier error is a separate source of uncertainty,

[9]Computed over months with at least 20 digital provisions.

assessed in Sections C2 and C3: stable misclassification transforms the series affinely — shifting its level and compressing its amplitude — while preserving the location of its turning points. A third source — whether a *different but equally defensible* encoder configuration would have produced a different series — is assessed in Section E5.

E2 Citation-network robustness

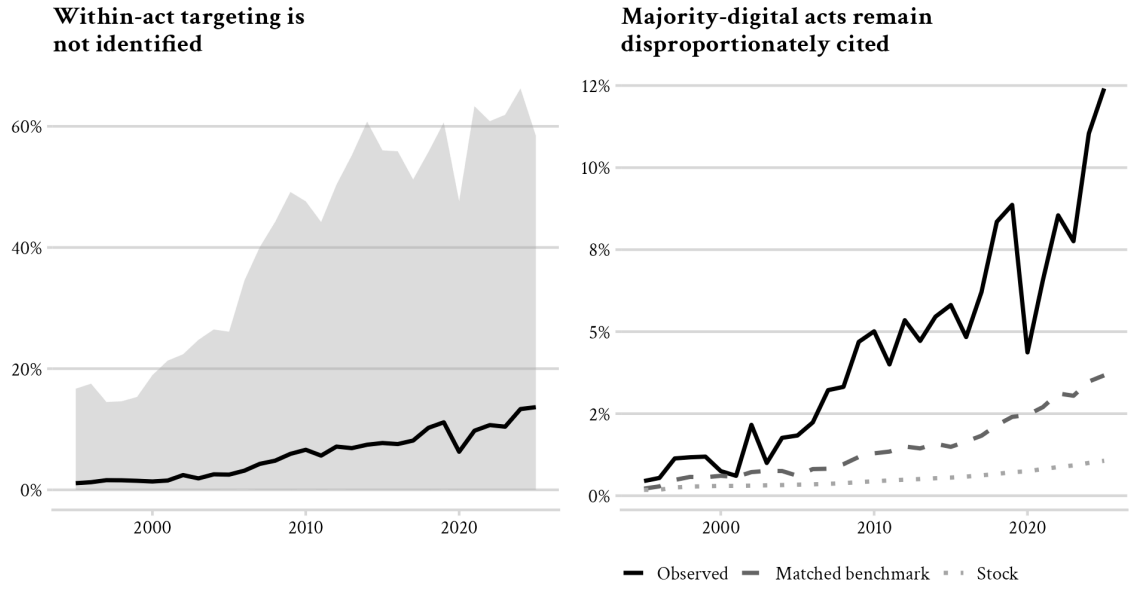


Figure 2 Robustness of the citation-network results, 1995–2025. The left panel gives the continuous traffic estimate and its logical bounds when citations to mixed acts are assigned adversarially to non-digital or digital provisions. The right panel replaces the continuous act share with an indicator for majority-digital acts (at least half of their provisions are digital) and compares observed citation traffic with the cumulative act stock and with a benchmark matched on the cited acts' vintages and instrument types.

Citations are recorded between acts, not provisions. The main text's citation figure therefore treats a cited act's digital share as the probability that a reference to it points to digital content. This section checks how much rests on that assumption. If every reference to a mixed act went to its non-digital provisions, the 2025 share would be 0.0%; if every one went to a digital provision, 58.5%. The plotted 13.6% is the value in between under even targeting, so the level is an estimate, not an observed rate. The same trend appears when we instead count only references to acts that are at least half digital: 12.4% of 2025 references go to digital law, against 3.7% under the matched benchmark and 1.1% from the stock (Figure 2). Majority-digital acts send 46.1% of their references to other majority-digital acts, against 2.7% for other acts, while still directing more than half of them outside digital law. Acts that are themselves essentially non-digital (at most 10% digital) cite digital content at about the rate the stock would predict — 4.4% of their 2025 references, against 4.3% from the stock — so the disproportion comes from acts that already carry digital content.

The temporary dip in 2020 mainly reflects fewer references from acts containing digital provisions and persists when explicitly COVID-related acts are excluded.

Which digital law does the rest of the acquis reach for? Among the 1,288 references from essentially non-digital acts (a digital share of at most 10%) into majority-digital acts, the four most-cited targets are all data-protection instruments: Regulation (EC) No 45/2001 and its successor Regulation (EU) 2018/1725 (data protection by the EU institutions, 272 and 220 references), the 1995 Data Protection Directive (146), and the GDPR (100). The data-protection lineage as a whole accounts for 57.8% of these references; most of the rest go to the digital systems through which other policies are administered — the customs information systems, the INSPIRE geodata infrastructure, the TRACES veterinary network, the e-Signatures framework.

E3 Regulatory tone with all articles included

The main text reports obligations and prohibitions as a share of tone-bearing articles, leaving descriptive articles (the *none* class) out of the denominator. If the classifier increasingly labeled permissions as descriptive, that share would rise without any change in the law. As a check, Figure 3 reports obligations and prohibitions as a share of *all* articles, which such relabelling cannot affect, alongside the *none* share itself.

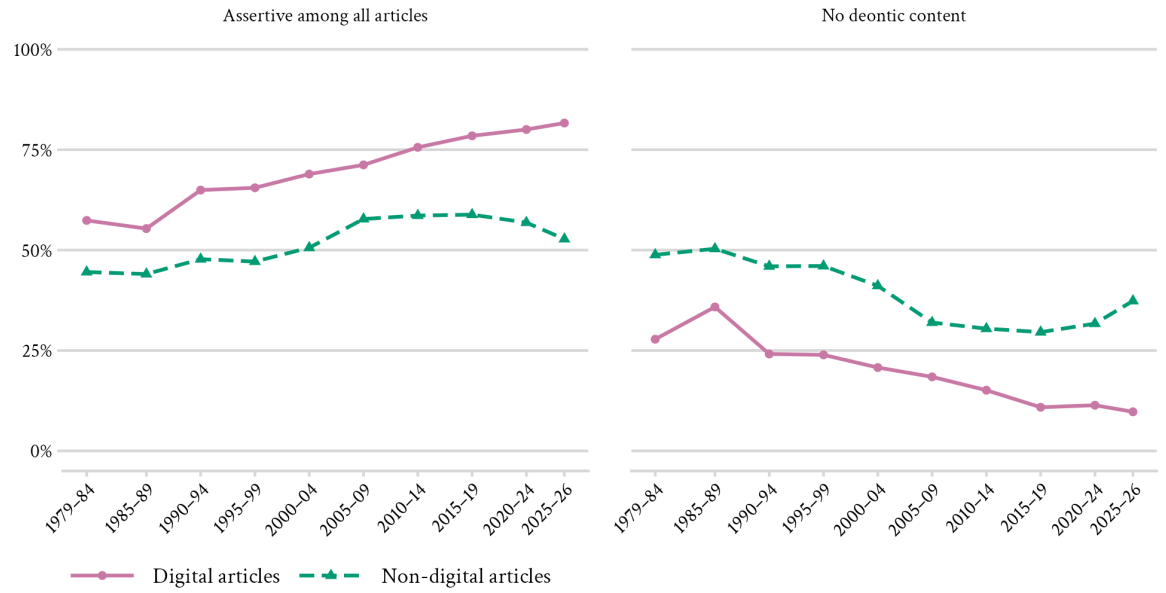


Figure 3 Regulatory tone with all articles retained in the denominator. Five-year periods pool article-level classifications; 1979–1984 is the first period and 2025–2026 is partial. The left panel reports obligations and prohibitions as a share of all articles — the boundary-invariant counterpart of the main-text measure. The right panel reports the share of articles classified as having no deontic content. Permissions among tone-bearing articles are not shown: they are the exact complement of the main-text assertive share.

Both the level and the trend survive. In 2020–2024, obligations and prohibitions account for 80.0% of *all* digital articles against 56.9% of non-digital ones, and the digital series rises from 57.4% in 1979–84 — a climb the non-digital series does not match (44.5% to 56.9%). The second panel adds a symptom check pointing the same way. If assertiveness were inflated by permissions being relabeled as descriptive, the pool of descriptive digital articles should be growing; instead it shrinks, from 27.8% to 11.4%, and stays

well below the non-digital rate (31.7%). Digital articles are at once more likely to carry deontic content and more likely to make that content an obligation or prohibition.

The trend remains when the none class is included in the denominator. It does not rule out every boundary problem. Drift in the opposite direction — descriptive articles increasingly read as obligations — would raise the assertive share and lower the none share in just the pattern observed here, and no measure computed from the same predictions can distinguish that from a real change. That is a question about the stability of classifier error over time. We can test it directly only for the digital-policy encoder (Section C6); the tone encoder has no era-specific audit, so this remains an assumption.

Whom the obligations and prohibitions address

Table 22 Addressees of articles imposing obligations or prohibitions, as a share of all such articles in each group.

Addressee	Digital articles	Digital articles, 2020–26	Non-digital articles
Member states	31.6%	27.8%	35.5%
EU institutions	30.4%	31.3%	25.0%
Private actors	28.9%	34.2%	19.2%
Other	8.1%	6.0%	19.2%
Individuals	1.0%	0.7%	1.1%

The main text's tone figure shows two addressees, private actors and member states. Table 22 gives the full breakdown of articles that impose an obligation or a prohibition. In digital law, member states, EU institutions and private actors each account for roughly three in ten such articles (31.6%, 30.4% and 28.9%), and individuals for almost none. Compared with non-digital law, digital law binds private actors more often (28.9% against 19.2%) and member states somewhat less; since 2020 private actors are the most frequent addressee of digital obligations and prohibitions (34.2%).

E4 Rights profile over time

This section adds the temporal dimension to the rights findings of the main text: the incidence of rights-invoking provisions among digital provisions over time (does the digital acquis invoke rights more often as it matures?), and the composition of the rights profile by period (does the civil-rights dominance predate the GDPR era or emerge with it?). It closes with the vocabulary evidence behind the main text's characterization of the three rights classes.

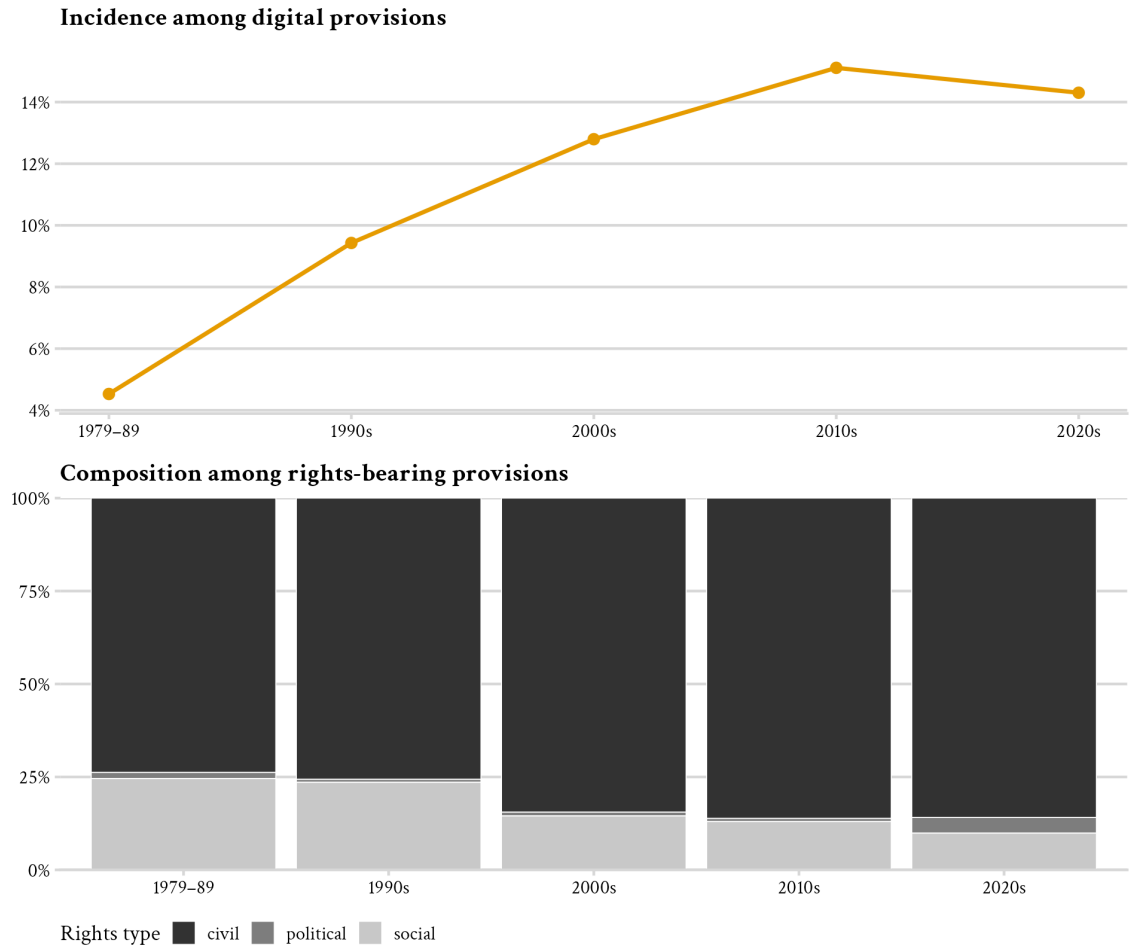


Figure 4 Rights in the digital acquis over time. The upper panel reports the share of all digital provisions classified as invoking a right. The lower panel reports the civil, political, and social composition among rights-bearing digital provisions. The first period pools 1979 with the 1980s; the 2020s run through April 2026.

Rights incidence rises during the early development of the digital acquis but does not increase monotonically thereafter. The share of digital provisions invoking a right grows from 4.5% in the 1980s to 9.4% in the 1990s, 12.8% in the 2000s, and 15.1% in the 2010s, before settling at 14.3% in the 2020s.

Civil-rights dominance clearly predates the GDPR. Civil rights already account for 75.6% of rights-bearing digital provisions in the 1990s and 84.5% in the 2000s, before the GDPR's adoption; the share is 86.2% in the 2010s and 85.9% in the 2020s. What changes more visibly is the residual profile: social rights fall from 23.6% in the 1990s to 9.9% in the 2020s, while political rights rise from 0.8% to 4.2%. Because political-rights provisions remain few and are concentrated in recent instruments, that last increase is descriptive rather than evidence of a smooth long-run trend.

What the rights classes talk about

The main text characterizes the three rights classes by their most distinctive vocabularies; this subsection shows the underlying evidence. For each class we select the twelve most distinctive bigrams by FREX — the harmonic mean of within-class frequency and cross-class exclusivity (Bischof and Airoldi 2012) — after removing stopwords, digits, and cross-reference formulae.

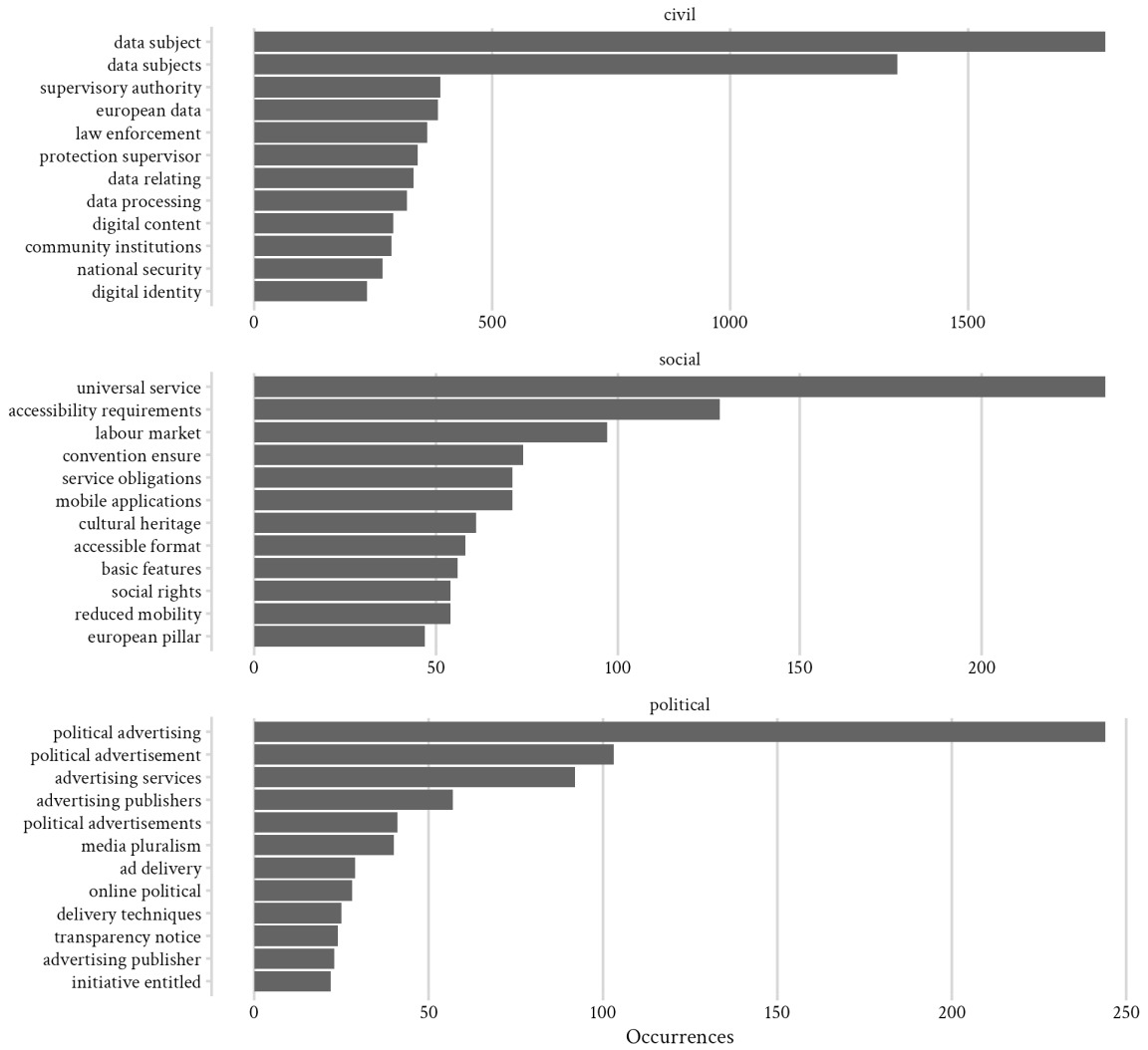


Figure 5 The twelve bigrams most distinctive of each rights class, selected by FREX (frequency x exclusivity); bar length shows the number of occurrences among rights-invoking digital provisions. Civil provisions speak the language of data protection; social provisions of universal service, accessibility, and labour; political provisions of political advertising and media pluralism.

E5 Encoder-configuration robustness

The hyperparameter sweep of Section C3 selects one encoder configuration, but several configurations reach nearly identical validation metrics, and classifiers that fit the training data equally well can still disagree on individual items (Marx, Calmon, and Ustun 2020). We therefore re-classified the corpus with the other configurations from the sweep and compare the findings.

We retrain the full 3 × 2 learning-rate/weight-decay grid of Table 8 from scratch on the production encoder's exact training data, and apply all six resulting encoders, alongside the deployed production encoder, to a fixed random sample of 300,000 provisions from the corpus. Because the six retrained encoders share their training data, tokenization, and selection protocol with the production model, any disagreement among them isolates the effect of hyperparameters and random initialization — the "researcher degrees of freedom" of the distillation step. The grid point that replicates the production

configuration (LR = 2×10^{-5}, WD = 0.1) doubles as a seed replicate: its disagreement with the deployed model measures pure training stochasticity.

Table 23 Predictive-multiplicity panel for the digital-policy encoder. Six encoders retrained on the production model's exact training data (one per grid point of the C3 sweep), plus the deployed production encoder, each applied to the same fixed 300,000-provision random sample of the corpus. Test F1 and human α are validation metrics of each retrained run (production row: deployed-model metadata); the last two columns report each encoder's digital share on the canonical-universe subset of the sample and its item-level disagreement with the deployed production encoder (full sample). The production configuration's grid point is a seed replicate of the deployed model.

Encoder	LR	WD	Test F1	Human α	Digital share	Disagr. vs. prod.
production (deployed)	2e-05	0.10	0.954	0.902	4.31%	—
run_00	1e-05	0.01	0.949	0.897	4.61%	1.04%
run_01	1e-05	0.10	0.951	0.887	4.60%	1.03%
run_02	2e-05	0.01	0.955	0.869	4.08%	0.78%
run_03	2e-05	0.10	0.957	0.902	4.59%	0.80%
run_04	5e-05	0.01	0.927	0.871	4.29%	1.28%
run_05	5e-05	0.10	0.913	0.833	4.94%	1.93%

Three results follow (Table 23). First, the encoders rarely disagree on individual provisions. All seven encoders agree unanimously on 96.6% of the sample; pairwise disagreement ranges from 0.78% to 2.18%. The structure of that range is informative: the largest disagreements all involve the LR = 5×10^{-5} configurations — precisely those the sweep identifies as degraded — while the four configurations with near-identical validation metrics disagree with the deployed model by at most 1.04%, barely above the seed-replicate baseline.

Second, the aggregate quantities the paper actually reports are insensitive to encoder choice. The corpus-level digital share ranges from 4.08% to 4.94% across all seven encoders (production: 4.31%, mid-range), and the yearly digital-share series of any two encoders correlate at 0.995 or higher, with a median cross-encoder spread of 1.0% per year. Every trend feature discussed in the main text — the step around 2018, the post-2020 plateau — is reproduced by every encoder in the panel; encoder choice contributes roughly 0.3 percentage points of level uncertainty within the viable configurations, comparable in kind to the superpopulation band of Section E1.

Third, individual labels are more likely to change when the production encoder assigns a probability between 0.1 and 0.9. A provision whose production-encoder probability falls in the uncertain band (0.1–0.9) flips under at least one sibling encoder with probability 84.7%; outside that band the flip rate is 2.9%. Because the band holds only 0.6% of the corpus, however, most flips (84.8%) occur on items where the deployed encoder looks confident — encoder confidence ranks flip risk extremely well but does

not guarantee any individual label. Because the substantive estimands are aggregate proportions and trends, the relevant multiplicity diagnostic is variation in those quantities across defensible encoders — which is small here even though some item labels change. Aggregation does not by itself eliminate classification error: classify-and-count estimates can remain biased despite high item-level accuracy (Hopkins and King 2010; Jerzak, King, and Strezhnev 2023).

E6 Are the dispersed digital provisions just cross-references?

The main text shows that most digital provisions lie outside the expert inventory of EU digital laws (Section D1), and that nearly half sit in acts that are less than half digital. One could object that these provisions are mostly formulaic: standard references to the data-protection rules, compliance reminders, and similar clauses, rather than digital rules in their own right. Such references are still digital content and still bind the acts that contain them, so classifying them as digital is not an error; but if the dispersed provisions were mostly of this kind, the finding would mean less than it appears. We therefore check what share of the digital provisions outside the inventory consists only of references to rules laid down elsewhere, in two ways: with text filters and with a model-based audit of a sample.

Filtering references and repeated text

The first filter identifies provisions that cite one of five data-protection instruments (Directive 95/46/EC, Regulation (EC) No 45/2001, Directive 2002/58/EC, Regulation (EU) 2016/679, Regulation (EU) 2018/1725) alongside phrases such as "in accordance with", "without prejudice to", and "in compliance with". These phrases refer to existing rules without adding to them. We also identify wording repeated in five or more acts after standardising numbers and punctuation, regardless of its subject. This captures compliance reminders, statements that a body was consulted, and other repeated passages. Together these rules form the *moderate* filter. A *broader* filter flags any provision citing one of the five data-protection instruments, whatever the surrounding wording, and any text repeated in three or more acts.

The broader filter flags 11.0% of digital provisions outside the expert-list acts and 4.9% inside them. Removing the flagged provisions changes the outside-list share little: it is 82.9% before filtering, 82.3% after the moderate filter removes 5.7% of all digital provisions, and 81.9% after the broader filter removes 10.0%. There are far more digital provisions outside the list (70,295) than inside it (14,524). To leave equal numbers inside and outside, we would have to remove 79% of the outside provisions — roughly seven times the share of outside provisions flagged by the broader filter. These filters cannot tell us how many of the remaining provisions contain rules specific to the act and how many are references they missed.

Checking a sample with language models

The filters miss references that neither use repeated wording nor cite a data-protection instrument. We therefore asked three open language models to classify a random sample of 1,500 digital provisions outside the expert-list acts. Two models came from our classification ensemble (Qwen3-235B, gpt-oss-120b); the third was a more recent open model (Meta's Muse Glimmer 30B). The coding instruction

is reproduced below (`scripts/audit_boilerplate.py`; the instruction names the category "boilerplate"). A provision is flagged only if all its digital content invokes, restates, or preserves rules laid down elsewhere. Any addition specific to the act excludes it from this category.

By majority vote, the models flag 7.3% of the sample: 12.4% of recitals and 2.7% of articles. They thus identify these references much more often in recitals; they classify almost all articles as containing something specific to the act. All three models agree on 89.9% of provisions, with pairwise agreement between 92.0% and 94.3%. Applying the sample's flagged share to all outside-list digital provisions and removing that share lowers the estimated outside-list share from 82.9% to 81.8% — within a percentage point of the filter results.

Both checks identify about a tenth of digital provisions outside the expert list as references or repeated text. Removing the provisions identified by either check changes the outside-list share by about one percentage point. The model audit has not been validated against human coding, and one of its three models is outside the validated ensemble of Section C1. Its results provide an additional check, not a validated estimate of how many provisions simply refer to rules elsewhere.

```
You are an expert annotator analyzing official EU policy texts that have been
classified as digital policy.

Task

Determine whether the text is digital BOILERPLATE: a formulaic, legally routine
reference to digital rules laid down elsewhere that adds nothing of its own to
the digital acquis. Return ONLY a JSON object:
{"boilerplate": "yes"} or {"boilerplate": "no"}

Definition

Code "yes" ONLY if the text's digital content consists entirely of invoking,
restating, or preserving digital rules that exist elsewhere and apply of their
own force — inserted as drafting routine, adding no rule, specification,
system, or reasoning specific to this act. Typical forms:
- saving clauses: 'without prejudice to Regulation (EU) 2016/679'; 'shall not
  affect Directive 2002/58/EC'
- compliance reminders: 'personal data shall be processed in accordance with
  Regulation (EU) 2018/1725' — where that regulation applies anyway and the
  text adds no specification
- procedural formulas: the European Data Protection Supervisor was consulted;
  a committee delivered an opinion
- background restatements: text that merely summarizes what another act's
  digital rules do, without drawing any consequence for this act
```

```
Code "no" in ALL other cases — whenever the text does anything digital of its
own, however modest: it creates, extends, or specifies any digital rule,
obligation, permission, or prohibition; designates responsibilities for data
or systems (for example, a controller); sets retention periods, data
categories, access rules, security or interoperability requirements;
establishes or governs a database, portal, or electronic procedure; or
explains a digital feature or rationale specific to this act.

Decision rule

The burden of proof is on "yes": if the text goes beyond formulaic reference in
ANY way — even a single specification, however small — code "no". If you are
unsure, code "no". Texts whose digital connection is thin or incidental are
also "no": they may be weakly digital, but they are not formulaic references to
other law.
```

E7 Interpreting and stress-testing the digital-policy encoder

The held-out comparisons with human coders in Section C2 tell us how often the encoder applies the coding scheme correctly. They do not tell us which parts of a provision drive a particular prediction, or whether the encoder has learned shortcuts that happen to work well on average. We therefore examine the digital-policy encoder in two additional ways. First, we remove short blocks of text from actual corpus provisions and measure how its predictions change. Second, we give it controlled pairs of nearly identical provisions that differ in one feature covered by the coding rules. The first exercise shows which text matters to individual predictions; the second tests whether the encoder responds to distinctions that the coding scheme says should matter.

Removing short blocks of text

We divide each provision into consecutive, non-overlapping blocks of five words; the final block can be shorter. Consider the first example below. The encoder sees the complete sentence—"All systems and software shall record and maintain audit logs containing the information set out in paragraph 2 and provide those logs upon request of the authorities"—and labels it digital. For each block, we then create two modified versions of the sentence. Taking "All systems and software shall" as an example, one version replaces these words with mask tokens and the other deletes them. We run both versions through the encoder again and observe how its prediction changes.

The encoder produces one raw score for "digital" and another for "not digital." We subtract the score for the alternative label from the score for the original label, producing what is conventionally called the *logit margin*. A large decline in this margin after a block is hidden means that the block supported the original prediction; an increase means that it worked against it. We use this margin rather than the displayed probability because probabilities compress changes in very confident predictions: a fall from 99.99% to 99.6%, for example, appears as 1.00 in both cases when shown to two decimal places,

despite representing a sizeable change in the underlying scores. Masking preserves sentence length but inserts artificial placeholders, whereas deletion avoids placeholders but changes the sentence around the missing words. The two methods produce only moderately similar block rankings within provisions (median Spearman's $\rho = 0.43$). We therefore carry both forward rather than treating either as definitive.

To cover easy and difficult predictions from different parts of the corpus, we sample separately by predicted probability (below 0.05, 0.05–0.50, 0.50–0.95, and at least 0.95), provision type (articles and recitals), and period (before 2000, 2000–2019, and 2020 onward). We draw 40 provisions from each of the resulting 24 groups. The 960 provisions come from 960 different acts, preventing any unusually long act from dominating the exercise. The pre-2000 stratum matters for the paper's temporal claims, and it is the era of vocabulary the classifier must handle: its confidently digital items include, for example, a 1989 recital recalling the Community programme on 'trade electronic data interchange systems' (TEDIS) — an early example of digital terminology — and the 1993 customs provisions on 'declarations made using a data-processing technique', both classified digital with near-certainty. The script `scripts/audit_encoder_interpretability.py` records the sample and attribution scores, together with hashes of the model and inputs needed to reproduce the run.

Figure 6 shows the deletion scores for three provisions. In the first, the references to systems, software, and audit logs support a clear digital prediction. The other two provisions are classified as non-digital. In one, "electronic form" and "searchable database" push toward digital, but the provision concerns the permitted format of periodical publications rather than regulating a digital system. In the other, "vascular diagnostic computer" pushes in the same direction, but the provision ultimately applies a customs tariff rather than regulating the technology. Orange always supports the prediction named above the panel; blue works against it. The examples make the scores visible, while the evidence below comes from all 960 sampled provisions and both ways of hiding text.

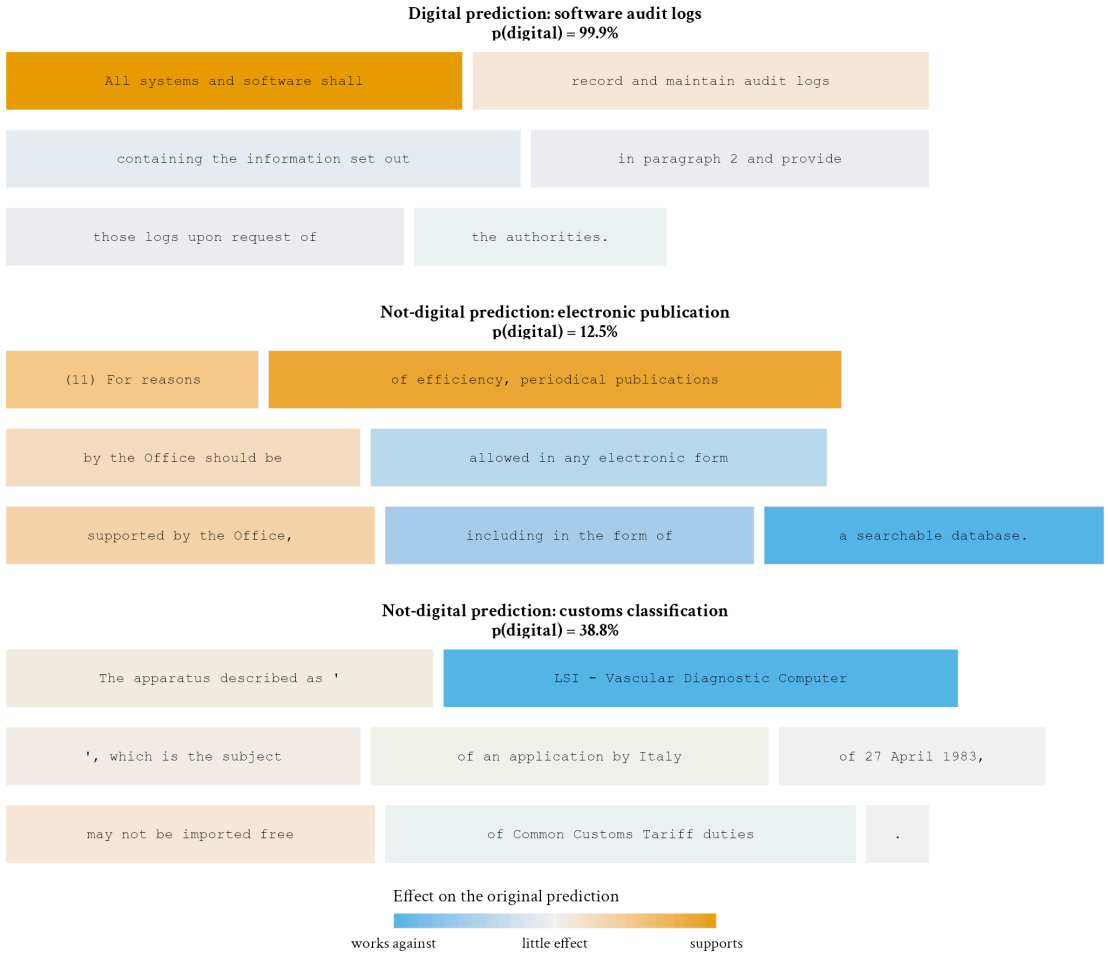


Figure 6 How deleting short blocks of text affects three encoder predictions. Each rectangle is one of the consecutive five-word blocks used in the audit (the final block can be shorter). Orange means that the block supports the encoder's original prediction: deleting it weakens that prediction. Blue means that the block works against the original prediction: deleting it strengthens that prediction. In the two non-digital examples, blue therefore marks text that pushes the encoder toward digital. Color intensity is scaled within each panel, so it ranks blocks within a provision but does not compare effect sizes across panels. These examples illustrate deletion; the aggregate results use deletion and masking across the full audit sample.

To test systematically whether the highlighted blocks matter to the encoder's prediction, we compare the blocks ranked most important with the same number of blocks chosen at random. Suppose a provision contains 20 blocks and we examine the top 10%, or two blocks. We first hide those two selected blocks. If the ranking is informative, this should weaken the original prediction more than hiding two randomly chosen blocks. We then do the reverse: keep only the two selected blocks and hide everything else. If the ranking is informative, those two blocks should preserve more of the original prediction than two randomly chosen blocks. We repeat the random comparison 20 times within each provision and run both tests at 5%, 10%, and 20% of its blocks.

We make the test harder by using one hiding method to choose the blocks and the other to test them. Blocks identified by deletion are evaluated with masking; blocks identified by masking are evaluated with deletion. This avoids giving a method an advantage by grading it with the exact operation that created its ranking. We summarize each comparison as the selected blocks' advantage over random blocks: selected minus random when blocks are removed, and random minus selected when blocks are

retained. Positive values therefore favor the selected blocks in both comparisons shown in Figure 7. The intervals come from 5,000 resamples that preserve the 24 groups in the audit design. They show how stable the comparison is across this deliberately balanced sample, not how common any kind of provision is in the corpus.

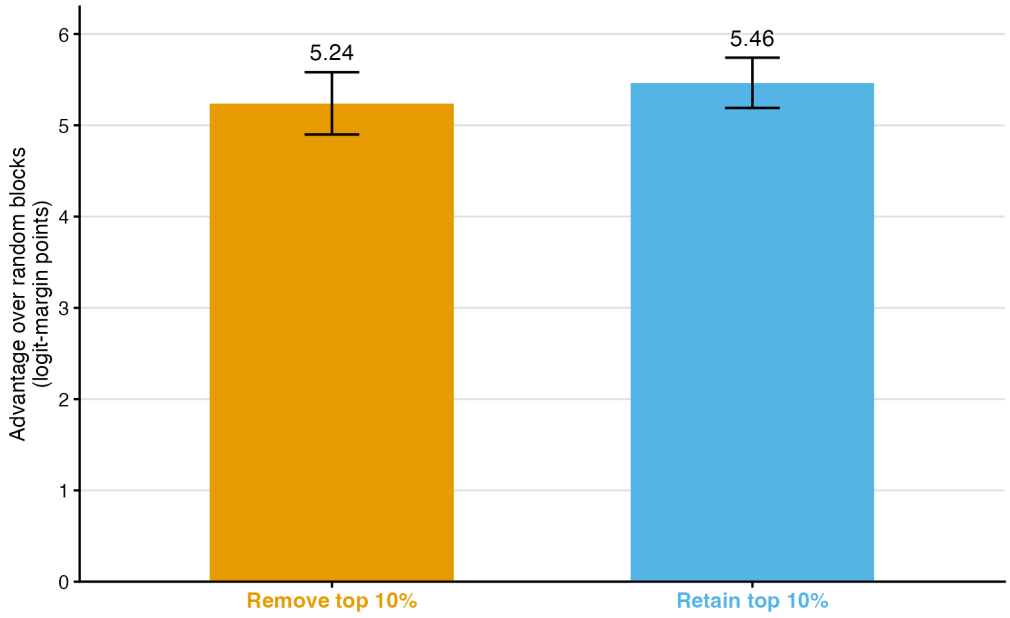


Figure 7 Do the top 10% of five-word blocks matter more than random blocks? Bars show the selected blocks' advantage over the same number of randomly chosen blocks; error bars are 95% intervals. Positive values favor the selected blocks. Rankings and tests use opposite removal methods and are averaged over both directions.

At the 10% cutoff, hiding the selected blocks reduces the original label's lead by 6.9 points, compared with 1.7 points for random blocks. Their difference is the 5.24-point advantage shown by the first bar: the blocks ranked important do much more damage when removed. In the reverse test, keeping only the selected blocks costs the model 1.6 points, compared with 7.1 points when only random blocks remain. This produces the 5.46-point advantage in the second bar: the selected blocks preserve much more of the prediction. The expected comparison holds provision by provision in 88.9% of removal tests and 83.6% of retention tests.[10]

These tests have two limits. First, the fixed five-word partition can split a meaningful phrase across adjacent blocks, so the exact boundaries should not be read too literally. Second, retaining a few scattered blocks produces an artificial fragment rather than a self-contained legal explanation. The retention test

[10] As a supplementary check, we recompute the block rankings after successively replacing fitted parts of the model, always using the same 960 provisions (following Adebayo et al. 2018). ModernBERT-large contains a 28-layer encoder, a pretrained prediction head that is further updated during fine-tuning, and the new two-class readout (Warner et al. 2024). Replacing only the readout or the prediction head leaves a seed-dependent resemblance: across three random draws, the signed median ρ ranges from −0.02 to 0.42 for the readout and from 0.06 to 0.46 when both are replaced. Once the upper half of the encoder is also replaced, the rankings become unrelated to those of the fitted model (pooled signed $\rho = -0.05$; absolute $\rho = 0.04$), as they do when the entire network is randomized (-0.00; 0.02). Restoring the original pretrained ModernBERT backbone while retaining the fitted head and readout likewise yields little similarity (signed $\rho = 0.12$; absolute $\rho = 0.04$). The rankings therefore depend on task-specific changes in the fitted encoder, especially its upper layers, rather than on the display procedure or generic pretrained representations alone. This check concerns the rankings, not label accuracy.

shows that selected blocks preserve more of the model's signal than random text of the same length; it does not show that they would be sufficient legal reasons on their own.

Controlled tests of the coding rules

The removal exercise shows which words affect a prediction. It does not show whether the encoder follows the distinctions in our coding rules. We examine that question with 80 pairs of short, constructed provisions. Within each pair, we change one feature and leave the rest of the sentence alone. This follows the logic of Ribeiro et al. (2020)'s CheckList tests.

Some pairs change the policy's substance. For example, "Providers shall establish rules for an interoperable electronic database" should receive a higher digital probability than the otherwise identical sentence about "a central paper archive." Other pairs introduce only a digital means of doing something: "Applicants shall submit the completed form by electronic means" and "Applicants shall submit the completed form by post" should both remain *not digital*. A third group changes details that should not determine the policy domain. Replacing an act number, replacing "the Commission" with "the Agency," or replacing *shall* with *may* should leave an otherwise digital provision classified as digital. The tests also cover older digital vocabulary, provisions about clearly single-purpose digital bodies, and the addition of a substantive digital rule.

We run every pair through the deployed encoder and through four similarly performing encoders from Section E5. This tells us whether a result is peculiar to the one model selected for production. Table 24 shows the eight types separately: a high overall average cannot conceal a rule that the encoders apply poorly.

Table 24 Controlled tests derived from the coding rules. Both labels expected requires both versions to fall on their specified sides of the classification threshold. Expected comparison asks whether directional pairs move in the specified direction and applies the same label test to the remaining pairs. Production is the deployed encoder; sibling range reports the expected-comparison result across four near-optimal configurations from Section E5.

Capability	Pairs	Both labels expected	Expected comparison	Sibling range
Digital object replaces analogue	12	83.3%	100.0%	100.0%–100.0%
Substantive digital rule added	12	91.7%	100.0%	100.0%–100.0%
Incidental digital channel	12	91.7%	91.7%	83.3%–100.0%
Named single-purpose digital body	12	75.0%	75.0%	58.3%–91.7%
Historical digital terminology	8	100.0%	100.0%	100.0%–100.0%
Act-number invariance	8	100.0%	100.0%	100.0%–100.0%
Institution-name invariance	8	100.0%	100.0%	100.0%–100.0%
Deontic-operator invariance	8	100.0%	100.0%	100.0%–100.0%

The production encoder gives both texts the expected labels in 91.2% of the 80 pairs and produces the expected comparison in 95.0%; the four sibling encoders range from 91.2% to 98.8% on the latter measure. It gets both labels right in 10 of 12 pairs when a digital object replaces a non-digital one and 11 of 12 when a substantive digital rule is added. In the three misses, the digital wording raises the score as expected, but the analogue version also remains above the digital threshold. The "both labels expected" measure counts these as failures, whereas the comparison measure records the intended movement. The production model gets every test of older digital terminology and every change to an act number, institution name, or deontic operator right. It also gets 11 of the 12 pairs contrasting a merely electronic channel with a non-digital channel right. Across these seven types of test, both labels are as expected in 94.1% of the pairs and the intended comparison holds in 98.5%.

The result is less consistent for provisions that name a body with an exclusively digital mandate. Under the coding rule used by the models and human coders, "ENISA shall submit its annual accounts to the Commission" counts as digital because ENISA is a single-purpose digital body; the otherwise identical sentence beginning "The Agency" does not. The production encoder draws the expected distinction in 9 of 12 pairs. The siblings range from 7 to 11. In every pair the generic agency remains *not digital*; the misses occur because a sparse administrative sentence naming ENISA or BEREC sometimes also remains below the digital threshold. For the ENISA annual-accounts pair, the body name raises the digital probability from 0.0% to 25.1%. In the pair "BEREC shall adopt its staff establishment plan under the Financial Regulation" versus the corresponding sentence about "the Agency," it rises from 0.0% to 2.3%. The body name therefore moves both predictions in the intended direction, but not far enough to cross the classification threshold. This is a specific limit of the distilled encoder, not evidence that the coding scheme as a whole was applied incorrectly. The controlled examples reveal the pattern but do not estimate how often it occurs in the corpus; the comparisons with naturally occurring, human-coded provisions in Section C2 remain the relevant accuracy test.

The encoder's behavior is less consistent for very sparse administrative provisions about named digital bodies. Neither exercise replaces the human validation in Section C2, which tells us how often the encoder classifies naturally occurring provisions correctly.

E8 The enforcement-decision exclusion and the composition by instrument

The main text excludes individual enforcement decisions from the analysis corpus (Section A2). Because the classification pipeline deliberately covers the full universe — every regulation, directive, and decision — this exclusion is an analysis-time choice, and we can ask directly what would change without it. The answer is: very little. With the 182,400 provisions of 1,691 enforcement decisions included, the corpus digital share moves from 4.33% to 4.41%; the only visible movement in the trend is concentrated in the early decades (the 1990s rise from 1.3% to 1.8%, reflecting the telecom competition cases of the liberalization era), while the 2020s change little (10.5% vs 10.2%). The dispersion finding becomes marginally *stronger* (enforcement decisions are not on the expert list, so the outside share rises from 82.9% to 84.6%). The adjudication findings are indifferent to the choice: among tone-bearing digital

articles, the assertive share is 88.8% filtered and 88.8% unfiltered; the share of digital obligations and prohibitions addressed to private actors is 28.9% vs 29.0%; and the rights profile shifts by at most a few tenths of a percentage point in any class. The exclusion removes a large, identifiable class of enforcement decisions and improves the corpus's fit to the paper's rulemaking concept; it does not carry any of the paper's results.

The retained decisions are a mixed group, and the mixture shows in the classifier's output. Decisions of the European Parliament and of the Council (1,325 acts, among them the Radio Spectrum Decision, the ISA² interoperability programme, and the Digital Decade Policy Programme) are as digital-intensive as any instrument in the corpus, with 10.8% of their provisions digital; Commission implementing decisions sit at almost the same rate (10.5%), Council implementing decisions lower (6.3%), and decisions carrying an explicit addressee formula ("Only the … text is authentic") contain almost none (0.9%), so their inclusion has little effect on the findings, though they are numerous and stay in the denominator.[11]

Table 25 The digital acquis by instrument type (canonical corpus). 'Carries' is each instrument's share of all digital provisions. The final row shows the excluded enforcement decisions for comparison.

Instrument	Acts	Provisions	Digital provisions	Digital share	Carries
Regulation	109,943	1,371,384	48,671	3.5%	57.4%
Directive	3,806	131,780	11,748	8.9%	13.9%
Decision	32,485	453,734	24,400	5.4%	28.8%
Enforcement (excluded)	1,691	182,400	9,591	5.3%	—

Two compositional facts stand out. Regulations carry the majority of the digital acquis (57.4% of all digital provisions), but directives are its densest instrument: 8.9% of directive provisions are digital, against 3.5% for regulations — the flagship digital directives of the 1990s and 2000s (data protection, e-commerce, electronic communications) and the 2010s' universal-service and accessibility instruments concentrate digital content in a comparatively small legislative form. Figure 8 traces this composition over time: the digital share rises within every instrument form, directives lead through the 1990s and 2000s, and the gap narrows as the 2020s' flagship regulations shift digital lawmaking toward directly applicable instruments.

[11] A further 185 retained decisions contribute no provision, because their text could not be split into recitals and articles (Section A2).

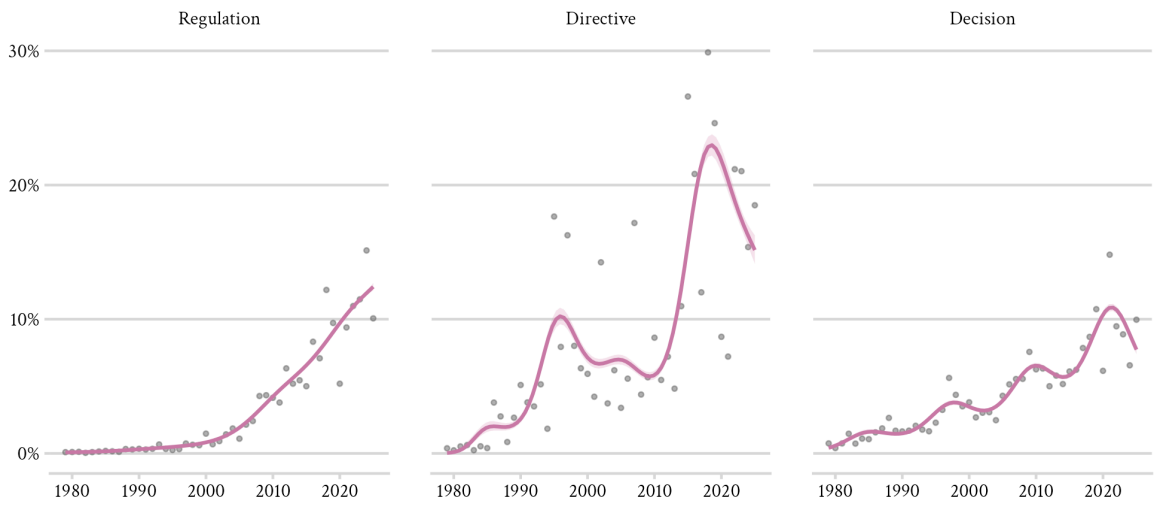


Figure 8 The digital share of newly adopted provisions by instrument form, canonical corpus. Points are annual shares; the curve is a binomial GAM weighted by each year's provision count. Panels share the y-scale. The partial year 2026 is omitted.

Part F — Disclosure of LLM use

This part reports our use of large language models following Version 1.1 of the GUIDE-LLM checklist (Feuerriegel et al. 2026). We distinguish two uses throughout. First, LLMs are part of the measurement design: they classify provisions, produce the market and rights ratings, and conduct one supplementary audit. Second, Claude and Codex served as interactive research assistants. The encoder models distilled from the LLM labels are documented in Part C. Conventional components of the corpus-building pipeline, including Docling for document extraction and Opus-MT for translation, are documented in Part A and are not repeated as research-assistant use below.

Core items

Scope of LLM use

Item A.1 — Purposes. The five-model ensemble classified sampled provisions on the four tasks defined in Part B; its consensus labels and vote fractions supervised the encoder models that classify the corpus (Sections C1–C3). A separate three-model panel produced the market and rights ratings (Section C7), and three open-weight models conducted the supplementary formulaic-reference audit (Section E6). Claude Code and Codex were used during research design, coding, debugging, robustness analysis, literature-related tasks involving selected materials supplied by the authors, interpretation, manuscript criticism, editing, and documentation. They were not research objects and did not interact with participants.

Item A.2 — Degree of automation and human oversight. LLM calls, response parsing, ensemble aggregation, the market and rights ratings, and encoder inference were automated; individual production classifications were not manually approved. The coding scheme and pipeline were developed under

author supervision and evaluated against independent coding by the two authors, Timo Seidl and Nir Kosti, who sometimes coded individually and sometimes both coded the same material (Sections B3 and C2). The paper's main text was written principally by the authors, with Claude and Codex used for criticism, revision suggestions, verification, and coding support. In the appendix, the assistants also produced some initial prose and code drafts, which the authors evaluated and edited. Substantive manuscript text, analysis code, analytical decisions, and reported results were reviewed by an author. Some ancillary workflow infrastructure, such as parts of the Label Studio setup, was instead checked through its operation and outputs rather than by line-by-line review. The authors used Claude and Codex to review each other's suggestions. For some supporting software, they checked the resulting workflow rather than reviewing every line of code. The authors made all substantive decisions and take responsibility for the complete manuscript.

Model and system details

Item B.1 — Models, versions, providers, and roles. Table 4 reports the exact identifiers, developers, providers, weight status, and numerical precision of the production classification ensemble: Qwen3-235B-A22B-Instruct, Llama-3.3-70B-Instruct, gpt-oss-120b, GPT-5.2, and DeepSeek-V4 Pro. Table 13 reports the three open-weight model–provider pairs used for scalar scoring. Section E6 identifies the models used in the formulaic-reference audit. Interactive assistance used Claude Code, principally with Claude Opus versions from 4.5 through 5 and Fable 5, and Codex with GPT-5.6 Sol. These assistants were used during preparation of the manuscript through September 2026; because model choice varied across interactive sessions and those sessions were not a measurement instrument, we do not treat them as a single fixed model configuration.

Item B.2 — Access and context mode. The measurement models were called programmatically through the provider routes reported in Sections C1, C7, and E6, using separate calls for each provision and task; calls were submitted synchronously or through provider batch facilities as available. Claude and GPT-5.6 Sol were accessed under ordinary subscriptions through the Claude Code and Codex coding harnesses. Both operated in ongoing conversations, could inspect and modify the repository, run terminal commands and document renders, inspect supplied documents, and access the web when requested.

Item B.3 — Configurations and repeated runs. The measurement scripts record model-specific configurations. In particular, model identifiers and numerical-precision constraints were pinned for the OpenRouter models; gpt-oss-120b used low reasoning effort; and thinking was disabled for DeepSeek-V4 Pro's short classification output. Other settings used provider defaults unless the scripts state otherwise. The classification ensemble contains five votes per item, the rating panel three scores per provision for each rating, and the supplementary audit three votes per item. Sampling seeds and the configurations of the encoder-training and multiplicity exercises are recorded in the corresponding scripts and model metadata; Section C4 reports repeated classifications used to assess prompt and run stability. For Claude Code and Codex, the authors did not manually set temperature, seed, token limits,

or other sampling parameters; model choice varied, while generation and reasoning settings generally remained at the harness defaults.

Item B.4 — Customization and tools. The production LLMs were not fine-tuned and did not use retrieval-augmented generation, automated prompt optimization, web search, or agentic tools during classification. They received the versioned coding instruction and one provision, and returned a short JSON-formatted answer. The downstream encoders were fine-tuned on ensemble-derived targets as documented in Section C3. Claude Code and Codex could access the repository, run terminal commands, use the web when requested, and follow saved project instructions and skills. Scholarly context was selective rather than retrieved from an unrestricted library: for particular tasks, an author sometimes supplied a paper, excerpts, or the author's notes. The assistants were also often used in reciprocal-review workflows.

Item B.5 — Persistent memory. The measurement calls were stateless. Interactive assistant conversations retained their own running context, while durable project context came from versioned or saved repository instructions and project files. Provider-level persistent memory was not enabled for this project.

Prompts

Item C.1 — Exact prompts. Section B2 reproduces the complete prompts for all four classification tasks and both rating prompts, generated from the version-controlled `config/label_config.yml`; Section E6 reproduces the supplementary audit prompt. The interactive assistants received task-specific conversational instructions. We do not reproduce those conversations because they were iterative research assistance rather than instruments that generated the observations analyzed in the paper.

Item C.2 — System-wide instructions. For measurement, each versioned coding instruction was supplied as the system prompt, together with a JSON-only response instruction; the provision text was the user input. Claude Code and Codex were additionally governed by their providers' and coding harnesses' system instructions, which are not controlled by the authors, and by repository-level instructions, saved project context, and custom skills created or selected by the authors. The latter materials accompany the working repository where they are part of the project.

Data inputs and privacy

Item D.1 — Personal or sensitive data. No participant data or personal, confidential, or sensitive information was supplied to any model. Measurement inputs were publicly available EU legal texts. Interactive assistants received the paper, code, public project data, and selected scholarly material or research notes made available for particular tasks. Credentials and API keys were kept outside prompts and release materials.

Validation and interpretation

Item E.1 — Human validation. The classification instruments were validated quantitatively against independent human coding by the two authors, both domain experts; the sampling, task gating,

independence of the coding, agreement measures, and comparisons of humans, ensemble, and encoders are reported in Sections B3 and C2. The rating design and its pairwise human validation are reported separately in Section C7. Prompt stability, configuration multiplicity, external comparisons, and behavioral diagnostics appear in Sections C4, D1–D3, and E5–E7. Interactive-assistant contributions were reviewed as described under Item A.2; they were not accepted automatically as evidence.

Item E.2 — Post-processing. Classification models were instructed to return a one-line JSON object. Responses were parsed programmatically, checked against the task's permitted values, and treated as missing when no valid value could be recovered; operational scripts support checkpointing, retrying, and auditing failed calls. Valid classifications were aggregated by majority vote, with the documented tie-break rule, while per-class vote fractions were retained for soft-label training (Section C1). The ratings were computed from renormalized digit-token probabilities rather than from free-form text (Section C7). Encoder probabilities and labels were produced by the versioned prediction scripts. Assistant-generated prose and code were edited, tested, rendered, or otherwise checked in proportion to their substantive role.

Reproducibility

Item F.1 — Shared materials. The replication package accompanying the preprint will include the analysis and LLM-calling code, versioned prompts and configurations, corpus snapshot and derived data needed for the reported analyses, human-validation labels, trained encoder weights and metadata, and documentation of the workflow. Complete Claude Code and Codex conversations will not be released: they are lengthy, interleaved working records rather than measurement instruments. Their material effects are represented in the reviewed manuscript, code, project instructions, and version history.

Competing interests

Item G.1 — Funding, support, and relationships. The authors received no dedicated provider funding, in-kind model or compute access, or special access for this project, and have no relevant professional relationship with the model providers. LLM services were accessed on ordinary commercial API terms or through ordinary subscriptions.

Optional items

Justification for model choice. The measurement design deliberately combines model families and providers rather than depending on one system. Four of the five production ensemble members are open-weight models, supporting inspectability and continued access, while one proprietary frontier model provides a performance anchor and tie-breaker. Final inclusion was based on validation performance, availability, operational reliability, and the value of model diversity; encoder selection was based on the comparisons in Section C3. Claude Code and Codex were selected for their model capabilities, repository and terminal integration, reciprocal-review value, and availability under existing subscriptions, not through a formal benchmark.

Rationale for prompt design. The classification prompts translate the conceptual coding scheme into a common structure: task and category definitions, inclusion and exclusion rules, edge-case guidance, and worked examples with rationales, followed by a constrained response format (Sections B1–B2). They were refined during the pilot round through disagreement and error analysis and frozen before the validation rounds (Sections B3 and C2). The same generated instruction text was used for models and human coders on the four classification tasks. Prompt variants and their stability are evaluated in Section C4; no automated prompt optimization was used.

Comparison against other methods or models. Yes. The appendix compares human coders, individual LLMs, ensemble consensus, and distilled encoders; hard- and soft-target training; prompt formulations and repeated runs; encoder hyperparameters; expert lists and EuroVoc indexing; and near-optimal sibling encoders on controlled behavioral tests. The relevant designs and metrics appear in Parts C–E.

Training-data leakage. The foundation models may have encountered EUR-Lex texts during pre-training, and such overlap cannot be ruled out. The task, however, asks models to apply a supplied coding scheme to individual provisions rather than retrieve pre-existing labels, and the study evaluates the resulting classifications against labels independently produced by the authors. This limits what overlap could establish, but does not make the models' unknown training corpora irrelevant.

Potential bias or systematic differences. The appendix examines prompt wording (Section C4), external indexing practices (Part D), encoder configuration (Section E5), and dependence on lexical or codebook shortcuts (Section E7); it also reports completed checks at natural prevalence (Section C5) and across periods (Section C6); Section A3 documents how the few non-English source texts were handled. The completed checks remain bounded by the available human-coded sample. In particular, average validation performance does not by itself establish identical error rates across periods or document types.

Conversation transcripts. Not shared. Exact research prompts, code, configurations, and outputs that generated analyzed observations are shared instead.

Ethical implications. The study uses public legal texts and involves no participants or consequential automated decisions. Its principal risks are epistemic: classification errors, opaque model behavior, or erroneous assistant suggestions could distort the empirical map. We address these risks through independent human validation, multiple models, robustness and interpretability checks, open-weight encoders, release of the measurement materials, and author review. These safeguards reduce but do not eliminate model dependence.

Computational resources. Model providers, serving configurations, and locally recoverable training and inference metadata are reported in the methods and accompanying files. Interactive Claude Code and Codex use occurred under subscriptions and was not separately metered in a form suitable for estimating tokens, energy use, or cost.

References

Abadie, Alberto, Susan Athey, Guido W. Imbens, and Jeffrey M. Wooldridge. 2020. "Sampling-Based Versus Design-Based Uncertainty in Regression Analysis." *Econometrica : Journal of the Econometric Society* 88 (1): 265–96. https://doi.org/10.3982/ECTA12675.

Adebayo, Julius, Justin Gilmer, Michael Muelly, Ian Goodfellow, Moritz Hardt, and Been Kim. 2018. "Sanity Checks for Saliency Maps." In *Advances in Neural Information Processing Systems 31*. Curran Associates, Inc. https://proceedings.neurips.cc/paper/2018/hash/294a8ed24b1ad22ec2e7efea049b8737-Abstract.html.

Barrie, Christopher, Elli Palaiologou, and Petter Törnberg. 2024. "Prompt Stability Scoring for Text Annotation with Large Language Models," July. https://doi.org/10.48550/arXiv.2407.02039.

Birkenmaier, Lukas, Clemens M. Lechner, and Claudia Wagner. 2024. "The Search for Solid Ground in Text as Data: A Systematic Review of Validation Practices and Practical Recommendations for Validation." *Communication Methods and Measures* 18 (3): 249–77. https://doi.org/10.1080/19312458.2023.2285765.

Bischof, Jonathan M., and Edoardo M. Airoldi. 2012. "Summarizing Topical Content with Word Frequency and Exclusivity." In *Proceedings of the 29th International Conference on Machine Learning (ICML 2012)*. Edinburgh, Scotland, UK.

Feuerriegel, Stefan, Christopher Barrie, M. J. Crockett, Laura K. Globig, Killian L. McLoughlin, Dan-Mircea Mirea, Arthur Spirling, et al. 2026. "A Reporting Checklist for Large Language Models in Behavioural Science." *Nature Human Behaviour*, June. https://doi.org/10.1038/s41562-026-02492-7.

Halterman, Andrew, and Katherine A. Keith. 2026. "Codebook LLMs: Evaluating LLMs as Measurement Tools for Political Science Concepts." *Political Analysis* 34 (2): 188–204. https://doi.org/10.1017/pan.2025.10017.

Hinton, Geoffrey, Oriol Vinyals, and Jeff Dean. 2015. "Distilling the Knowledge in a Neural Network." arXiv. https://doi.org/10.48550/arXiv.1503.02531.

Hopkins, Daniel J., and Gary King. 2010. "A Method of Automated Nonparametric Content Analysis for Social Science." *American Journal of Political Science* 54 (1): 229–47.

Jerzak, Connor T., Gary King, and Anton Strezhnev. 2023. "An Improved Method of Automated Nonparametric Content Analysis for Social Science." *Political Analysis* 31 (1): 42–58. https://doi.org/10.1017/pan.2021.36.

Licht, Hauke, Rupak Sarkar, Patrick Y. Wu, Pranav Goel, Niklas Stoehr, Elliott Ash, and Alexander Miserlis Hoyle. 2025. "Measuring Scalar Constructs in Social Science with LLMs." In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, edited by Christos Christodoulopou-

los, Tanmoy Chakraborty, Carolyn Rose, and Violet Peng, 32144–71. Suzhou, China: Association for Computational Linguistics. https://doi.org/10.18653/v1/2025.emnlp-main.1635.

Marx, Charles, Flavio Calmon, and Berk Ustun. 2020. "Predictive Multiplicity in Classification." In *Proceedings of the 37th International Conference on Machine Learning*, 6765–74. PMLR.

Peterson, Joshua C., Ruairidh M. Battleday, Thomas L. Griffiths, and Olga Russakovsky. 2019. "Human Uncertainty Makes Classification More Robust." arXiv. https://doi.org/10.48550/arXiv.1908.07086.

Ribeiro, Marco Tulio, Tongshuang Wu, Carlos Guestrin, and Sameer Singh. 2020. "Beyond Accuracy: Behavioral Testing of NLP Models with CheckList." In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, 4902–12. Association for Computational Linguistics. https://doi.org/10.18653/v1/2020.acl-main.442.

Warner, Benjamin, Antoine Chaffin, Benjamin Clavié, Orion Weller, Oskar Hallström, Said Taghadouini, Alexis Gallagher, et al. 2024. "Smarter, Better, Faster, Longer: A Modern Bidirectional Encoder for Fast, Memory Efficient, and Long Context Finetuning and Inference." https://doi.org/10.48550/arXiv.2412.13663.

Yang, Eddie, Zoey Wang, Carl Zhou, and Yaosheng Xu. 2025. "Data Annotation with Large Language Models: Lessons from a Large Empirical Evaluation." Working Paper. https://www.eddieyang.net/research/llm_annotation.pdf.

Zenner, Kai, Scott J. Marcus, and Kamil Sekut. 2025. "A Dataset of EU Legal and Policy Instruments for the Digital World." *CEPS*.